

TỐI ƯU HÓA DANH MỤC ĐẦU TƯ CỔ PHIẾU NGÂN HÀNG VIỆT NAM

Sử dụng Mô hình Black-Litterman kết hợp Phân cụm K-means và Tín hiệu Momentum Tương đối (Idiosyncratic Momentum)

Black-Litterman Portfolio Optimization with K-means Clustering and Idiosyncratic Momentum Signals: Evidence from Vietnamese Banking Stocks

Dữ liệu	25 cổ phiếu ngân hàng niêm yết HOSE/HNX, tháng 6/2014 – tháng 5/2026
Phương pháp	K-means (k=4) + Black-Litterman + Rolling window 36 tháng
Tín hiệu	Idiosyncratic momentum 6-1 tháng + Low-vol composite
Kết quả chính	BL Sharpe = 0.9396 > EW (1/N) Sharpe = 0.8373
Drawdown tối đa	BL: -30.7% vs EW: -36.6% vs TAN: -51.5%
Tham chiếu chính	Black & Litterman (1992), Jegadeesh & Titman (1993), DeMiguel et al. (2009), Baker et al. (2011)

1. Tóm tắt Hiệu suất (6 Danh mục, OOS annualised)

So sánh 6 danh mục đầu tư theo 7 chỉ số rủi ro-lợi nhuận. BL = Black-Litterman (mô hình đề xuất), EW = Equal-Weight (chuẩn 1/N của DeMiguel et al. 2009), MKT = Market-cap, TAN = Tangency, MV = Min-Variance, RP = Risk-Parity. Kết quả OOS trên 106 tháng (tháng 8/2017 – tháng 5/2026), lãi suất phi rủi ro = VN10Y/12 (time-varying).

Portfolio	Ann_Return	Ann_Vol	Sharpe	Sortino	MDD	Calmar	Win_Rate
MKT	0.234033	0.270493	0.728858	1.1842	-0.325098	0.719883	0.613208
TAN	0.191990	0.278101	0.557740	0.872662	-0.515448	0.372472	0.622642
MV	0.162408	0.253993	0.494209	0.801307	-0.384445	0.422448	0.575472
BL	0.301188	0.281283	0.939644	1.6281	-0.306594	0.982370	0.613208
EW	0.266157	0.273821	0.837317	1.4530	-0.365473	0.728253	0.622642
RP	0.256658	0.271436	0.809677	1.3900	-0.357157	0.718613	0.622642

1b. Kiểm định t-test (BL vs các danh mục khác)

Kiểm định t-test theo cặp (paired t-test) cho lợi nhuận tháng OOS. H_0 : trung bình lợi nhuận BL = trung bình danh mục so sánh.

Pair	t_statistic	p_value	is_significant_5pct
BL vs MKT	1.1420	0.256054	False
BL vs TAN	2.0206	0.045872	True
BL vs MV	2.7305	0.007418	True
BL vs EW	0.774217	0.440543	False
BL vs RP	0.992437	0.323266	False

1c. Kiểm định Jobson-Korkie (so sánh Sharpe ratio)

Kiểm định Jobson-Korkie (1981): H_0 : $SR_{BL} = SR_{danh_muc_khac}$. z-statistic dương = BL có Sharpe cao hơn. $p < 0.05$ = có ý nghĩa thống kê.

Pair	z_statistic	p_value	is_significant_5pct
BL vs MKT	0.965171	0.334459	False
BL vs TAN	1.9395	0.052441	False
BL vs MV	2.3255	0.020046	True

Pair	z_statistic	p_value	is_significant_5pct
BL vs EW	0.606804	0.543981	False
BL vs RP	0.775014	0.438331	False

2. Phân tích Tính vững theo Giai đoạn (Sub-period Robustness)

Chia giai đoạn OOS làm hai nửa bằng nhau để kiểm tra tính vững của kết quả. Period 1 (2017–2021): giai đoạn tăng trưởng mạnh, thị trường bull. Period 2 (2022–2026): giai đoạn biến động cao (hậu COVID, lãi suất tăng, thị trường điều chỉnh). BL thắng EW rõ ràng ở Period 2 (Sharpe 0.86 vs 0.38), xác nhận mô hình hoạt động tốt nhất khi cần nhất.

Period	Period_Start	Period_End	N_months	Portfolio	Ann_Return	Ann_Vol	Sharpe	Sortino	MDD	Calmar
Full	2017-08-31	2026-05-31	106	BL	0.301188	0.281283	0.939644	1.6281	-0.306594	0.982370
Full	2017-08-31	2026-05-31	106	EW	0.266157	0.273821	0.837317	1.4530	-0.365473	0.728253
Period_1	2017-08-31	2021-12-31	53	BL	0.414678	0.351477	1.0696	1.9065	-0.306594	1.3525
Period_1	2017-08-31	2021-12-31	53	EW	0.433110	0.320610	1.2301	2.2541	-0.303570	1.4267
Period_2	2022-01-31	2026-05-31	53	BL	0.196101	0.187020	0.861236	1.3701	-0.222165	0.882679
Period_2	2022-01-31	2026-05-31	53	EW	0.117209	0.214168	0.383702	0.593285	-0.365473	0.320704

3. Phân tích Information Coefficient (IC) của Tín hiệu K-means

IC (Spearman rank correlation) đo mức độ tín hiệu idiosyncratic momentum dự báo đúng chiều lợi nhuận thực tế tháng kế tiếp. IC > 0: tín hiệu xếp hạng đúng chiều. Hit rate: tỷ lệ bước rebalancing mà cluster Long thực sự vượt cluster Short. Cả hai chỉ số > 50% xác nhận tín hiệu có giá trị dự báo dương.

Chỉ số	Giá trị
Mean IC	+0.0667
IC > 0 (%)	50.0%
Hit Rate (%)	50.9%
Std IC	0.3618
Số bước OOS	106

IC theo từng bước rebalancing:

Date	IC	Hit	N_stocks
2017-08-31	-0.357143	1	7
2017-09-30	-0.071429	0	7
2017-10-31	-0.107143	0	7
2017-11-30	-0.321429	0	7
2017-12-31	-0.464286	0	7
2018-01-31	0.071429	0	7
2018-02-28	0.535714	1	7
2018-03-31	0.178571	1	7
2018-04-30	0.000000	1	7
2018-05-31	-0.357143	1	7
2018-06-30	-0.250000	0	7
2018-07-31	-0.178571	0	7
2018-08-31	0.392857	1	7
2018-09-30	-0.821429	0	7
2018-10-31	0.357143	1	7

Date	IC	Hit	N_stocks
2018-11-30	0.750000	1	7
2018-12-31	0.500000	1	7
2019-01-31	0.607143	1	7
2019-02-28	0.142857	1	7
2019-03-31	0.000000	1	7
2019-04-30	-0.018019	0	7
2019-05-31	0.535714	1	7
2019-06-30	0.892857	1	7
2019-07-31	0.642857	1	7
2019-08-31	0.321429	0	7
2019-09-30	0.821429	1	7
2019-10-31	0.678571	1	7
2019-11-30	0.857143	1	7
2019-12-31	-0.035714	0	7
2020-01-31	-0.642857	0	7
2020-02-29	-0.964286	0	7
2020-03-31	0.214286	0	8
2020-04-30	-0.071429	0	8
2020-05-31	-0.142857	0	8
2020-06-30	-0.738095	0	8
2020-07-31	-0.095238	0	8
2020-08-31	-0.150000	0	9
2020-09-30	0.466667	1	9
2020-10-31	0.406061	0	10
2020-11-30	-0.260606	0	10
2020-12-31	-0.072727	0	11

Date	IC	Hit	N_stocks
2021-01-31	0.763636	1	11
2021-02-28	0.482517	1	12
2021-03-31	0.252747	1	13
2021-04-30	0.615385	1	13
2021-05-31	0.489011	1	13
2021-06-30	-0.358242	0	14
2021-07-31	0.037363	1	14
2021-08-31	-0.142857	0	15
2021-09-30	-0.342857	0	15
2021-10-31	-0.135714	0	15
2021-11-30	0.050000	0	15
2021-12-31	-0.150000	1	15
2022-01-31	-0.010714	0	15
2022-02-28	0.428571	0	15
2022-03-31	0.210714	0	15
2022-04-30	-0.182143	0	15
2022-05-31	-0.203571	1	15
2022-06-30	-0.221429	0	15
2022-07-31	-0.150000	1	15
2022-08-31	0.428571	1	15
2022-09-30	0.564706	1	16
2022-10-31	0.229412	1	16
2022-11-30	-0.079412	1	16
2022-12-31	0.258824	1	16
2023-01-31	-0.208824	0	16
2023-02-28	0.085294	0	16

Date	IC	Hit	N_stocks
2023-03-31	0.282353	1	16
2023-04-30	-0.573529	0	16
2023-05-31	-0.044118	1	16
2023-06-30	-0.282353	0	16
2023-07-31	0.250000	1	16
2023-08-31	0.023529	0	16
2023-09-30	-0.247549	1	17
2023-10-31	-0.058824	0	17
2023-11-30	0.061275	0	17
2023-12-31	-0.091747	1	19
2024-01-31	0.277193	1	19
2024-02-29	0.113495	0	22
2024-03-31	0.250556	1	23
2024-04-30	-0.170949	1	23
2024-05-31	-0.030435	0	24
2024-06-30	-0.049608	1	24
2024-07-31	0.374783	1	24
2024-08-31	0.320870	0	24
2024-09-30	-0.008462	0	25
2024-10-31	0.166154	1	25
2024-11-30	0.135385	0	25
2024-12-31	0.373846	1	25
2025-01-31	-0.030390	1	25
2025-02-28	-0.227692	0	25
2025-03-31	-0.395845	0	25
2025-04-30	-0.026154	1	25

Date	IC	Hit	N_stocks
2025-05-31	0.258462	0	25
2025-06-30	0.160769	1	25
2025-07-31	-0.259231	0	25
2025-08-31	-0.230769	0	25
2025-09-30	0.390769	1	25
2025-10-31	0.214615	1	25
2025-11-30	0.056549	1	25
2025-12-31	-0.056923	1	25
2026-01-31	-0.414695	0	25
2026-02-28	0.448932	1	25
2026-03-31	-0.039231	0	25
2026-04-30	0.136923	1	25
2026-05-31	0.049231	0	25

4. Kiểm định Số cụm K tối ưu (Elbow + Silhouette)

Xác định k tối ưu cho K-means bằng hai phương pháp độc lập: (1) Elbow method: inertia (tổng bình phương trong cụm) giảm và tìm 'điểm gãy'; (2) Silhouette score (Rousseeuw 1987): đo mức độ tách biệt giữa các cụm (cao hơn = tách biệt tốt hơn). Sensitivity analysis OOS xác nhận k=4 tối ưu về Sharpe ratio (BL Sharpe 0.9396, cao nhất trong các k được test với cùng mẫu).

k	Mean_Inertia	Mean_Silhouette	Std_Silhouette
2	19.4828	0.420100	0.089200
3	11.0918	0.405700	0.053100
4	7.3568	0.389100	0.048000
5	5.0307	0.382500	0.060200
6	3.5828	0.371600	0.066600

5. Phân tích Độ nhạy (One-at-a-time Parameter Sweep)

Thay đổi một tham số tại một thời điểm trong khi giữ nguyên các tham số còn lại. Baseline: lookback=36, max_weight=30%, k=4, rf=VN10Y dynamic. Kết quả cho thấy BL ổn định qua các giá trị tham số hợp lý.

Tham số: lookback

Value	Is_Baseline	Ann_Return	Ann_Vol	Sharpe	MDD
24		0.237854	0.282611	0.702925	-0.374048
36	★	0.301188	0.281283	0.939644	-0.306594
48		0.268895	0.264060	0.885409	-0.309510

Tham số: max_weight

Value	Is_Baseline	Ann_Return	Ann_Vol	Sharpe	MDD
20%		0.271627	0.269711	0.870359	-0.340115
30%	★	0.301188	0.281283	0.939644	-0.306594
40%		0.292378	0.283401	0.901537	-0.337344

Tham số: k_clusters

Value	Is_Baseline	Ann_Return	Ann_Vol	Sharpe	MDD
2		0.240639	0.281538	0.723727	-0.420598
3		0.274545	0.285040	0.833787	-0.328046
4	★	0.301188	0.281283	0.939644	-0.306594
5		0.303473	0.283770	0.939461	-0.308210
6		0.318746	0.274950	1.0400	-0.247620

Tham số: rf_rate

Value	Is_Baseline	Ann_Return	Ann_Vol	Sharpe	MDD
4% fixed		0.304884	0.282740	0.936849	-0.306046
5% fixed		0.308893	0.284987	0.908438	-0.306493
6% fixed		0.310200	0.286957	0.871907	-0.306994

Value	Is_Baseline	Ann_Return	Ann_Vol	Sharpe	MDD
VN10Y (dynamic)	★	0.301188	0.281283	0.939644	-0.306594

6. Phân tích Drawdown Chi tiết

6a. Tóm tắt Drawdown theo Danh mục

Portfolio	N_Drawdowns	Max_DD_pct	Avg_DD_Depth_pct	Avg_Recovery_Months	N_Months_Below_Neg20pct
MKT	9	-32.5100	-15.8300	6.9000	19
TAN	5	-51.5400	-24.9000	9.6000	51
MV	5	-38.4400	-21.3200	11.8000	26
BL	12	-30.6600	-9.4800	3.8000	21
EW	7	-36.5500	-15.3500	9.0000	34
RP	7	-35.7200	-15.6100	6.5000	34

6b. Các sự kiện Drawdown từng Danh mục

Liệt kê từng đợt drawdown: ngày bắt đầu, đáy, phục hồi, mức giảm tối đa, số tháng phục hồi.

Portfolio	DD_Start	DD_Trough	DD_Recovery	Depth_pct	Months_to_Trough	Recovery_Months	Total_Months
MKT	2018-04-30	2018-06-30	2019-12-31 00:00:00	-26.9400	2	18.0000	20
MKT	2020-02-29	2020-03-31	2020-11-30 00:00:00	-31.0700	1	8.0000	9
MKT	2021-01-31	2021-01-31	2021-02-28 00:00:00	-7.8800	0	1.0000	1
MKT	2021-07-31	2021-08-31	2022-01-31 00:00:00	-14.3800	1	5.0000	6
MKT	2022-02-28	2022-10-31	2024-02-29 00:00:00	-32.5100	8	16.0000	24
MKT	2024-04-30	2024-06-30	2024-08-31 00:00:00	-6.9900	2	2.0000	4
MKT	2024-10-31	2024-11-30	2025-01-31 00:00:00	-2.8100	1	2.0000	3
MKT	2025-04-30	2025-04-30	2025-07-31 00:00:00	-7.6900	0	3.0000	3
MKT	2025-09-30	2026-03-31	Not recovered	-12.2300	6	nan	8
TAN	2018-04-30	2020-03-31	2020-12-31 00:00:00	-38.0500	23	9.0000	32
TAN	2021-01-31	2021-01-31	2021-02-28 00:00:00	-8.0800	0	1.0000	1
TAN	2021-07-31	2021-08-31	2021-12-31 00:00:00	-15.6200	1	4.0000	5
TAN	2022-02-28	2022-12-31	2025-08-31 00:00:00	-51.5400	10	32.0000	42

Portfolio	DD_Start	DD_Trough	DD_Recovery	Depth_pct	Months_to_Trough	Recovery_Months	Total_Months
TAN	2025-09-30	2026-03-31	2026-05-31 00:00:00	-11.1900	6	2.0000	8
MV	2018-04-30	2020-03-31	2020-12-31 00:00:00	-38.4400	23	9.0000	32
MV	2021-01-31	2021-01-31	2021-03-31 00:00:00	-8.0800	0	2.0000	2
MV	2021-06-30	2021-08-31	2022-01-31 00:00:00	-13.0300	2	5.0000	7
MV	2022-02-28	2022-12-31	2025-07-31 00:00:00	-31.6700	10	31.0000	41
MV	2025-09-30	2026-03-31	Not recovered	-15.3700	6	nan	8
BL	2017-10-31	2017-10-31	2017-11-30 00:00:00	-2.0500	0	1.0000	1
BL	2017-12-31	2017-12-31	2018-01-31 00:00:00	-1.1400	0	1.0000	1
BL	2018-04-30	2019-01-31	2020-01-31 00:00:00	-30.4700	9	12.0000	21
BL	2020-02-29	2020-03-31	2020-09-30 00:00:00	-30.6600	1	6.0000	7
BL	2021-01-31	2021-01-31	2021-02-28 00:00:00	-3.2400	0	1.0000	1
BL	2021-06-30	2022-10-31	2024-01-31 00:00:00	-25.4300	16	15.0000	31
BL	2024-06-30	2024-06-30	2024-07-31 00:00:00	-3.9000	0	1.0000	1
BL	2024-11-30	2024-11-30	2024-12-31 00:00:00	-0.850000	0	1.0000	1
BL	2025-03-31	2025-04-30	2025-06-30 00:00:00	-5.9300	1	2.0000	3
BL	2025-09-30	2025-09-30	2025-10-31 00:00:00	-2.0600	0	1.0000	1
BL	2025-11-30	2025-11-30	2025-12-31 00:00:00	-1.4700	0	1.0000	1
BL	2026-01-31	2026-03-31	Not recovered	-6.5600	2	nan	4
EW	2018-04-30	2018-12-31	2020-09-30 00:00:00	-30.3600	8	21.0000	29
EW	2021-01-31	2021-01-31	2021-02-28 00:00:00	-6.2900	0	1.0000	1
EW	2021-06-30	2021-08-31	2022-01-31 00:00:00	-13.5800	2	5.0000	7
EW	2022-02-28	2022-10-31	2024-09-30 00:00:00	-36.5500	8	23.0000	31
EW	2024-10-31	2024-11-30	2025-02-28 00:00:00	-4.6700	1	3.0000	4
EW	2025-04-30	2025-04-30	2025-05-31 00:00:00	-4.8700	0	1.0000	1
EW	2025-09-30	2026-03-31	Not recovered	-11.1000	6	nan	8
RP	2018-04-30	2020-03-31	2020-09-30 00:00:00	-32.1400	23	6.0000	29

Portfolio	DD_Start	DD_Trough	DD_Recovery	Depth_pct	Months_to_Trough	Recovery_Months	Total_Months
RP	2021-01-31	2021-01-31	2021-02-28 00:00:00	-6.0000	0	1.0000	1
RP	2021-06-30	2021-08-31	2022-01-31 00:00:00	-13.7700	2	5.0000	7
RP	2022-02-28	2022-10-31	2024-09-30 00:00:00	-35.7200	8	23.0000	31
RP	2024-10-31	2024-11-30	2025-02-28 00:00:00	-4.5500	1	3.0000	4
RP	2025-04-30	2025-04-30	2025-05-31 00:00:00	-5.3300	0	1.0000	1
RP	2025-09-30	2026-03-31	Not recovered	-11.7400	6	nan	8

7. Kiểm định Phân phối Lợi nhuận

7a. Kiểm định Jarque-Bera (Chuẩn hóa)

H_0 : lợi nhuận có phân phối chuẩn. $p < 0.05$ = bác bỏ chuẩn hóa. Hầu hết danh mục cổ phiếu đều có fat tail và skewness.

Portfolio	JB_stat	JB_pval	Skewness	Excess_Kurtosis	is_normal_5pct
MKT	10.1416	0.006277	-0.175466	1.6061	False
TAN	16.2824	0.000291	-0.212219	2.0242	False
MV	83.7728	0.000000	0.143990	4.6169	False
BL	16.2666	0.000294	-0.013640	2.0714	False
EW	7.2799	0.026253	0.198406	1.3413	False
RP	10.7793	0.004564	0.150547	1.6675	False

7b. Kiểm định Ljung-Box (Tự tương quan)

H_0 : lợi nhuận không có tự tương quan (white noise). $p < 0.05$ = tồn tại tự tương quan (có thể khai thác được).

Portfolio	LB5_stat	LB5_pval	LB5_no_autocorr	LB10_stat	LB10_pval	LB10_no_autocorr	LB20_stat	LB20_pval	LB20_no_autocorr
MKT	8.8677	0.114457	True	19.7177	0.032039	False	34.2652	0.024385	False
TAN	4.5857	0.468501	True	11.5802	0.314135	True	25.5470	0.181299	True
MV	2.0715	0.839169	True	16.2845	0.091772	True	25.7963	0.172645	True
BL	5.9811	0.308063	True	12.4184	0.258027	True	22.7773	0.299862	True
EW	3.0475	0.692660	True	12.8746	0.230769	True	23.9440	0.244848	True
RP	2.6747	0.749978	True	13.3277	0.205920	True	24.2006	0.233738	True

8. Monte Carlo Robustness (Nhiều trong View BL)

Kiểm tra tính ổn định khi thêm nhiễu Gaussian vào vector view q của BL. Delta Noise từ -10% đến +10%. Mô hình BL khai thác tính tuyến tính $\mu_{BL}(q') = \text{base_vec} + A_mat @ q'$ để tránh tối ưu lại 2000 lần mỗi bước.

Delta_Noise	Port_Type	Mean_Sharpe	Std_Sharpe
-0.100000	BL	0.828200	0.527900
-0.100000	TAN	1.0271	0.398100
-0.050000	BL	0.831400	0.530900
-0.050000	TAN	1.0271	0.398100
0.000000	BL	0.831100	0.530600
0.000000	TAN	1.0271	0.398100
0.050000	BL	0.830700	0.530200
0.050000	TAN	1.0271	0.398100
0.100000	BL	0.830400	0.530000
0.100000	TAN	1.0271	0.398100

9. Chuỗi Lợi nhuận Out-of-Sample (106 tháng)

Lợi nhuận tháng (simple return) của 6 danh mục và lãi suất phi rủi ro cho từng tháng OOS (tháng 8/2017 – tháng 5/2026).

Date	RF_monthly	MKT	TAN	MV	BL	EW	RP
2017-08-31	0.004833	-0.026592	0.021803	0.000532	-0.050765	-0.024700	-0.020210
2017-09-30	0.004833	0.008600	0.017614	0.012287	0.007288	0.014080	0.012381
2017-10-31	0.004833	0.059921	0.050255	0.029226	-0.020522	0.017326	0.018877
2017-11-30	0.004833	0.184409	0.152730	0.163976	0.175780	0.187071	0.182169
2017-12-31	0.004833	0.048090	0.050632	0.034358	-0.011364	0.018884	0.020052
2018-01-31	0.003817	0.243752	0.230535	0.294043	0.267976	0.267098	0.266837
2018-02-28	0.003817	0.120337	0.099277	0.077586	0.134487	0.098473	0.096365
2018-03-31	0.003817	0.016710	0.017842	0.015931	0.041644	0.020543	0.021760
2018-04-30	0.003817	-0.146733	-0.126264	-0.123802	-0.138059	-0.131849	-0.131825
2018-05-31	0.003817	-0.114886	-0.073255	-0.076191	-0.114640	-0.111764	-0.103784
2018-06-30	0.003817	-0.032586	-0.053069	-0.064188	-0.063714	-0.069962	-0.069599
2018-07-31	0.003817	0.007452	0.022622	0.022947	-0.005468	0.006366	0.009336
2018-08-31	0.003817	0.123037	0.060966	0.055417	0.172304	0.095768	0.090940
2018-09-30	0.003817	0.018549	0.002697	0.004246	0.012967	0.040664	0.033908
2018-10-31	0.003817	-0.121796	-0.095982	-0.095374	-0.126462	-0.115741	-0.111643
2018-11-30	0.003817	0.007542	-0.013142	-0.013195	-0.004620	-0.005717	-0.007262
2018-12-31	0.003817	-0.018578	-0.045817	-0.044849	-0.051438	-0.037550	-0.042356
2019-01-31	0.003817	0.012980	0.048089	0.056572	-0.000935	0.021086	0.028444
2019-02-28	0.003817	0.041454	0.019705	0.016662	0.034640	0.024012	0.023248
2019-03-31	0.003817	0.087474	0.063028	0.052068	0.074795	0.052151	0.052914
2019-04-30	0.003817	-0.009915	-0.002335	-0.001570	-0.007965	-0.014232	-0.012657
2019-05-31	0.003817	-0.040294	-0.036642	-0.028837	-0.033866	-0.045280	-0.042339
2019-06-30	0.003817	0.018816	0.015486	0.011442	0.004369	-0.006345	-0.003536

Date	RF_monthly	MKT	TAN	MV	BL	EW	RP
2019-07-31	0.003817	0.103573	0.079358	0.061666	0.099161	0.045768	0.049044
2019-08-31	0.003817	0.003178	0.004126	-0.007868	0.014018	-0.003207	-0.004538
2019-09-30	0.003817	0.044382	0.054395	0.054182	0.053843	0.042613	0.044720
2019-10-31	0.003817	0.047669	0.035522	0.034822	0.044811	0.028980	0.030364
2019-11-30	0.003817	-0.019920	-0.028705	-0.038766	-0.023996	-0.042676	-0.042495
2019-12-31	0.003817	0.067450	0.011788	-0.001566	0.045882	0.032970	0.026890
2020-01-31	0.002500	0.040052	0.009322	-0.000004	0.081208	0.064014	0.053729
2020-02-29	0.002500	-0.047792	-0.041175	0.029620	-0.026162	0.089194	0.078052
2020-03-31	0.002500	-0.276107	-0.300868	-0.301494	-0.287965	-0.240255	-0.256221
2020-04-30	0.002500	0.137555	0.150589	0.155972	0.134515	0.178490	0.171579
2020-05-31	0.002500	0.168756	0.089504	0.089163	0.068182	0.104038	0.112812
2020-06-30	0.002500	-0.046926	-0.055452	-0.035664	-0.015210	-0.018955	-0.017534
2020-07-31	0.002500	-0.040333	-0.053315	-0.048023	-0.051459	-0.035201	-0.033178
2020-08-31	0.002500	0.121801	0.114106	0.125604	0.146489	0.146584	0.145434
2020-09-30	0.002500	0.034012	0.054371	0.061032	0.130943	0.097577	0.096664
2020-10-31	0.002500	-0.000661	0.013751	0.034318	0.017359	0.034211	0.037935
2020-11-30	0.002500	0.114471	0.091426	0.111259	0.109581	0.107254	0.107803
2020-12-31	0.002500	0.100638	0.157318	0.165052	0.128925	0.139943	0.145595
2021-01-31	0.002108	-0.078755	-0.080811	-0.080838	-0.032355	-0.062929	-0.059956
2021-02-28	0.002108	0.128424	0.104445	0.056616	0.132551	0.144913	0.127255
2021-03-31	0.002108	0.043929	0.195558	0.224080	0.238249	0.147558	0.164876
2021-04-30	0.002108	0.069686	0.087668	0.039030	0.155218	0.089302	0.062330
2021-05-31	0.002108	0.154716	0.220957	0.144040	0.201190	0.233709	0.228602
2021-06-30	0.002108	0.044947	0.004665	-0.018138	-0.068264	-0.004647	-0.006477
2021-07-31	0.002108	-0.113116	-0.123328	-0.111084	-0.094478	-0.091124	-0.092756
2021-08-31	0.002108	-0.034625	-0.037530	-0.003530	-0.038708	-0.044764	-0.043336

Date	RF_monthly	MKT	TAN	MV	BL	EW	RP
2021-09-30	0.002108	0.000996	0.011400	0.025518	-0.028558	0.004185	0.007465
2021-10-31	0.002108	0.030575	0.031964	0.028788	0.032399	0.040120	0.038113
2021-11-30	0.002108	0.032911	0.057504	0.030239	0.059137	0.052475	0.051667
2021-12-31	0.002108	0.020766	0.083291	0.030334	0.031570	0.032872	0.034591
2022-01-31	0.003333	0.104257	0.031341	0.043366	0.035040	0.076226	0.074845
2022-02-28	0.003333	-0.043408	-0.019388	-0.027255	-0.020250	-0.038160	-0.037485
2022-03-31	0.003333	-0.020615	-0.022193	-0.016968	0.063683	0.004651	0.002219
2022-04-30	0.003333	-0.082787	-0.147377	-0.099271	-0.107490	-0.109177	-0.105956
2022-05-31	0.003333	-0.072449	-0.103551	-0.044184	-0.041574	-0.081414	-0.075345
2022-06-30	0.003333	-0.068432	-0.136007	-0.070779	-0.082785	-0.087762	-0.087492
2022-07-31	0.003333	0.044292	0.090316	0.018105	0.065715	0.059262	0.055490
2022-08-31	0.003333	0.074039	0.013024	0.046948	0.088153	0.050773	0.050525
2022-09-30	0.003333	-0.143466	-0.127780	-0.109730	-0.138716	-0.136735	-0.133303
2022-10-31	0.003333	-0.053868	-0.057153	-0.033293	-0.007417	-0.084503	-0.080754
2022-11-30	0.003333	0.083200	-0.084619	0.002064	0.029794	0.028596	0.027381
2022-12-31	0.003333	-0.006743	-0.079643	-0.028363	0.009177	-0.010396	-0.011879
2023-01-31	0.002917	0.148118	0.157561	0.143657	0.149694	0.152493	0.153030
2023-02-28	0.002917	-0.047656	-0.080255	0.005233	-0.052600	-0.061735	-0.054593
2023-03-31	0.002917	0.047184	0.070172	-0.007017	0.050817	0.052515	0.047303
2023-04-30	0.002917	-0.020110	-0.037522	-0.006488	-0.021901	-0.006776	-0.007503
2023-05-31	0.002917	0.021645	0.031663	0.037142	0.019517	0.023716	0.024116
2023-06-30	0.002917	0.046573	0.048782	0.045352	0.045279	0.056366	0.052440
2023-07-31	0.002917	0.072933	0.058160	0.042184	0.044361	0.064792	0.063650
2023-08-31	0.002917	-0.002617	0.011854	-0.017438	0.029900	0.016439	0.012578
2023-09-30	0.002917	-0.031677	-0.030950	-0.019040	-0.039045	-0.045869	-0.041493
2023-10-31	0.002917	-0.059835	-0.003531	-0.033969	-0.054354	-0.075824	-0.070516

Date	RF_monthly	MKT	TAN	MV	BL	EW	RP
2023-11-30	0.002917	0.024465	0.009890	0.007350	0.018884	0.044460	0.040951
2023-12-31	0.002917	0.023527	0.008199	0.017215	0.029754	0.036214	0.037548
2024-01-31	0.002550	0.091943	0.075801	0.078301	0.099873	0.072711	0.076635
2024-02-29	0.002550	0.095842	0.043038	0.063421	0.051506	0.065798	0.070645
2024-03-31	0.002550	0.003849	0.003879	-0.000978	0.006890	0.004252	0.004793
2024-04-30	0.002550	-0.048663	-0.013038	-0.042120	0.017273	-0.029587	-0.033917
2024-05-31	0.002550	-0.003662	0.002642	0.010818	0.010148	0.022505	0.020859
2024-06-30	0.002550	-0.018775	-0.022561	-0.020244	-0.039007	-0.012736	-0.012871
2024-07-31	0.002550	0.048634	0.052790	0.019355	0.047735	0.037203	0.038247
2024-08-31	0.002550	0.025287	0.051975	0.001160	0.033135	0.018478	0.016333
2024-09-30	0.002550	0.026921	0.012138	-0.015353	0.036651	0.039055	0.034229
2024-10-31	0.002550	-0.008231	-0.015224	-0.011221	0.010314	-0.024802	-0.022269
2024-11-30	0.002550	-0.020013	-0.021911	-0.013213	-0.008478	-0.022484	-0.023717
2024-12-31	0.002550	0.018309	0.093132	0.011599	0.040576	0.030247	0.029882
2025-01-31	0.002583	0.018841	0.049506	0.037886	0.063690	0.011546	0.010500
2025-02-28	0.002583	0.028168	0.014633	0.032052	0.039368	0.045352	0.044752
2025-03-31	0.002583	0.000756	-0.039288	0.006031	-0.024999	0.000248	0.001343
2025-04-30	0.002583	-0.076861	-0.054662	-0.063103	-0.035208	-0.048695	-0.053265
2025-05-31	0.002583	0.042136	0.026155	0.015781	0.040975	0.077463	0.068187
2025-06-30	0.002583	0.038606	0.018894	0.012406	0.063788	0.029487	0.029292
2025-07-31	0.002583	0.101024	0.086195	0.093231	0.115972	0.156467	0.153496
2025-08-31	0.002583	0.191455	0.174109	0.093140	0.092556	0.155768	0.152875
2025-09-30	0.002583	-0.061845	-0.026854	-0.084397	-0.020579	-0.059003	-0.065239
2025-10-31	0.002583	-0.042226	-0.006632	-0.027556	0.024278	-0.019916	-0.022683
2025-11-30	0.002583	-0.019884	-0.018711	-0.037211	-0.014664	-0.016465	-0.017814
2025-12-31	0.002583	0.025236	-0.006659	-0.001797	0.053382	0.016349	0.014466

Date	RF_monthly	MKT	TAN	MV	BL	EW	RP
2026-01-31	0.003625	0.098177	0.000622	0.047368	-0.015152	0.034101	0.042349
2026-02-28	0.003625	-0.023014	0.007064	-0.013919	0.013093	-0.004318	-0.007500
2026-03-31	0.003625	-0.093967	-0.064750	-0.042369	-0.063529	-0.063432	-0.062731
2026-04-30	0.003625	0.029591	0.091541	0.029153	0.033291	0.036212	0.033699
2026-05-31	0.003625	0.029712	0.079321	0.011041	0.010291	0.011828	0.009975

10. Tín hiệu Phân cụm K-means (k=4)

Nhãn cụm và vị thế view (Long / Short / Neutral) của từng cổ phiếu tại mỗi kỳ tái cân bằng. Phân cụm dựa trên tín hiệu tổng hợp: idiosyncratic momentum 6-1 tháng (loại bỏ yếu tố thị trường) + low-volatility feature (âm hóa vol). K-means (k=4) trên không gian chuẩn hóa 2 chiều.

Tổng số quan sát: 1,561 dòng

Date	Ticker	Cluster_Label	Assigned_View_Position
2017-08-31	VCB	0	Short
2017-08-31	BID	2	Neutral
2017-08-31	CTG	2	Neutral
2017-08-31	MBB	3	Long
2017-08-31	ACB	1	Neutral
2017-08-31	STB	1	Neutral
2017-08-31	SHB	1	Neutral
2017-09-30	VCB	2	Short
2017-09-30	BID	0	Neutral
2017-09-30	CTG	0	Neutral
2017-09-30	MBB	1	Long
2017-09-30	ACB	3	Neutral
2017-09-30	STB	0	Neutral
2017-09-30	SHB	3	Neutral
2017-10-31	VCB	2	Short
2017-10-31	BID	0	Neutral
2017-10-31	CTG	0	Neutral
2017-10-31	MBB	3	Long
2017-10-31	ACB	2	Short
2017-10-31	STB	0	Neutral
2017-10-31	SHB	1	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2017-11-30	VCB	0	Short
2017-11-30	BID	2	Neutral
2017-11-30	CTG	2	Neutral
2017-11-30	MBB	3	Long
2017-11-30	ACB	1	Neutral
2017-11-30	STB	2	Neutral
2017-11-30	SHB	1	Neutral
2017-12-31	VCB	2	Short
2017-12-31	BID	0	Neutral
2017-12-31	CTG	0	Neutral
2017-12-31	MBB	3	Long
2017-12-31	ACB	1	Neutral
2017-12-31	STB	0	Neutral
2017-12-31	SHB	1	Neutral
2018-01-31	VCB	0	Neutral
2018-01-31	BID	3	Neutral
2018-01-31	CTG	1	Short
2018-01-31	MBB	2	Long
2018-01-31	ACB	0	Neutral
2018-01-31	STB	1	Short
2018-01-31	SHB	1	Short
2018-02-28	VCB	1	Neutral
2018-02-28	BID	0	Neutral
2018-02-28	CTG	0	Neutral
2018-02-28	MBB	2	Long
2018-02-28	ACB	1	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2018-02-28	STB	3	Short
2018-02-28	SHB	1	Neutral
2018-03-31	VCB	2	Long
2018-03-31	BID	0	Neutral
2018-03-31	CTG	0	Neutral
2018-03-31	MBB	3	Neutral
2018-03-31	ACB	2	Long
2018-03-31	STB	1	Short
2018-03-31	SHB	0	Neutral
2018-04-30	VCB	0	Long
2018-04-30	BID	2	Neutral
2018-04-30	CTG	1	Short
2018-04-30	MBB	3	Neutral
2018-04-30	ACB	0	Long
2018-04-30	STB	1	Short
2018-04-30	SHB	2	Neutral
2018-05-31	VCB	2	Long
2018-05-31	BID	0	Neutral
2018-05-31	CTG	0	Neutral
2018-05-31	MBB	3	Neutral
2018-05-31	ACB	3	Neutral
2018-05-31	STB	1	Short
2018-05-31	SHB	0	Neutral
2018-06-30	VCB	0	Neutral
2018-06-30	BID	3	Long
2018-06-30	CTG	1	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2018-06-30	MBB	0	Neutral
2018-06-30	ACB	0	Neutral
2018-06-30	STB	2	Short
2018-06-30	SHB	1	Neutral
2018-07-31	VCB	0	Neutral
2018-07-31	BID	1	Long
2018-07-31	CTG	1	Long
2018-07-31	MBB	2	Neutral
2018-07-31	ACB	0	Neutral
2018-07-31	STB	3	Short
2018-07-31	SHB	1	Long
2018-08-31	VCB	2	Neutral
2018-08-31	BID	0	Neutral
2018-08-31	CTG	0	Neutral
2018-08-31	MBB	2	Neutral
2018-08-31	ACB	3	Long
2018-08-31	STB	1	Short
2018-08-31	SHB	0	Neutral
2018-09-30	VCB	2	Long
2018-09-30	BID	0	Neutral
2018-09-30	CTG	0	Neutral
2018-09-30	MBB	2	Long
2018-09-30	ACB	2	Long
2018-09-30	STB	3	Short
2018-09-30	SHB	1	Neutral
2018-10-31	VCB	2	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2018-10-31	BID	3	Neutral
2018-10-31	CTG	0	Neutral
2018-10-31	MBB	2	Long
2018-10-31	ACB	2	Long
2018-10-31	STB	3	Neutral
2018-10-31	SHB	1	Short
2018-11-30	VCB	2	Long
2018-11-30	BID	3	Neutral
2018-11-30	CTG	0	Neutral
2018-11-30	MBB	2	Long
2018-11-30	ACB	2	Long
2018-11-30	STB	1	Short
2018-11-30	SHB	1	Short
2018-12-31	VCB	2	Long
2018-12-31	BID	3	Neutral
2018-12-31	CTG	0	Neutral
2018-12-31	MBB	0	Neutral
2018-12-31	ACB	2	Long
2018-12-31	STB	3	Neutral
2018-12-31	SHB	1	Short
2019-01-31	VCB	0	Long
2019-01-31	BID	3	Neutral
2019-01-31	CTG	2	Neutral
2019-01-31	MBB	0	Long
2019-01-31	ACB	2	Neutral
2019-01-31	STB	3	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2019-01-31	SHB	1	Short
2019-02-28	VCB	2	Long
2019-02-28	BID	3	Neutral
2019-02-28	CTG	0	Short
2019-02-28	MBB	1	Neutral
2019-02-28	ACB	1	Neutral
2019-02-28	STB	3	Neutral
2019-02-28	SHB	0	Short
2019-03-31	VCB	0	Neutral
2019-03-31	BID	2	Long
2019-03-31	CTG	1	Short
2019-03-31	MBB	0	Neutral
2019-03-31	ACB	0	Neutral
2019-03-31	STB	3	Neutral
2019-03-31	SHB	1	Short
2019-04-30	VCB	0	Neutral
2019-04-30	BID	2	Long
2019-04-30	CTG	1	Short
2019-04-30	MBB	0	Neutral
2019-04-30	ACB	0	Neutral
2019-04-30	STB	3	Neutral
2019-04-30	SHB	1	Short
2019-05-31	VCB	2	Neutral
2019-05-31	BID	3	Neutral
2019-05-31	CTG	0	Short
2019-05-31	MBB	2	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2019-05-31	ACB	2	Neutral
2019-05-31	STB	1	Long
2019-05-31	SHB	3	Neutral
2019-06-30	VCB	0	Long
2019-06-30	BID	2	Neutral
2019-06-30	CTG	1	Short
2019-06-30	MBB	3	Neutral
2019-06-30	ACB	3	Neutral
2019-06-30	STB	1	Short
2019-06-30	SHB	1	Short
2019-07-31	VCB	2	Long
2019-07-31	BID	3	Neutral
2019-07-31	CTG	0	Short
2019-07-31	MBB	1	Neutral
2019-07-31	ACB	1	Neutral
2019-07-31	STB	0	Short
2019-07-31	SHB	0	Short
2019-08-31	VCB	2	Long
2019-08-31	BID	3	Neutral
2019-08-31	CTG	0	Short
2019-08-31	MBB	1	Neutral
2019-08-31	ACB	1	Neutral
2019-08-31	STB	0	Short
2019-08-31	SHB	3	Neutral
2019-09-30	VCB	2	Long
2019-09-30	BID	3	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2019-09-30	CTG	0	Neutral
2019-09-30	MBB	1	Neutral
2019-09-30	ACB	1	Neutral
2019-09-30	STB	0	Neutral
2019-09-30	SHB	3	Short
2019-10-31	VCB	2	Long
2019-10-31	BID	0	Neutral
2019-10-31	CTG	0	Neutral
2019-10-31	MBB	1	Neutral
2019-10-31	ACB	1	Neutral
2019-10-31	STB	3	Short
2019-10-31	SHB	3	Short
2019-11-30	VCB	0	Long
2019-11-30	BID	2	Neutral
2019-11-30	CTG	1	Short
2019-11-30	MBB	3	Neutral
2019-11-30	ACB	3	Neutral
2019-11-30	STB	1	Short
2019-11-30	SHB	1	Short
2019-12-31	VCB	0	Long
2019-12-31	BID	2	Neutral
2019-12-31	CTG	1	Short
2019-12-31	MBB	0	Long
2019-12-31	ACB	3	Neutral
2019-12-31	STB	1	Short
2019-12-31	SHB	1	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2020-01-31	VCB	0	Long
2020-01-31	BID	1	Neutral
2020-01-31	CTG	1	Neutral
2020-01-31	MBB	2	Neutral
2020-01-31	ACB	2	Neutral
2020-01-31	STB	3	Short
2020-01-31	SHB	3	Short
2020-02-29	VCB	1	Neutral
2020-02-29	BID	2	Long
2020-02-29	CTG	0	Neutral
2020-02-29	MBB	1	Neutral
2020-02-29	ACB	1	Neutral
2020-02-29	STB	3	Short
2020-02-29	SHB	0	Neutral
2020-03-31	VCB	0	Neutral
2020-03-31	BID	2	Long
2020-03-31	CTG	3	Short
2020-03-31	MBB	1	Neutral
2020-03-31	ACB	1	Neutral
2020-03-31	VIB	0	Neutral
2020-03-31	STB	1	Neutral
2020-03-31	SHB	3	Short
2020-04-30	VCB	0	Neutral
2020-04-30	BID	2	Long
2020-04-30	CTG	0	Neutral
2020-04-30	MBB	1	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2020-04-30	ACB	1	Short
2020-04-30	VIB	0	Neutral
2020-04-30	STB	1	Short
2020-04-30	SHB	3	Neutral
2020-05-31	VCB	2	Neutral
2020-05-31	BID	0	Neutral
2020-05-31	CTG	0	Neutral
2020-05-31	MBB	1	Short
2020-05-31	ACB	2	Neutral
2020-05-31	VIB	1	Short
2020-05-31	STB	1	Short
2020-05-31	SHB	3	Long
2020-06-30	VCB	2	Neutral
2020-06-30	BID	0	Neutral
2020-06-30	CTG	0	Neutral
2020-06-30	MBB	3	Short
2020-06-30	ACB	2	Neutral
2020-06-30	VIB	3	Short
2020-06-30	STB	0	Neutral
2020-06-30	SHB	1	Long
2020-07-31	VCB	2	Neutral
2020-07-31	BID	0	Neutral
2020-07-31	CTG	0	Neutral
2020-07-31	MBB	3	Short
2020-07-31	ACB	2	Neutral
2020-07-31	VIB	3	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2020-07-31	STB	0	Neutral
2020-07-31	SHB	1	Long
2020-08-31	VCB	3	Neutral
2020-08-31	BID	2	Neutral
2020-08-31	CTG	2	Neutral
2020-08-31	MBB	0	Short
2020-08-31	ACB	3	Neutral
2020-08-31	VIB	0	Short
2020-08-31	STB	2	Neutral
2020-08-31	SHB	1	Long
2020-08-31	KLB	3	Neutral
2020-09-30	VCB	0	Neutral
2020-09-30	BID	2	Short
2020-09-30	CTG	3	Neutral
2020-09-30	MBB	0	Neutral
2020-09-30	ACB	0	Neutral
2020-09-30	VIB	3	Neutral
2020-09-30	STB	3	Neutral
2020-09-30	SHB	1	Long
2020-09-30	KLB	0	Neutral
2020-10-31	VCB	0	Neutral
2020-10-31	BID	1	Short
2020-10-31	CTG	1	Short
2020-10-31	MBB	0	Neutral
2020-10-31	ACB	0	Neutral
2020-10-31	VPB	1	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2020-10-31	VIB	3	Neutral
2020-10-31	STB	3	Neutral
2020-10-31	SHB	2	Long
2020-10-31	KLB	0	Neutral
2020-11-30	VCB	0	Neutral
2020-11-30	BID	1	Short
2020-11-30	CTG	1	Short
2020-11-30	MBB	0	Neutral
2020-11-30	ACB	0	Neutral
2020-11-30	VPB	1	Short
2020-11-30	VIB	2	Neutral
2020-11-30	STB	1	Short
2020-11-30	SHB	3	Long
2020-11-30	KLB	0	Neutral
2020-12-31	VCB	1	Neutral
2020-12-31	BID	2	Neutral
2020-12-31	CTG	2	Neutral
2020-12-31	MBB	1	Neutral
2020-12-31	ACB	1	Neutral
2020-12-31	VPB	2	Neutral
2020-12-31	VIB	0	Long
2020-12-31	STB	0	Long
2020-12-31	SHB	2	Neutral
2020-12-31	LPB	0	Long
2020-12-31	KLB	3	Short
2021-01-31	VCB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2021-01-31	BID	1	Short
2021-01-31	CTG	2	Neutral
2021-01-31	MBB	0	Neutral
2021-01-31	ACB	0	Neutral
2021-01-31	VPB	1	Short
2021-01-31	VIB	3	Long
2021-01-31	STB	2	Neutral
2021-01-31	SHB	1	Short
2021-01-31	LPB	2	Neutral
2021-01-31	KLB	0	Neutral
2021-02-28	VCB	1	Short
2021-02-28	BID	1	Short
2021-02-28	CTG	0	Neutral
2021-02-28	MBB	0	Neutral
2021-02-28	ACB	0	Neutral
2021-02-28	VPB	1	Short
2021-02-28	VIB	3	Long
2021-02-28	STB	0	Neutral
2021-02-28	SHB	1	Short
2021-02-28	LPB	0	Neutral
2021-02-28	BAB	2	Neutral
2021-02-28	KLB	0	Neutral
2021-03-31	VCB	2	Short
2021-03-31	BID	2	Short
2021-03-31	CTG	3	Neutral
2021-03-31	MBB	3	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2021-03-31	ACB	3	Neutral
2021-03-31	VPB	3	Neutral
2021-03-31	HDB	3	Neutral
2021-03-31	VIB	0	Long
2021-03-31	STB	3	Neutral
2021-03-31	SHB	2	Short
2021-03-31	LPB	3	Neutral
2021-03-31	BAB	1	Neutral
2021-03-31	KLB	0	Long
2021-04-30	VCB	1	Short
2021-04-30	BID	1	Short
2021-04-30	CTG	2	Neutral
2021-04-30	MBB	2	Neutral
2021-04-30	ACB	2	Neutral
2021-04-30	VPB	2	Neutral
2021-04-30	HDB	2	Neutral
2021-04-30	VIB	0	Long
2021-04-30	STB	2	Neutral
2021-04-30	SHB	1	Short
2021-04-30	LPB	2	Neutral
2021-04-30	BAB	3	Neutral
2021-04-30	KLB	2	Neutral
2021-05-31	VCB	2	Neutral
2021-05-31	BID	1	Short
2021-05-31	CTG	0	Neutral
2021-05-31	MBB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2021-05-31	ACB	0	Neutral
2021-05-31	VPB	3	Long
2021-05-31	HDB	0	Neutral
2021-05-31	VIB	3	Long
2021-05-31	STB	0	Neutral
2021-05-31	SHB	1	Short
2021-05-31	LPB	0	Neutral
2021-05-31	BAB	2	Neutral
2021-05-31	KLB	0	Neutral
2021-06-30	VCB	0	Short
2021-06-30	BID	0	Short
2021-06-30	CTG	1	Neutral
2021-06-30	MBB	1	Neutral
2021-06-30	ACB	3	Neutral
2021-06-30	VPB	2	Long
2021-06-30	HDB	1	Neutral
2021-06-30	VIB	2	Long
2021-06-30	TPB	3	Neutral
2021-06-30	STB	1	Neutral
2021-06-30	SHB	2	Long
2021-06-30	LPB	1	Neutral
2021-06-30	BAB	3	Neutral
2021-06-30	KLB	1	Neutral
2021-07-31	VCB	2	Neutral
2021-07-31	BID	1	Short
2021-07-31	CTG	1	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2021-07-31	MBB	3	Neutral
2021-07-31	ACB	2	Neutral
2021-07-31	VPB	0	Long
2021-07-31	HDB	1	Short
2021-07-31	VIB	3	Neutral
2021-07-31	TPB	2	Neutral
2021-07-31	STB	3	Neutral
2021-07-31	SHB	0	Long
2021-07-31	LPB	3	Neutral
2021-07-31	BAB	2	Neutral
2021-07-31	KLB	3	Neutral
2021-08-31	VCB	0	Neutral
2021-08-31	BID	2	Short
2021-08-31	CTG	2	Short
2021-08-31	MBB	1	Long
2021-08-31	TCB	1	Long
2021-08-31	ACB	0	Neutral
2021-08-31	VPB	3	Neutral
2021-08-31	HDB	2	Short
2021-08-31	VIB	1	Long
2021-08-31	TPB	0	Neutral
2021-08-31	STB	1	Long
2021-08-31	SHB	3	Neutral
2021-08-31	LPB	1	Long
2021-08-31	BAB	0	Neutral
2021-08-31	KLB	1	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2021-09-30	VCB	0	Neutral
2021-09-30	BID	1	Short
2021-09-30	CTG	1	Short
2021-09-30	MBB	2	Long
2021-09-30	TCB	1	Short
2021-09-30	ACB	0	Neutral
2021-09-30	VPB	3	Neutral
2021-09-30	HDB	1	Short
2021-09-30	VIB	2	Long
2021-09-30	TPB	0	Neutral
2021-09-30	STB	3	Neutral
2021-09-30	SHB	3	Neutral
2021-09-30	LPB	2	Long
2021-09-30	BAB	0	Neutral
2021-09-30	KLB	1	Short
2021-10-31	VCB	0	Neutral
2021-10-31	BID	2	Short
2021-10-31	CTG	2	Short
2021-10-31	MBB	1	Neutral
2021-10-31	TCB	1	Neutral
2021-10-31	ACB	1	Neutral
2021-10-31	VPB	3	Long
2021-10-31	HDB	1	Neutral
2021-10-31	VIB	1	Neutral
2021-10-31	TPB	0	Neutral
2021-10-31	STB	3	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2021-10-31	SHB	3	Long
2021-10-31	LPB	1	Neutral
2021-10-31	BAB	0	Neutral
2021-10-31	KLB	2	Short
2021-11-30	VCB	3	Neutral
2021-11-30	BID	0	Short
2021-11-30	CTG	2	Neutral
2021-11-30	MBB	2	Neutral
2021-11-30	TCB	2	Neutral
2021-11-30	ACB	3	Neutral
2021-11-30	VPB	1	Long
2021-11-30	HDB	2	Neutral
2021-11-30	VIB	2	Neutral
2021-11-30	TPB	3	Neutral
2021-11-30	STB	1	Long
2021-11-30	SHB	1	Long
2021-11-30	LPB	1	Long
2021-11-30	BAB	3	Neutral
2021-11-30	KLB	2	Neutral
2021-12-31	VCB	1	Short
2021-12-31	BID	3	Neutral
2021-12-31	CTG	3	Neutral
2021-12-31	MBB	0	Long
2021-12-31	TCB	0	Long
2021-12-31	ACB	0	Long
2021-12-31	VPB	2	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2021-12-31	HDB	0	Long
2021-12-31	VIB	3	Neutral
2021-12-31	TPB	0	Long
2021-12-31	STB	2	Neutral
2021-12-31	SHB	2	Neutral
2021-12-31	LPB	0	Long
2021-12-31	BAB	1	Short
2021-12-31	KLB	3	Neutral
2022-01-31	VCB	1	Short
2022-01-31	BID	0	Neutral
2022-01-31	CTG	0	Neutral
2022-01-31	MBB	3	Neutral
2022-01-31	TCB	3	Neutral
2022-01-31	ACB	3	Neutral
2022-01-31	VPB	0	Neutral
2022-01-31	HDB	3	Neutral
2022-01-31	VIB	0	Neutral
2022-01-31	TPB	2	Long
2022-01-31	STB	0	Neutral
2022-01-31	SHB	0	Neutral
2022-01-31	LPB	3	Neutral
2022-01-31	BAB	1	Short
2022-01-31	KLB	0	Neutral
2022-02-28	VCB	0	Long
2022-02-28	BID	1	Neutral
2022-02-28	CTG	1	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2022-02-28	MBB	0	Long
2022-02-28	TCB	1	Neutral
2022-02-28	ACB	0	Long
2022-02-28	VPB	2	Neutral
2022-02-28	HDB	0	Long
2022-02-28	VIB	1	Neutral
2022-02-28	TPB	0	Long
2022-02-28	STB	1	Neutral
2022-02-28	SHB	2	Neutral
2022-02-28	LPB	1	Neutral
2022-02-28	BAB	3	Short
2022-02-28	KLB	1	Neutral
2022-03-31	VCB	0	Neutral
2022-03-31	BID	1	Neutral
2022-03-31	CTG	2	Short
2022-03-31	MBB	2	Short
2022-03-31	TCB	1	Neutral
2022-03-31	ACB	0	Neutral
2022-03-31	VPB	1	Neutral
2022-03-31	HDB	1	Neutral
2022-03-31	VIB	1	Neutral
2022-03-31	TPB	3	Long
2022-03-31	STB	1	Neutral
2022-03-31	SHB	1	Neutral
2022-03-31	LPB	2	Short
2022-03-31	BAB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2022-03-31	KLB	1	Neutral
2022-04-30	VCB	0	Neutral
2022-04-30	BID	1	Long
2022-04-30	CTG	2	Neutral
2022-04-30	MBB	2	Neutral
2022-04-30	TCB	2	Neutral
2022-04-30	ACB	0	Neutral
2022-04-30	VPB	3	Short
2022-04-30	HDB	2	Neutral
2022-04-30	VIB	2	Neutral
2022-04-30	TPB	1	Long
2022-04-30	STB	3	Short
2022-04-30	SHB	3	Short
2022-04-30	LPB	2	Neutral
2022-04-30	BAB	0	Neutral
2022-04-30	KLB	1	Long
2022-05-31	VCB	0	Neutral
2022-05-31	BID	2	Neutral
2022-05-31	CTG	1	Short
2022-05-31	MBB	2	Neutral
2022-05-31	TCB	1	Short
2022-05-31	ACB	0	Neutral
2022-05-31	VPB	1	Short
2022-05-31	HDB	1	Short
2022-05-31	VIB	2	Neutral
2022-05-31	TPB	3	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2022-05-31	STB	2	Neutral
2022-05-31	SHB	1	Short
2022-05-31	LPB	1	Short
2022-05-31	BAB	0	Neutral
2022-05-31	KLB	2	Neutral
2022-06-30	VCB	0	Neutral
2022-06-30	BID	1	Neutral
2022-06-30	CTG	2	Short
2022-06-30	MBB	1	Neutral
2022-06-30	TCB	2	Short
2022-06-30	ACB	0	Neutral
2022-06-30	VPB	2	Short
2022-06-30	HDB	1	Neutral
2022-06-30	VIB	1	Neutral
2022-06-30	TPB	1	Neutral
2022-06-30	STB	1	Neutral
2022-06-30	SHB	2	Short
2022-06-30	LPB	2	Short
2022-06-30	BAB	0	Neutral
2022-06-30	KLB	3	Long
2022-07-31	VCB	2	Neutral
2022-07-31	BID	0	Long
2022-07-31	CTG	3	Neutral
2022-07-31	MBB	2	Neutral
2022-07-31	TCB	3	Neutral
2022-07-31	ACB	2	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2022-07-31	VPB	3	Neutral
2022-07-31	HDB	3	Neutral
2022-07-31	VIB	0	Long
2022-07-31	TPB	2	Neutral
2022-07-31	STB	0	Long
2022-07-31	SHB	1	Short
2022-07-31	LPB	3	Neutral
2022-07-31	BAB	2	Neutral
2022-07-31	KLB	0	Long
2022-08-31	VCB	0	Neutral
2022-08-31	BID	2	Long
2022-08-31	CTG	3	Neutral
2022-08-31	MBB	2	Long
2022-08-31	TCB	3	Neutral
2022-08-31	ACB	0	Neutral
2022-08-31	VPB	3	Neutral
2022-08-31	HDB	3	Neutral
2022-08-31	VIB	3	Neutral
2022-08-31	TPB	2	Long
2022-08-31	STB	3	Neutral
2022-08-31	SHB	1	Short
2022-08-31	LPB	3	Neutral
2022-08-31	BAB	0	Neutral
2022-08-31	KLB	2	Long
2022-09-30	VCB	2	Long
2022-09-30	BID	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2022-09-30	CTG	3	Neutral
2022-09-30	MBB	0	Neutral
2022-09-30	TCB	3	Neutral
2022-09-30	ACB	2	Long
2022-09-30	VPB	0	Neutral
2022-09-30	HDB	3	Neutral
2022-09-30	VIB	1	Short
2022-09-30	TPB	3	Neutral
2022-09-30	STB	1	Short
2022-09-30	SHB	1	Short
2022-09-30	LPB	3	Neutral
2022-09-30	BAB	2	Long
2022-09-30	KLB	0	Neutral
2022-09-30	VBB	1	Short
2022-10-31	VCB	0	Long
2022-10-31	BID	2	Neutral
2022-10-31	CTG	2	Neutral
2022-10-31	MBB	2	Neutral
2022-10-31	TCB	2	Neutral
2022-10-31	ACB	0	Long
2022-10-31	VPB	2	Neutral
2022-10-31	HDB	2	Neutral
2022-10-31	VIB	2	Neutral
2022-10-31	TPB	1	Short
2022-10-31	STB	3	Neutral
2022-10-31	SHB	3	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2022-10-31	LPB	1	Short
2022-10-31	BAB	0	Long
2022-10-31	KLB	2	Neutral
2022-10-31	VBB	1	Short
2022-11-30	VCB	0	Long
2022-11-30	BID	3	Neutral
2022-11-30	CTG	3	Neutral
2022-11-30	MBB	3	Neutral
2022-11-30	TCB	1	Neutral
2022-11-30	ACB	0	Long
2022-11-30	VPB	1	Neutral
2022-11-30	HDB	3	Neutral
2022-11-30	VIB	1	Neutral
2022-11-30	TPB	2	Short
2022-11-30	STB	1	Neutral
2022-11-30	SHB	1	Neutral
2022-11-30	LPB	1	Neutral
2022-11-30	BAB	0	Long
2022-11-30	KLB	3	Neutral
2022-11-30	VBB	2	Short
2022-12-31	VCB	0	Long
2022-12-31	BID	2	Neutral
2022-12-31	CTG	2	Neutral
2022-12-31	MBB	2	Neutral
2022-12-31	TCB	1	Neutral
2022-12-31	ACB	0	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2022-12-31	VPB	1	Neutral
2022-12-31	HDB	2	Neutral
2022-12-31	VIB	1	Neutral
2022-12-31	TPB	3	Short
2022-12-31	STB	1	Neutral
2022-12-31	SHB	1	Neutral
2022-12-31	LPB	1	Neutral
2022-12-31	BAB	0	Long
2022-12-31	KLB	1	Neutral
2022-12-31	VBB	3	Short
2023-01-31	VCB	0	Long
2023-01-31	BID	2	Neutral
2023-01-31	CTG	2	Neutral
2023-01-31	MBB	3	Neutral
2023-01-31	TCB	1	Short
2023-01-31	ACB	0	Long
2023-01-31	VPB	1	Short
2023-01-31	HDB	2	Neutral
2023-01-31	VIB	1	Short
2023-01-31	TPB	3	Neutral
2023-01-31	STB	1	Short
2023-01-31	SHB	1	Short
2023-01-31	LPB	3	Neutral
2023-01-31	BAB	0	Long
2023-01-31	KLB	1	Short
2023-01-31	VBB	3	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2023-02-28	VCB	3	Neutral
2023-02-28	BID	0	Long
2023-02-28	CTG	0	Long
2023-02-28	MBB	1	Short
2023-02-28	TCB	1	Short
2023-02-28	ACB	3	Neutral
2023-02-28	VPB	2	Neutral
2023-02-28	HDB	1	Short
2023-02-28	VIB	2	Neutral
2023-02-28	TPB	1	Short
2023-02-28	STB	2	Neutral
2023-02-28	SHB	2	Neutral
2023-02-28	LPB	0	Long
2023-02-28	BAB	3	Neutral
2023-02-28	KLB	2	Neutral
2023-02-28	VBB	1	Short
2023-03-31	VCB	3	Neutral
2023-03-31	BID	1	Long
2023-03-31	CTG	1	Long
2023-03-31	MBB	0	Neutral
2023-03-31	TCB	0	Neutral
2023-03-31	ACB	3	Neutral
2023-03-31	VPB	1	Long
2023-03-31	HDB	0	Neutral
2023-03-31	VIB	1	Long
2023-03-31	TPB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2023-03-31	STB	1	Long
2023-03-31	SHB	1	Long
2023-03-31	LPB	1	Long
2023-03-31	BAB	3	Neutral
2023-03-31	KLB	2	Short
2023-03-31	VBB	0	Neutral
2023-04-30	VCB	0	Long
2023-04-30	BID	2	Neutral
2023-04-30	CTG	2	Neutral
2023-04-30	MBB	2	Neutral
2023-04-30	TCB	1	Neutral
2023-04-30	ACB	0	Long
2023-04-30	VPB	1	Neutral
2023-04-30	HDB	2	Neutral
2023-04-30	VIB	1	Neutral
2023-04-30	TPB	2	Neutral
2023-04-30	STB	1	Neutral
2023-04-30	SHB	1	Neutral
2023-04-30	LPB	2	Neutral
2023-04-30	BAB	0	Long
2023-04-30	KLB	3	Short
2023-04-30	VBB	2	Neutral
2023-05-31	VCB	0	Long
2023-05-31	BID	0	Long
2023-05-31	CTG	0	Long
2023-05-31	MBB	3	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2023-05-31	TCB	3	Neutral
2023-05-31	ACB	0	Long
2023-05-31	VPB	3	Neutral
2023-05-31	HDB	0	Long
2023-05-31	VIB	1	Neutral
2023-05-31	TPB	3	Neutral
2023-05-31	STB	1	Neutral
2023-05-31	SHB	3	Neutral
2023-05-31	LPB	1	Neutral
2023-05-31	BAB	0	Long
2023-05-31	KLB	2	Short
2023-05-31	VBB	1	Neutral
2023-06-30	VCB	3	Neutral
2023-06-30	BID	2	Long
2023-06-30	CTG	2	Long
2023-06-30	MBB	3	Neutral
2023-06-30	TCB	0	Neutral
2023-06-30	ACB	3	Neutral
2023-06-30	VPB	2	Long
2023-06-30	HDB	3	Neutral
2023-06-30	VIB	0	Neutral
2023-06-30	TPB	0	Neutral
2023-06-30	STB	0	Neutral
2023-06-30	SHB	0	Neutral
2023-06-30	LPB	2	Long
2023-06-30	BAB	3	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2023-06-30	KLB	1	Short
2023-06-30	VBB	0	Neutral
2023-07-31	VCB	0	Neutral
2023-07-31	BID	2	Neutral
2023-07-31	CTG	2	Neutral
2023-07-31	MBB	0	Neutral
2023-07-31	TCB	2	Neutral
2023-07-31	ACB	0	Neutral
2023-07-31	VPB	2	Neutral
2023-07-31	HDB	0	Neutral
2023-07-31	VIB	1	Long
2023-07-31	TPB	2	Neutral
2023-07-31	STB	1	Long
2023-07-31	SHB	1	Long
2023-07-31	LPB	1	Long
2023-07-31	BAB	0	Neutral
2023-07-31	KLB	3	Short
2023-07-31	VBB	1	Long
2023-08-31	VCB	0	Long
2023-08-31	BID	2	Neutral
2023-08-31	CTG	2	Neutral
2023-08-31	MBB	2	Neutral
2023-08-31	TCB	1	Neutral
2023-08-31	ACB	0	Long
2023-08-31	VPB	1	Neutral
2023-08-31	HDB	0	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2023-08-31	VIB	1	Neutral
2023-08-31	TPB	1	Neutral
2023-08-31	STB	1	Neutral
2023-08-31	SHB	1	Neutral
2023-08-31	LPB	1	Neutral
2023-08-31	BAB	2	Neutral
2023-08-31	KLB	3	Short
2023-08-31	VBB	1	Neutral
2023-09-30	VCB	2	Long
2023-09-30	BID	0	Neutral
2023-09-30	CTG	0	Neutral
2023-09-30	MBB	2	Long
2023-09-30	TCB	1	Neutral
2023-09-30	ACB	2	Long
2023-09-30	VPB	0	Neutral
2023-09-30	HDB	2	Long
2023-09-30	VIB	1	Neutral
2023-09-30	TPB	1	Neutral
2023-09-30	STB	1	Neutral
2023-09-30	SHB	1	Neutral
2023-09-30	LPB	0	Neutral
2023-09-30	BAB	0	Neutral
2023-09-30	KLB	3	Short
2023-09-30	VBB	1	Neutral
2023-09-30	BVB	1	Neutral
2023-10-31	VCB	1	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2023-10-31	BID	3	Short
2023-10-31	CTG	3	Short
2023-10-31	MBB	1	Neutral
2023-10-31	TCB	1	Neutral
2023-10-31	ACB	3	Short
2023-10-31	VPB	1	Neutral
2023-10-31	HDB	1	Neutral
2023-10-31	VIB	0	Neutral
2023-10-31	TPB	1	Neutral
2023-10-31	STB	0	Neutral
2023-10-31	SHB	2	Long
2023-10-31	LPB	0	Neutral
2023-10-31	BAB	3	Short
2023-10-31	KLB	0	Neutral
2023-10-31	VBB	0	Neutral
2023-10-31	BVB	0	Neutral
2023-11-30	VCB	1	Neutral
2023-11-30	BID	1	Neutral
2023-11-30	CTG	0	Neutral
2023-11-30	MBB	0	Neutral
2023-11-30	TCB	0	Neutral
2023-11-30	ACB	1	Neutral
2023-11-30	VPB	0	Neutral
2023-11-30	HDB	1	Neutral
2023-11-30	VIB	0	Neutral
2023-11-30	TPB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2023-11-30	STB	2	Long
2023-11-30	SHB	2	Long
2023-11-30	LPB	2	Long
2023-11-30	BAB	1	Neutral
2023-11-30	KLB	3	Short
2023-11-30	VBB	3	Short
2023-11-30	BVB	3	Short
2023-12-31	VCB	0	Neutral
2023-12-31	BID	0	Neutral
2023-12-31	CTG	0	Neutral
2023-12-31	MBB	0	Neutral
2023-12-31	TCB	2	Neutral
2023-12-31	ACB	0	Neutral
2023-12-31	VPB	2	Neutral
2023-12-31	HDB	0	Neutral
2023-12-31	VIB	2	Neutral
2023-12-31	TPB	2	Neutral
2023-12-31	STB	2	Neutral
2023-12-31	SHB	2	Neutral
2023-12-31	LPB	2	Neutral
2023-12-31	NAB	3	Long
2023-12-31	BAB	0	Neutral
2023-12-31	KLB	1	Short
2023-12-31	VBB	1	Short
2023-12-31	SGB	2	Neutral
2023-12-31	BVB	1	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-01-31	VCB	0	Neutral
2024-01-31	BID	3	Neutral
2024-01-31	CTG	3	Neutral
2024-01-31	MBB	0	Neutral
2024-01-31	TCB	3	Neutral
2024-01-31	ACB	0	Neutral
2024-01-31	VPB	3	Neutral
2024-01-31	HDB	0	Neutral
2024-01-31	VIB	1	Short
2024-01-31	TPB	3	Neutral
2024-01-31	STB	1	Short
2024-01-31	SHB	1	Short
2024-01-31	LPB	2	Long
2024-01-31	NAB	2	Long
2024-01-31	BAB	3	Neutral
2024-01-31	KLB	1	Short
2024-01-31	VBB	1	Short
2024-01-31	SGB	3	Neutral
2024-01-31	BVB	1	Short
2024-02-29	VCB	0	Neutral
2024-02-29	BID	3	Neutral
2024-02-29	CTG	0	Neutral
2024-02-29	MBB	0	Neutral
2024-02-29	TCB	3	Neutral
2024-02-29	ACB	0	Neutral
2024-02-29	VPB	3	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-02-29	HDB	0	Neutral
2024-02-29	VIB	3	Neutral
2024-02-29	TPB	3	Neutral
2024-02-29	STB	1	Short
2024-02-29	SHB	1	Short
2024-02-29	LPB	2	Long
2024-02-29	MSB	0	Neutral
2024-02-29	NAB	2	Long
2024-02-29	BAB	3	Neutral
2024-02-29	KLB	1	Short
2024-02-29	PGB	1	Short
2024-02-29	VBB	1	Short
2024-02-29	SGB	3	Neutral
2024-02-29	ABB	3	Neutral
2024-02-29	BVB	1	Short
2024-03-31	VCB	2	Neutral
2024-03-31	BID	2	Neutral
2024-03-31	CTG	2	Neutral
2024-03-31	MBB	2	Neutral
2024-03-31	TCB	0	Neutral
2024-03-31	ACB	2	Neutral
2024-03-31	VPB	0	Neutral
2024-03-31	HDB	1	Long
2024-03-31	VIB	0	Neutral
2024-03-31	TPB	0	Neutral
2024-03-31	STB	3	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-03-31	SHB	0	Neutral
2024-03-31	LPB	1	Long
2024-03-31	MSB	2	Neutral
2024-03-31	OCB	2	Neutral
2024-03-31	NAB	1	Long
2024-03-31	BAB	2	Neutral
2024-03-31	KLB	3	Short
2024-03-31	PGB	3	Short
2024-03-31	VBB	3	Short
2024-03-31	SGB	0	Neutral
2024-03-31	ABB	0	Neutral
2024-03-31	BVB	3	Short
2024-04-30	VCB	0	Neutral
2024-04-30	BID	2	Neutral
2024-04-30	CTG	2	Neutral
2024-04-30	MBB	2	Neutral
2024-04-30	TCB	0	Neutral
2024-04-30	ACB	2	Neutral
2024-04-30	VPB	0	Neutral
2024-04-30	HDB	2	Neutral
2024-04-30	VIB	0	Neutral
2024-04-30	TPB	0	Neutral
2024-04-30	STB	1	Short
2024-04-30	SHB	1	Short
2024-04-30	LPB	3	Long
2024-04-30	MSB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-04-30	OCB	2	Neutral
2024-04-30	NAB	3	Long
2024-04-30	BAB	0	Neutral
2024-04-30	KLB	1	Short
2024-04-30	PGB	1	Short
2024-04-30	VBB	1	Short
2024-04-30	SGB	0	Neutral
2024-04-30	ABB	0	Neutral
2024-04-30	BVB	1	Short
2024-05-31	VCB	0	Neutral
2024-05-31	BID	2	Long
2024-05-31	CTG	2	Long
2024-05-31	MBB	2	Long
2024-05-31	TCB	2	Long
2024-05-31	ACB	0	Neutral
2024-05-31	VPB	3	Neutral
2024-05-31	HDB	2	Long
2024-05-31	VIB	2	Long
2024-05-31	TPB	3	Neutral
2024-05-31	STB	1	Short
2024-05-31	SHB	3	Neutral
2024-05-31	LPB	1	Short
2024-05-31	SSB	3	Neutral
2024-05-31	MSB	2	Long
2024-05-31	OCB	2	Long
2024-05-31	NAB	2	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-05-31	BAB	0	Neutral
2024-05-31	KLB	1	Short
2024-05-31	PGB	1	Short
2024-05-31	VBB	1	Short
2024-05-31	SGB	3	Neutral
2024-05-31	ABB	3	Neutral
2024-05-31	BVB	1	Short
2024-06-30	VCB	0	Neutral
2024-06-30	BID	2	Long
2024-06-30	CTG	2	Long
2024-06-30	MBB	2	Long
2024-06-30	TCB	2	Long
2024-06-30	ACB	0	Neutral
2024-06-30	VPB	3	Neutral
2024-06-30	HDB	2	Long
2024-06-30	VIB	2	Long
2024-06-30	TPB	3	Neutral
2024-06-30	STB	1	Short
2024-06-30	SHB	3	Neutral
2024-06-30	LPB	2	Long
2024-06-30	SSB	3	Neutral
2024-06-30	MSB	3	Neutral
2024-06-30	OCB	3	Neutral
2024-06-30	NAB	2	Long
2024-06-30	BAB	0	Neutral
2024-06-30	KLB	1	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-06-30	PGB	1	Short
2024-06-30	VBB	1	Short
2024-06-30	SGB	3	Neutral
2024-06-30	ABB	3	Neutral
2024-06-30	BVB	1	Short
2024-07-31	VCB	0	Neutral
2024-07-31	BID	1	Long
2024-07-31	CTG	1	Long
2024-07-31	MBB	1	Long
2024-07-31	TCB	1	Long
2024-07-31	ACB	3	Neutral
2024-07-31	VPB	0	Neutral
2024-07-31	HDB	1	Long
2024-07-31	VIB	1	Long
2024-07-31	TPB	0	Neutral
2024-07-31	STB	2	Short
2024-07-31	SHB	0	Neutral
2024-07-31	LPB	1	Long
2024-07-31	SSB	0	Neutral
2024-07-31	MSB	0	Neutral
2024-07-31	OCB	0	Neutral
2024-07-31	NAB	1	Long
2024-07-31	BAB	3	Neutral
2024-07-31	KLB	2	Short
2024-07-31	PGB	2	Short
2024-07-31	VBB	1	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-07-31	SGB	3	Neutral
2024-07-31	ABB	0	Neutral
2024-07-31	BVB	0	Neutral
2024-08-31	VCB	3	Neutral
2024-08-31	BID	0	Neutral
2024-08-31	CTG	0	Neutral
2024-08-31	MBB	0	Neutral
2024-08-31	TCB	2	Long
2024-08-31	ACB	0	Neutral
2024-08-31	VPB	3	Neutral
2024-08-31	HDB	0	Neutral
2024-08-31	VIB	0	Neutral
2024-08-31	TPB	0	Neutral
2024-08-31	STB	1	Short
2024-08-31	SHB	0	Neutral
2024-08-31	LPB	2	Long
2024-08-31	SSB	3	Neutral
2024-08-31	MSB	0	Neutral
2024-08-31	OCB	0	Neutral
2024-08-31	NAB	0	Neutral
2024-08-31	BAB	3	Neutral
2024-08-31	KLB	1	Short
2024-08-31	PGB	1	Short
2024-08-31	VBB	0	Neutral
2024-08-31	SGB	3	Neutral
2024-08-31	ABB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-08-31	BVB	0	Neutral
2024-09-30	VCB	3	Neutral
2024-09-30	BID	0	Neutral
2024-09-30	CTG	0	Neutral
2024-09-30	MBB	0	Neutral
2024-09-30	TCB	1	Long
2024-09-30	ACB	3	Neutral
2024-09-30	VPB	0	Neutral
2024-09-30	HDB	0	Neutral
2024-09-30	VIB	0	Neutral
2024-09-30	TPB	0	Neutral
2024-09-30	STB	2	Short
2024-09-30	SHB	0	Neutral
2024-09-30	LPB	1	Long
2024-09-30	SSB	3	Neutral
2024-09-30	MSB	0	Neutral
2024-09-30	OCB	0	Neutral
2024-09-30	NAB	0	Neutral
2024-09-30	BAB	3	Neutral
2024-09-30	KLB	2	Short
2024-09-30	PGB	2	Short
2024-09-30	VAB	1	Long
2024-09-30	VBB	0	Neutral
2024-09-30	SGB	3	Neutral
2024-09-30	ABB	0	Neutral
2024-09-30	BVB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-10-31	VCB	1	Neutral
2024-10-31	BID	3	Neutral
2024-10-31	CTG	1	Neutral
2024-10-31	MBB	2	Long
2024-10-31	TCB	2	Long
2024-10-31	ACB	1	Neutral
2024-10-31	VPB	1	Neutral
2024-10-31	HDB	2	Long
2024-10-31	VIB	3	Neutral
2024-10-31	TPB	3	Neutral
2024-10-31	STB	3	Neutral
2024-10-31	SHB	3	Neutral
2024-10-31	LPB	2	Long
2024-10-31	SSB	1	Neutral
2024-10-31	MSB	1	Neutral
2024-10-31	OCB	3	Neutral
2024-10-31	NAB	2	Long
2024-10-31	BAB	1	Neutral
2024-10-31	KLB	3	Neutral
2024-10-31	PGB	0	Short
2024-10-31	VAB	2	Long
2024-10-31	VBB	3	Neutral
2024-10-31	SGB	1	Neutral
2024-10-31	ABB	1	Neutral
2024-10-31	BVB	2	Long
2024-11-30	VCB	3	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-11-30	BID	0	Neutral
2024-11-30	CTG	0	Neutral
2024-11-30	MBB	0	Neutral
2024-11-30	TCB	0	Neutral
2024-11-30	ACB	3	Neutral
2024-11-30	VPB	0	Neutral
2024-11-30	HDB	0	Neutral
2024-11-30	VIB	0	Neutral
2024-11-30	TPB	0	Neutral
2024-11-30	STB	0	Neutral
2024-11-30	SHB	0	Neutral
2024-11-30	LPB	2	Long
2024-11-30	SSB	3	Neutral
2024-11-30	MSB	0	Neutral
2024-11-30	OCB	0	Neutral
2024-11-30	NAB	0	Neutral
2024-11-30	BAB	3	Neutral
2024-11-30	KLB	0	Neutral
2024-11-30	PGB	1	Short
2024-11-30	VAB	0	Neutral
2024-11-30	VBB	0	Neutral
2024-11-30	SGB	3	Neutral
2024-11-30	ABB	0	Neutral
2024-11-30	BVB	0	Neutral
2024-12-31	VCB	2	Neutral
2024-12-31	BID	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-12-31	CTG	0	Neutral
2024-12-31	MBB	0	Neutral
2024-12-31	TCB	0	Neutral
2024-12-31	ACB	2	Neutral
2024-12-31	VPB	0	Neutral
2024-12-31	HDB	0	Neutral
2024-12-31	VIB	0	Neutral
2024-12-31	TPB	0	Neutral
2024-12-31	STB	0	Neutral
2024-12-31	SHB	0	Neutral
2024-12-31	LPB	1	Long
2024-12-31	SSB	2	Neutral
2024-12-31	MSB	0	Neutral
2024-12-31	OCB	0	Neutral
2024-12-31	NAB	0	Neutral
2024-12-31	BAB	2	Neutral
2024-12-31	KLB	0	Neutral
2024-12-31	PGB	3	Short
2024-12-31	VAB	0	Neutral
2024-12-31	VBB	0	Neutral
2024-12-31	SGB	2	Neutral
2024-12-31	ABB	0	Neutral
2024-12-31	BVB	0	Neutral
2025-01-31	VCB	3	Neutral
2025-01-31	BID	0	Short
2025-01-31	CTG	2	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-01-31	MBB	2	Neutral
2025-01-31	TCB	0	Short
2025-01-31	ACB	3	Neutral
2025-01-31	VPB	2	Neutral
2025-01-31	HDB	2	Neutral
2025-01-31	VIB	0	Short
2025-01-31	TPB	2	Neutral
2025-01-31	STB	1	Long
2025-01-31	SHB	0	Short
2025-01-31	LPB	1	Long
2025-01-31	SSB	3	Neutral
2025-01-31	MSB	2	Neutral
2025-01-31	OCB	0	Short
2025-01-31	NAB	2	Neutral
2025-01-31	BAB	3	Neutral
2025-01-31	KLB	0	Short
2025-01-31	PGB	0	Short
2025-01-31	VAB	0	Short
2025-01-31	VBB	0	Short
2025-01-31	SGB	3	Neutral
2025-01-31	ABB	3	Neutral
2025-01-31	BVB	2	Neutral
2025-02-28	VCB	1	Neutral
2025-02-28	BID	0	Neutral
2025-02-28	CTG	3	Long
2025-02-28	MBB	3	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-02-28	TCB	0	Neutral
2025-02-28	ACB	1	Neutral
2025-02-28	VPB	1	Neutral
2025-02-28	HDB	3	Long
2025-02-28	VIB	0	Neutral
2025-02-28	TPB	3	Long
2025-02-28	STB	3	Long
2025-02-28	SHB	2	Short
2025-02-28	LPB	3	Long
2025-02-28	SSB	2	Short
2025-02-28	MSB	1	Neutral
2025-02-28	OCB	2	Short
2025-02-28	NAB	3	Long
2025-02-28	BAB	1	Neutral
2025-02-28	KLB	0	Neutral
2025-02-28	PGB	0	Neutral
2025-02-28	VAB	2	Short
2025-02-28	VBB	0	Neutral
2025-02-28	SGB	2	Short
2025-02-28	ABB	2	Short
2025-02-28	BVB	2	Short
2025-03-31	VCB	2	Neutral
2025-03-31	BID	1	Neutral
2025-03-31	CTG	3	Long
2025-03-31	MBB	1	Neutral
2025-03-31	TCB	1	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-03-31	ACB	2	Neutral
2025-03-31	VPB	1	Neutral
2025-03-31	HDB	3	Long
2025-03-31	VIB	1	Neutral
2025-03-31	TPB	3	Long
2025-03-31	STB	3	Long
2025-03-31	SHB	1	Neutral
2025-03-31	LPB	3	Long
2025-03-31	SSB	2	Neutral
2025-03-31	MSB	1	Neutral
2025-03-31	OCB	1	Neutral
2025-03-31	NAB	3	Long
2025-03-31	BAB	2	Neutral
2025-03-31	KLB	0	Short
2025-03-31	PGB	0	Short
2025-03-31	VAB	1	Neutral
2025-03-31	VBB	3	Long
2025-03-31	SGB	2	Neutral
2025-03-31	ABB	1	Neutral
2025-03-31	BVB	1	Neutral
2025-04-30	VCB	2	Neutral
2025-04-30	BID	0	Neutral
2025-04-30	CTG	0	Neutral
2025-04-30	MBB	0	Neutral
2025-04-30	TCB	0	Neutral
2025-04-30	ACB	2	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-04-30	VPB	0	Neutral
2025-04-30	HDB	0	Neutral
2025-04-30	VIB	0	Neutral
2025-04-30	TPB	0	Neutral
2025-04-30	STB	3	Long
2025-04-30	SHB	1	Short
2025-04-30	LPB	3	Long
2025-04-30	SSB	2	Neutral
2025-04-30	MSB	0	Neutral
2025-04-30	OCB	1	Short
2025-04-30	NAB	0	Neutral
2025-04-30	BAB	2	Neutral
2025-04-30	KLB	1	Short
2025-04-30	PGB	1	Short
2025-04-30	VAB	1	Short
2025-04-30	VBB	0	Neutral
2025-04-30	SGB	2	Neutral
2025-04-30	ABB	1	Short
2025-04-30	BVB	0	Neutral
2025-05-31	VCB	2	Neutral
2025-05-31	BID	0	Neutral
2025-05-31	CTG	1	Long
2025-05-31	MBB	0	Neutral
2025-05-31	TCB	1	Long
2025-05-31	ACB	2	Neutral
2025-05-31	VPB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-05-31	HDB	0	Neutral
2025-05-31	VIB	1	Long
2025-05-31	TPB	1	Long
2025-05-31	STB	1	Long
2025-05-31	SHB	0	Neutral
2025-05-31	LPB	1	Long
2025-05-31	SSB	2	Neutral
2025-05-31	MSB	0	Neutral
2025-05-31	OCB	0	Neutral
2025-05-31	NAB	0	Neutral
2025-05-31	BAB	2	Neutral
2025-05-31	KLB	3	Short
2025-05-31	PGB	3	Short
2025-05-31	VAB	1	Long
2025-05-31	VBB	1	Long
2025-05-31	SGB	2	Neutral
2025-05-31	ABB	0	Neutral
2025-05-31	BVB	1	Long
2025-06-30	VCB	2	Neutral
2025-06-30	BID	0	Neutral
2025-06-30	CTG	1	Long
2025-06-30	MBB	1	Long
2025-06-30	TCB	1	Long
2025-06-30	ACB	2	Neutral
2025-06-30	VPB	0	Neutral
2025-06-30	HDB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-06-30	VIB	1	Long
2025-06-30	TPB	0	Neutral
2025-06-30	STB	1	Long
2025-06-30	SHB	1	Long
2025-06-30	LPB	1	Long
2025-06-30	SSB	2	Neutral
2025-06-30	MSB	0	Neutral
2025-06-30	OCB	0	Neutral
2025-06-30	NAB	0	Neutral
2025-06-30	BAB	2	Neutral
2025-06-30	KLB	3	Short
2025-06-30	PGB	3	Short
2025-06-30	VAB	1	Long
2025-06-30	VBB	1	Long
2025-06-30	SGB	2	Neutral
2025-06-30	ABB	0	Neutral
2025-06-30	BVB	1	Long
2025-07-31	VCB	1	Neutral
2025-07-31	BID	0	Short
2025-07-31	CTG	1	Neutral
2025-07-31	MBB	2	Long
2025-07-31	TCB	2	Long
2025-07-31	ACB	1	Neutral
2025-07-31	VPB	0	Short
2025-07-31	HDB	0	Short
2025-07-31	VIB	0	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-07-31	TPB	0	Short
2025-07-31	STB	2	Long
2025-07-31	SHB	2	Long
2025-07-31	LPB	2	Long
2025-07-31	SSB	1	Neutral
2025-07-31	MSB	0	Short
2025-07-31	OCB	0	Short
2025-07-31	NAB	2	Long
2025-07-31	BAB	1	Neutral
2025-07-31	KLB	3	Neutral
2025-07-31	PGB	3	Neutral
2025-07-31	VAB	2	Long
2025-07-31	VBB	2	Long
2025-07-31	SGB	1	Neutral
2025-07-31	ABB	1	Neutral
2025-07-31	BVB	2	Long
2025-08-31	VCB	2	Neutral
2025-08-31	BID	1	Short
2025-08-31	CTG	1	Short
2025-08-31	MBB	1	Short
2025-08-31	TCB	0	Neutral
2025-08-31	ACB	2	Neutral
2025-08-31	VPB	1	Short
2025-08-31	HDB	1	Short
2025-08-31	VIB	1	Short
2025-08-31	TPB	1	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-08-31	STB	0	Neutral
2025-08-31	SHB	3	Long
2025-08-31	LPB	0	Neutral
2025-08-31	SSB	2	Neutral
2025-08-31	MSB	1	Short
2025-08-31	OCB	1	Short
2025-08-31	NAB	1	Short
2025-08-31	BAB	2	Neutral
2025-08-31	KLB	3	Long
2025-08-31	PGB	0	Neutral
2025-08-31	VAB	3	Long
2025-08-31	VBB	0	Neutral
2025-08-31	SGB	2	Neutral
2025-08-31	ABB	1	Short
2025-08-31	BVB	1	Short
2025-09-30	VCB	3	Neutral
2025-09-30	BID	2	Short
2025-09-30	CTG	0	Neutral
2025-09-30	MBB	0	Neutral
2025-09-30	TCB	1	Long
2025-09-30	ACB	3	Neutral
2025-09-30	VPB	2	Short
2025-09-30	HDB	2	Short
2025-09-30	VIB	2	Short
2025-09-30	TPB	2	Short
2025-09-30	STB	1	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-09-30	SHB	1	Long
2025-09-30	LPB	0	Neutral
2025-09-30	SSB	3	Neutral
2025-09-30	MSB	0	Neutral
2025-09-30	OCB	0	Neutral
2025-09-30	NAB	0	Neutral
2025-09-30	BAB	3	Neutral
2025-09-30	KLB	1	Long
2025-09-30	PGB	2	Short
2025-09-30	VAB	1	Long
2025-09-30	VBB	1	Long
2025-09-30	SGB	3	Neutral
2025-09-30	ABB	0	Neutral
2025-09-30	BVB	0	Neutral
2025-10-31	VCB	1	Short
2025-10-31	BID	0	Neutral
2025-10-31	CTG	0	Neutral
2025-10-31	MBB	0	Neutral
2025-10-31	TCB	3	Neutral
2025-10-31	ACB	1	Short
2025-10-31	VPB	3	Neutral
2025-10-31	HDB	0	Neutral
2025-10-31	VIB	0	Neutral
2025-10-31	TPB	0	Neutral
2025-10-31	STB	3	Neutral
2025-10-31	SHB	2	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-10-31	LPB	0	Neutral
2025-10-31	SSB	1	Short
2025-10-31	MSB	0	Neutral
2025-10-31	OCB	0	Neutral
2025-10-31	NAB	0	Neutral
2025-10-31	BAB	1	Short
2025-10-31	KLB	2	Long
2025-10-31	PGB	3	Neutral
2025-10-31	VAB	2	Long
2025-10-31	VBB	3	Neutral
2025-10-31	SGB	1	Short
2025-10-31	ABB	2	Long
2025-10-31	BVB	0	Neutral
2025-11-30	VCB	3	Neutral
2025-11-30	BID	2	Short
2025-11-30	CTG	3	Neutral
2025-11-30	MBB	0	Neutral
2025-11-30	TCB	0	Neutral
2025-11-30	ACB	3	Neutral
2025-11-30	VPB	1	Long
2025-11-30	HDB	0	Neutral
2025-11-30	VIB	0	Neutral
2025-11-30	TPB	0	Neutral
2025-11-30	STB	0	Neutral
2025-11-30	SHB	1	Long
2025-11-30	LPB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-11-30	SSB	3	Neutral
2025-11-30	MSB	0	Neutral
2025-11-30	OCB	0	Neutral
2025-11-30	NAB	2	Short
2025-11-30	BAB	3	Neutral
2025-11-30	KLB	1	Long
2025-11-30	PGB	2	Short
2025-11-30	VAB	1	Long
2025-11-30	VBB	0	Neutral
2025-11-30	SGB	3	Neutral
2025-11-30	ABB	1	Long
2025-11-30	BVB	2	Short
2025-12-31	VCB	1	Short
2025-12-31	BID	1	Short
2025-12-31	CTG	1	Short
2025-12-31	MBB	0	Neutral
2025-12-31	TCB	0	Neutral
2025-12-31	ACB	1	Short
2025-12-31	VPB	2	Neutral
2025-12-31	HDB	0	Neutral
2025-12-31	VIB	0	Neutral
2025-12-31	TPB	0	Neutral
2025-12-31	STB	0	Neutral
2025-12-31	SHB	2	Neutral
2025-12-31	LPB	2	Neutral
2025-12-31	SSB	1	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-12-31	MSB	0	Neutral
2025-12-31	OCB	0	Neutral
2025-12-31	NAB	1	Short
2025-12-31	BAB	1	Short
2025-12-31	KLB	3	Long
2025-12-31	PGB	0	Neutral
2025-12-31	VAB	2	Neutral
2025-12-31	VBB	0	Neutral
2025-12-31	SGB	1	Short
2025-12-31	ABB	2	Neutral
2025-12-31	BVB	1	Short
2026-01-31	VCB	0	Short
2026-01-31	BID	0	Short
2026-01-31	CTG	2	Neutral
2026-01-31	MBB	2	Neutral
2026-01-31	TCB	2	Neutral
2026-01-31	ACB	0	Short
2026-01-31	VPB	1	Neutral
2026-01-31	HDB	1	Neutral
2026-01-31	VIB	2	Neutral
2026-01-31	TPB	1	Neutral
2026-01-31	STB	2	Neutral
2026-01-31	SHB	1	Neutral
2026-01-31	LPB	1	Neutral
2026-01-31	SSB	0	Short
2026-01-31	MSB	2	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2026-01-31	OCB	2	Neutral
2026-01-31	NAB	2	Neutral
2026-01-31	BAB	0	Short
2026-01-31	KLB	3	Long
2026-01-31	PGB	1	Neutral
2026-01-31	VAB	1	Neutral
2026-01-31	VBB	1	Neutral
2026-01-31	SGB	0	Short
2026-01-31	ABB	1	Neutral
2026-01-31	BVB	0	Short
2026-02-28	VCB	0	Short
2026-02-28	BID	0	Short
2026-02-28	CTG	2	Neutral
2026-02-28	MBB	2	Neutral
2026-02-28	TCB	2	Neutral
2026-02-28	ACB	0	Short
2026-02-28	VPB	1	Long
2026-02-28	HDB	1	Long
2026-02-28	VIB	2	Neutral
2026-02-28	TPB	1	Long
2026-02-28	STB	2	Neutral
2026-02-28	SHB	1	Long
2026-02-28	LPB	1	Long
2026-02-28	SSB	0	Short
2026-02-28	MSB	1	Long
2026-02-28	OCB	2	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2026-02-28	NAB	2	Neutral
2026-02-28	BAB	0	Short
2026-02-28	KLB	3	Neutral
2026-02-28	PGB	3	Neutral
2026-02-28	VAB	2	Neutral
2026-02-28	VBB	3	Neutral
2026-02-28	SGB	0	Short
2026-02-28	ABB	1	Long
2026-02-28	BVB	0	Short
2026-03-31	VCB	2	Short
2026-03-31	BID	0	Neutral
2026-03-31	CTG	0	Neutral
2026-03-31	MBB	3	Neutral
2026-03-31	TCB	0	Neutral
2026-03-31	ACB	2	Short
2026-03-31	VPB	1	Long
2026-03-31	HDB	1	Long
2026-03-31	VIB	0	Neutral
2026-03-31	TPB	3	Neutral
2026-03-31	STB	3	Neutral
2026-03-31	SHB	3	Neutral
2026-03-31	LPB	3	Neutral
2026-03-31	SSB	2	Short
2026-03-31	MSB	3	Neutral
2026-03-31	OCB	0	Neutral
2026-03-31	NAB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2026-03-31	BAB	2	Short
2026-03-31	KLB	1	Long
2026-03-31	PGB	3	Neutral
2026-03-31	VAB	0	Neutral
2026-03-31	VBB	0	Neutral
2026-03-31	SGB	2	Short
2026-03-31	ABB	1	Long
2026-03-31	BVB	0	Neutral
2026-04-30	VCB	1	Long
2026-04-30	BID	1	Long
2026-04-30	CTG	1	Long
2026-04-30	MBB	1	Long
2026-04-30	TCB	0	Neutral
2026-04-30	ACB	3	Short
2026-04-30	VPB	2	Neutral
2026-04-30	HDB	1	Long
2026-04-30	VIB	3	Short
2026-04-30	TPB	0	Neutral
2026-04-30	STB	1	Long
2026-04-30	SHB	0	Neutral
2026-04-30	LPB	2	Neutral
2026-04-30	SSB	3	Short
2026-04-30	MSB	0	Neutral
2026-04-30	OCB	3	Short
2026-04-30	NAB	3	Short
2026-04-30	BAB	3	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2026-04-30	KLB	2	Neutral
2026-04-30	PGB	2	Neutral
2026-04-30	VAB	0	Neutral
2026-04-30	VBB	0	Neutral
2026-04-30	SGB	3	Short
2026-04-30	ABB	2	Neutral
2026-04-30	BVB	3	Short
2026-05-31	VCB	0	Neutral
2026-05-31	BID	3	Neutral
2026-05-31	CTG	2	Long
2026-05-31	MBB	2	Long
2026-05-31	TCB	1	Short
2026-05-31	ACB	0	Neutral
2026-05-31	VPB	1	Short
2026-05-31	HDB	3	Neutral
2026-05-31	VIB	0	Neutral
2026-05-31	TPB	1	Short
2026-05-31	STB	2	Long
2026-05-31	SHB	1	Short
2026-05-31	LPB	1	Short
2026-05-31	SSB	0	Neutral
2026-05-31	MSB	1	Short
2026-05-31	OCB	0	Neutral
2026-05-31	NAB	0	Neutral
2026-05-31	BAB	0	Neutral
2026-05-31	KLB	3	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2026-05-31	PGB	1	Short
2026-05-31	VAB	1	Short
2026-05-31	VBB	0	Neutral
2026-05-31	SGB	0	Neutral
2026-05-31	ABB	3	Neutral
2026-05-31	BVB	0	Neutral

11. Lịch sử Tỷ trọng Danh mục

Tỷ trọng của 6 danh mục cho từng cổ phiếu tại mỗi kỳ tái cân bằng. w_mkt = market-cap; w_TAN = Tangency; w_MV = Min-Variance; w_BL = Black-Litterman; w_EW = Equal-Weight (1/N); w_RP = Risk-Parity.

Tổng số quan sát: 1,561 dòng

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2017-08-31	VCB	0.373349	0.300000	0.228374	0.000000	0.142857	0.143445
2017-08-31	BID	0.235950	0.078146	0.000000	0.100000	0.142857	0.116484
2017-08-31	CTG	0.138339	0.000000	0.000000	0.000000	0.142857	0.126106
2017-08-31	MBB	0.085414	0.300000	0.300000	0.300000	0.142857	0.200617
2017-08-31	ACB	0.067046	0.300000	0.180592	0.000000	0.142857	0.148225
2017-08-31	STB	0.070691	0.000000	0.000000	0.300000	0.142857	0.126525
2017-08-31	SHB	0.029211	0.021854	0.291034	0.300000	0.142857	0.138600
2017-09-30	VCB	0.376914	0.300000	0.237058	0.000000	0.142857	0.144085
2017-09-30	BID	0.233311	0.035950	0.000000	0.100000	0.142857	0.116089
2017-09-30	CTG	0.132644	0.000000	0.000000	0.000000	0.142857	0.126566
2017-09-30	MBB	0.087568	0.300000	0.300000	0.300000	0.142857	0.200615
2017-09-30	ACB	0.076128	0.300000	0.176242	0.000000	0.142857	0.148797
2017-09-30	STB	0.064861	0.000000	0.005906	0.300000	0.142857	0.126844
2017-09-30	SHB	0.028574	0.064050	0.280794	0.300000	0.142857	0.137005
2017-10-31	VCB	0.387099	0.300000	0.246125	0.000000	0.142857	0.146093
2017-10-31	BID	0.219039	0.079333	0.000000	0.100000	0.142857	0.116211
2017-10-31	CTG	0.132609	0.000000	0.000000	0.000000	0.142857	0.125730
2017-10-31	MBB	0.083441	0.300000	0.300000	0.300000	0.142857	0.201082
2017-10-31	ACB	0.080421	0.300000	0.168708	0.000000	0.142857	0.147171
2017-10-31	STB	0.068705	0.000000	0.000000	0.300000	0.142857	0.124942
2017-10-31	SHB	0.028686	0.020667	0.285167	0.300000	0.142857	0.138770

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2017-11-30	VCB	0.401989	0.300000	0.258089	0.000000	0.142857	0.148904
2017-11-30	BID	0.230218	0.034857	0.000000	0.100000	0.142857	0.116873
2017-11-30	CTG	0.126737	0.000000	0.000000	0.000000	0.142857	0.126059
2017-11-30	MBB	0.081745	0.300000	0.300000	0.300000	0.142857	0.198075
2017-11-30	ACB	0.075875	0.300000	0.160293	0.000000	0.142857	0.147259
2017-11-30	STB	0.057042	0.000000	0.000000	0.300000	0.142857	0.123783
2017-11-30	SHB	0.026393	0.065143	0.281618	0.300000	0.142857	0.139047
2017-12-31	VCB	0.397617	0.300000	0.248045	0.000000	0.142857	0.148158
2017-12-31	BID	0.228292	0.070882	0.000000	0.120614	0.142857	0.116260
2017-12-31	CTG	0.138682	0.000000	0.000000	0.000000	0.142857	0.126748
2017-12-31	MBB	0.078230	0.300000	0.300000	0.300000	0.142857	0.198885
2017-12-31	ACB	0.073008	0.300000	0.167856	0.000000	0.142857	0.147921
2017-12-31	STB	0.057355	0.000000	0.000000	0.279386	0.142857	0.121493
2017-12-31	SHB	0.026816	0.029118	0.284100	0.300000	0.142857	0.140535
2018-01-31	VCB	0.424734	0.300000	0.234821	0.300000	0.142857	0.148700
2018-01-31	BID	0.216572	0.070629	0.000000	0.300000	0.142857	0.119427
2018-01-31	CTG	0.131843	0.000000	0.000000	0.000000	0.142857	0.120997
2018-01-31	MBB	0.074099	0.300000	0.300000	0.300000	0.142857	0.201702
2018-01-31	ACB	0.073756	0.300000	0.208134	0.100000	0.142857	0.151086
2018-01-31	STB	0.053679	0.000000	0.000000	0.000000	0.142857	0.120734
2018-01-31	SHB	0.025316	0.029371	0.257046	0.000000	0.142857	0.137354
2018-02-28	VCB	0.418904	0.300000	0.232274	0.198273	0.142857	0.150310
2018-02-28	BID	0.233422	0.033103	0.000000	0.300000	0.142857	0.119047
2018-02-28	CTG	0.119111	0.000000	0.000000	0.201727	0.142857	0.120776
2018-02-28	MBB	0.076429	0.300000	0.300000	0.300000	0.142857	0.200325
2018-02-28	ACB	0.066989	0.300000	0.209142	0.000000	0.142857	0.150444

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2018-02-28	STB	0.054579	0.000000	0.000000	0.000000	0.142857	0.120510
2018-02-28	SHB	0.030568	0.066897	0.258584	0.000000	0.142857	0.138587
2018-03-31	VCB	0.414303	0.300000	0.266719	0.300000	0.142857	0.151868
2018-03-31	BID	0.237604	0.022044	0.000000	0.187710	0.142857	0.129991
2018-03-31	CTG	0.131338	0.000000	0.000000	0.212290	0.142857	0.140028
2018-03-31	MBB	0.073556	0.300000	0.300000	0.000000	0.142857	0.181390
2018-03-31	ACB	0.067390	0.300000	0.300000	0.300000	0.142857	0.159768
2018-03-31	STB	0.048267	0.000000	0.000000	0.000000	0.142857	0.119400
2018-03-31	SHB	0.027543	0.077956	0.133281	0.000000	0.142857	0.117555
2018-04-30	VCB	0.389328	0.300000	0.254140	0.300000	0.142857	0.152819
2018-04-30	BID	0.260056	0.019358	0.000000	0.300000	0.142857	0.129780
2018-04-30	CTG	0.132959	0.000000	0.000000	0.100000	0.142857	0.133971
2018-04-30	MBB	0.072920	0.300000	0.300000	0.000000	0.142857	0.183512
2018-04-30	ACB	0.072388	0.300000	0.300000	0.300000	0.142857	0.159655
2018-04-30	STB	0.045704	0.000000	0.000000	0.000000	0.142857	0.121077
2018-04-30	SHB	0.026645	0.080642	0.145860	0.000000	0.142857	0.119185
2018-05-31	VCB	0.388273	0.300000	0.261662	0.300000	0.142857	0.154197
2018-05-31	BID	0.253551	0.056793	0.000000	0.300000	0.142857	0.128754
2018-05-31	CTG	0.132002	0.000000	0.000000	0.300000	0.142857	0.136729
2018-05-31	MBB	0.073715	0.300000	0.300000	0.000000	0.142857	0.183561
2018-05-31	ACB	0.077021	0.300000	0.300000	0.100000	0.142857	0.158183
2018-05-31	STB	0.047862	0.000000	0.000000	0.000000	0.142857	0.119996
2018-05-31	SHB	0.027575	0.043207	0.138338	0.000000	0.142857	0.118579
2018-06-30	VCB	0.395582	0.300000	0.242855	0.247144	0.142857	0.151310
2018-06-30	BID	0.232078	0.021900	0.000000	0.300000	0.142857	0.127641
2018-06-30	CTG	0.137903	0.000000	0.000000	0.300000	0.142857	0.136343

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2018-06-30	MBB	0.081008	0.300000	0.300000	0.000000	0.142857	0.177291
2018-06-30	ACB	0.082125	0.300000	0.300000	0.152856	0.142857	0.160742
2018-06-30	STB	0.045875	0.000000	0.000000	0.000000	0.142857	0.124615
2018-06-30	SHB	0.025429	0.078100	0.157145	0.000000	0.142857	0.122059
2018-07-31	VCB	0.438421	0.300000	0.300000	0.181967	0.142857	0.155666
2018-07-31	BID	0.213290	0.000000	0.000000	0.300000	0.142857	0.122501
2018-07-31	CTG	0.127604	0.020642	0.000000	0.300000	0.142857	0.136709
2018-07-31	MBB	0.075376	0.300000	0.300000	0.000000	0.142857	0.177940
2018-07-31	ACB	0.075545	0.300000	0.300000	0.009195	0.142857	0.167837
2018-07-31	STB	0.046613	0.000000	0.000000	0.000000	0.142857	0.121973
2018-07-31	SHB	0.023152	0.079358	0.100000	0.208838	0.142857	0.117373
2018-08-31	VCB	0.438906	0.300000	0.300000	0.100000	0.142857	0.161942
2018-08-31	BID	0.214099	0.000000	0.000000	0.300000	0.142857	0.122746
2018-08-31	CTG	0.123236	0.047303	0.000000	0.300000	0.142857	0.135822
2018-08-31	MBB	0.079746	0.300000	0.300000	0.000000	0.142857	0.175114
2018-08-31	ACB	0.075422	0.300000	0.300000	0.300000	0.142857	0.163707
2018-08-31	STB	0.045868	0.000000	0.012860	0.000000	0.142857	0.123873
2018-08-31	SHB	0.022722	0.052697	0.087140	0.000000	0.142857	0.116795
2018-09-30	VCB	0.417540	0.300000	0.300000	0.100000	0.142857	0.160807
2018-09-30	BID	0.249364	0.000000	0.000000	0.300000	0.142857	0.124611
2018-09-30	CTG	0.126297	0.000000	0.000000	0.300000	0.142857	0.139271
2018-09-30	MBB	0.071702	0.300000	0.300000	0.000000	0.142857	0.173356
2018-09-30	ACB	0.073380	0.300000	0.300000	0.300000	0.142857	0.162861
2018-09-30	STB	0.040486	0.000000	0.013247	0.000000	0.142857	0.123124
2018-09-30	SHB	0.021232	0.100000	0.086753	0.000000	0.142857	0.115971
2018-10-31	VCB	0.413195	0.300000	0.300000	0.300000	0.142857	0.167176

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2018-10-31	BID	0.250562	0.000000	0.000000	0.100000	0.142857	0.113130
2018-10-31	CTG	0.125792	0.000000	0.000000	0.300000	0.142857	0.137628
2018-10-31	MBB	0.068582	0.300000	0.300000	0.000000	0.142857	0.173066
2018-10-31	ACB	0.072234	0.300000	0.300000	0.300000	0.142857	0.170543
2018-10-31	STB	0.047278	0.000000	0.007476	0.000000	0.142857	0.122569
2018-10-31	SHB	0.022356	0.100000	0.092524	0.000000	0.142857	0.115889
2018-11-30	VCB	0.420658	0.300000	0.300000	0.300000	0.142857	0.167815
2018-11-30	BID	0.239121	0.000000	0.000000	0.100000	0.142857	0.113567
2018-11-30	CTG	0.121656	0.000000	0.000000	0.300000	0.142857	0.138167
2018-11-30	MBB	0.074092	0.300000	0.300000	0.000000	0.142857	0.174392
2018-11-30	ACB	0.073125	0.300000	0.300000	0.300000	0.142857	0.171472
2018-11-30	STB	0.049847	0.000000	0.013261	0.000000	0.142857	0.118590
2018-11-30	SHB	0.021501	0.100000	0.086739	0.000000	0.142857	0.115998
2018-12-31	VCB	0.415930	0.300000	0.300000	0.300000	0.142857	0.168719
2018-12-31	BID	0.252662	0.000000	0.000000	0.100000	0.142857	0.112393
2018-12-31	CTG	0.118716	0.000000	0.000000	0.300000	0.142857	0.136075
2018-12-31	MBB	0.072385	0.300000	0.300000	0.000000	0.142857	0.178076
2018-12-31	ACB	0.071391	0.300000	0.300000	0.300000	0.142857	0.168932
2018-12-31	STB	0.048089	0.000000	0.039476	0.000000	0.142857	0.120820
2018-12-31	SHB	0.020828	0.100000	0.060524	0.000000	0.142857	0.114985
2019-01-31	VCB	0.409325	0.300000	0.300000	0.300000	0.142857	0.170686
2019-01-31	BID	0.281927	0.073694	0.000000	0.300000	0.142857	0.113303
2019-01-31	CTG	0.101434	0.000000	0.000000	0.300000	0.142857	0.136392
2019-01-31	MBB	0.066575	0.300000	0.300000	0.000000	0.142857	0.174570
2019-01-31	ACB	0.072193	0.300000	0.300000	0.100000	0.142857	0.167835
2019-01-31	STB	0.048193	0.000000	0.077855	0.000000	0.142857	0.124695

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2019-01-31	SHB	0.020352	0.026306	0.022145	0.000000	0.142857	0.112519
2019-02-28	VCB	0.428960	0.300000	0.300000	0.300000	0.142857	0.171268
2019-02-28	BID	0.254887	0.100000	0.000000	0.300000	0.142857	0.113703
2019-02-28	CTG	0.102289	0.000000	0.000000	0.100000	0.142857	0.131303
2019-02-28	MBB	0.073970	0.300000	0.300000	0.000000	0.142857	0.170248
2019-02-28	ACB	0.070834	0.300000	0.300000	0.000000	0.142857	0.169107
2019-02-28	STB	0.048968	0.000000	0.093891	0.300000	0.142857	0.130938
2019-02-28	SHB	0.020091	0.000000	0.006109	0.000000	0.142857	0.113433
2019-03-31	VCB	0.438656	0.300000	0.300000	0.300000	0.142857	0.171248
2019-03-31	BID	0.252572	0.100000	0.000000	0.300000	0.142857	0.117281
2019-03-31	CTG	0.103662	0.000000	0.000000	0.000000	0.142857	0.131273
2019-03-31	MBB	0.069664	0.300000	0.300000	0.100000	0.142857	0.165414
2019-03-31	ACB	0.068640	0.300000	0.300000	0.300000	0.142857	0.168791
2019-03-31	STB	0.047019	0.000000	0.100000	0.000000	0.142857	0.133603
2019-03-31	SHB	0.019786	0.000000	0.000000	0.000000	0.142857	0.112391
2019-04-30	VCB	0.448710	0.300000	0.300000	0.300000	0.142857	0.170996
2019-04-30	BID	0.255822	0.100000	0.000000	0.300000	0.142857	0.117509
2019-04-30	CTG	0.102834	0.000000	0.000000	0.000000	0.142857	0.131118
2019-04-30	MBB	0.066792	0.300000	0.300000	0.100000	0.142857	0.165002
2019-04-30	ACB	0.064460	0.300000	0.300000	0.300000	0.142857	0.168978
2019-04-30	STB	0.042886	0.000000	0.100000	0.000000	0.142857	0.133895
2019-04-30	SHB	0.018498	0.000000	0.000000	0.000000	0.142857	0.112502
2019-05-31	VCB	0.457295	0.300000	0.300000	0.283801	0.142857	0.167831
2019-05-31	BID	0.251111	0.100000	0.000000	0.000000	0.142857	0.117203
2019-05-31	CTG	0.098402	0.000000	0.000000	0.000000	0.142857	0.131355
2019-05-31	MBB	0.067460	0.300000	0.300000	0.121459	0.142857	0.164976

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2019-05-31	ACB	0.064621	0.300000	0.300000	0.000000	0.142857	0.172002
2019-05-31	STB	0.042428	0.000000	0.100000	0.300000	0.142857	0.134375
2019-05-31	SHB	0.018683	0.000000	0.000000	0.294740	0.142857	0.112260
2019-06-30	VCB	0.471576	0.300000	0.300000	0.300000	0.142857	0.170680
2019-06-30	BID	0.236853	0.100000	0.000000	0.300000	0.142857	0.116605
2019-06-30	CTG	0.097897	0.000000	0.000000	0.300000	0.142857	0.130236
2019-06-30	MBB	0.067179	0.300000	0.300000	0.000000	0.142857	0.164749
2019-06-30	ACB	0.064915	0.300000	0.300000	0.100000	0.142857	0.171922
2019-06-30	STB	0.043469	0.000000	0.100000	0.000000	0.142857	0.133870
2019-06-30	SHB	0.018111	0.000000	0.000000	0.000000	0.142857	0.111938
2019-07-31	VCB	0.485559	0.300000	0.300000	0.300000	0.142857	0.171102
2019-07-31	BID	0.233966	0.100000	0.000000	0.300000	0.142857	0.115394
2019-07-31	CTG	0.092262	0.000000	0.000000	0.300000	0.142857	0.131000
2019-07-31	MBB	0.066174	0.300000	0.300000	0.000000	0.142857	0.164099
2019-07-31	ACB	0.063519	0.300000	0.300000	0.100000	0.142857	0.171285
2019-07-31	STB	0.041214	0.000000	0.100000	0.000000	0.142857	0.135196
2019-07-31	SHB	0.017307	0.000000	0.000000	0.000000	0.142857	0.111924
2019-08-31	VCB	0.500510	0.300000	0.300000	0.300000	0.142857	0.170530
2019-08-31	BID	0.237375	0.100000	0.000000	0.300000	0.142857	0.115587
2019-08-31	CTG	0.089039	0.000000	0.000000	0.300000	0.142857	0.131112
2019-08-31	MBB	0.063691	0.300000	0.300000	0.100000	0.142857	0.165396
2019-08-31	ACB	0.058992	0.300000	0.300000	0.000000	0.142857	0.170782
2019-08-31	STB	0.035207	0.000000	0.100000	0.000000	0.142857	0.134846
2019-08-31	SHB	0.015186	0.000000	0.000000	0.000000	0.142857	0.111747
2019-09-30	VCB	0.483369	0.300000	0.300000	0.300000	0.142857	0.170916
2019-09-30	BID	0.258367	0.100000	0.000000	0.000000	0.142857	0.114564

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2019-09-30	CTG	0.086328	0.000000	0.000000	0.300000	0.142857	0.131017
2019-09-30	MBB	0.066037	0.300000	0.300000	0.100000	0.142857	0.164890
2019-09-30	ACB	0.057285	0.300000	0.300000	0.300000	0.142857	0.172767
2019-09-30	STB	0.034111	0.000000	0.100000	0.000000	0.142857	0.134191
2019-09-30	SHB	0.014501	0.000000	0.000000	0.000000	0.142857	0.111655
2019-10-31	VCB	0.489136	0.300000	0.300000	0.300000	0.142857	0.169621
2019-10-31	BID	0.251482	0.100000	0.000000	0.300000	0.142857	0.114901
2019-10-31	CTG	0.086596	0.000000	0.000000	0.300000	0.142857	0.130658
2019-10-31	MBB	0.067399	0.300000	0.300000	0.000000	0.142857	0.164797
2019-10-31	ACB	0.057760	0.300000	0.300000	0.100000	0.142857	0.171762
2019-10-31	STB	0.033133	0.000000	0.100000	0.000000	0.142857	0.135085
2019-10-31	SHB	0.014494	0.000000	0.000000	0.000000	0.142857	0.113176
2019-11-30	VCB	0.498635	0.300000	0.300000	0.300000	0.142857	0.169133
2019-11-30	BID	0.247408	0.100000	0.000000	0.300000	0.142857	0.114349
2019-11-30	CTG	0.085986	0.000000	0.000000	0.300000	0.142857	0.129952
2019-11-30	MBB	0.064914	0.300000	0.284686	0.000000	0.142857	0.163545
2019-11-30	ACB	0.056848	0.300000	0.300000	0.100000	0.142857	0.170605
2019-11-30	STB	0.032375	0.000000	0.115314	0.000000	0.142857	0.136952
2019-11-30	SHB	0.013835	0.000000	0.000000	0.000000	0.142857	0.115465
2019-12-31	VCB	0.494942	0.300000	0.300000	0.300000	0.142857	0.170360
2019-12-31	BID	0.260411	0.100000	0.000000	0.300000	0.142857	0.113886
2019-12-31	CTG	0.081548	0.000000	0.000000	0.100000	0.142857	0.129036
2019-12-31	MBB	0.063759	0.300000	0.280297	0.300000	0.142857	0.162529
2019-12-31	ACB	0.055586	0.300000	0.300000	0.000000	0.142857	0.169617
2019-12-31	STB	0.030892	0.000000	0.119703	0.000000	0.142857	0.136794
2019-12-31	SHB	0.012863	0.000000	0.000000	0.000000	0.142857	0.117778

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2020-01-31	VCB	0.494655	0.300000	0.290018	0.300000	0.142857	0.169474
2020-01-31	BID	0.277999	0.100000	0.000000	0.300000	0.142857	0.113946
2020-01-31	CTG	0.078027	0.000000	0.000000	0.300000	0.142857	0.127837
2020-01-31	MBB	0.056113	0.300000	0.269295	0.000000	0.142857	0.163685
2020-01-31	ACB	0.051371	0.300000	0.300000	0.100000	0.142857	0.167361
2020-01-31	STB	0.028797	0.000000	0.140687	0.000000	0.142857	0.141447
2020-01-31	SHB	0.013039	0.000000	0.000000	0.000000	0.142857	0.116250
2020-02-29	VCB	0.467669	0.300000	0.276845	0.300000	0.142857	0.167074
2020-02-29	BID	0.298281	0.255886	0.000000	0.300000	0.142857	0.112808
2020-02-29	CTG	0.087657	0.000000	0.000000	0.150484	0.142857	0.127413
2020-02-29	MBB	0.053952	0.280408	0.237207	0.015377	0.142857	0.159390
2020-02-29	ACB	0.049843	0.163706	0.300000	0.234139	0.142857	0.164978
2020-02-29	STB	0.028101	0.000000	0.185949	0.000000	0.142857	0.153602
2020-02-29	SHB	0.014497	0.000000	0.000000	0.000000	0.142857	0.114736
2020-03-31	VCB	0.448047	0.300000	0.293476	0.300000	0.125000	0.141392
2020-03-31	BID	0.271740	0.300000	0.000000	0.300000	0.125000	0.095978
2020-03-31	CTG	0.095336	0.000000	0.000000	0.000000	0.125000	0.112612
2020-03-31	MBB	0.054536	0.300000	0.139178	0.000000	0.125000	0.133415
2020-03-31	ACB	0.056280	0.100000	0.300000	0.297898	0.125000	0.172055
2020-03-31	VIB	0.018004	0.000000	0.061800	0.102102	0.125000	0.120064
2020-03-31	STB	0.032958	0.000000	0.205545	0.000000	0.125000	0.130132
2020-03-31	SHB	0.023100	0.000000	0.000000	0.000000	0.125000	0.094351
2020-04-30	VCB	0.465684	0.300000	0.300000	0.300000	0.125000	0.143503
2020-04-30	BID	0.254497	0.191172	0.000000	0.300000	0.125000	0.096175
2020-04-30	CTG	0.087985	0.000000	0.000000	0.180847	0.125000	0.115229
2020-04-30	MBB	0.050055	0.192232	0.147313	0.000000	0.125000	0.136300

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2020-04-30	ACB	0.054967	0.251651	0.300000	0.031903	0.125000	0.174011
2020-04-30	VIB	0.017614	0.000000	0.062582	0.187249	0.125000	0.122770
2020-04-30	STB	0.028651	0.000000	0.190105	0.000000	0.125000	0.129179
2020-04-30	SHB	0.040545	0.064945	0.000000	0.000000	0.125000	0.082833
2020-05-31	VCB	0.448980	0.300000	0.300000	0.143694	0.125000	0.148615
2020-05-31	BID	0.258161	0.300000	0.000000	0.000000	0.125000	0.102299
2020-05-31	CTG	0.089926	0.000000	0.000000	0.233272	0.125000	0.116232
2020-05-31	MBB	0.051820	0.000000	0.129241	0.000000	0.125000	0.128187
2020-05-31	ACB	0.055653	0.100000	0.300000	0.137009	0.125000	0.157648
2020-05-31	VIB	0.017556	0.000000	0.003073	0.186025	0.125000	0.126971
2020-05-31	STB	0.031397	0.000000	0.000000	0.000000	0.125000	0.120569
2020-05-31	SHB	0.046506	0.300000	0.267685	0.300000	0.125000	0.099479
2020-06-30	VCB	0.481368	0.300000	0.300000	0.094670	0.125000	0.150942
2020-06-30	BID	0.248812	0.300000	0.000000	0.000000	0.125000	0.103320
2020-06-30	CTG	0.086612	0.000000	0.000000	0.005330	0.125000	0.115924
2020-06-30	MBB	0.047778	0.000000	0.092300	0.000000	0.125000	0.126739
2020-06-30	ACB	0.053168	0.100000	0.300000	0.300000	0.125000	0.156363
2020-06-30	VIB	0.016841	0.000000	0.056661	0.000000	0.125000	0.128923
2020-06-30	STB	0.030406	0.000000	0.000000	0.300000	0.125000	0.116151
2020-06-30	SHB	0.035016	0.300000	0.251040	0.300000	0.125000	0.101638
2020-07-31	VCB	0.475964	0.300000	0.300000	0.000000	0.125000	0.143114
2020-07-31	BID	0.246758	0.300000	0.000000	0.000000	0.125000	0.104278
2020-07-31	CTG	0.086994	0.000000	0.000000	0.100000	0.125000	0.116403
2020-07-31	MBB	0.048426	0.000000	0.110347	0.000000	0.125000	0.131954
2020-07-31	ACB	0.055532	0.100000	0.300000	0.300000	0.125000	0.156527
2020-07-31	VIB	0.018894	0.000000	0.023637	0.000000	0.125000	0.130081

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2020-07-31	STB	0.033297	0.000000	0.000000	0.300000	0.125000	0.116346
2020-07-31	SHB	0.034135	0.300000	0.266016	0.300000	0.125000	0.101298
2020-08-31	VCB	0.470276	0.300000	0.239297	0.000000	0.111111	0.123321
2020-08-31	BID	0.247840	0.255999	0.000000	0.000000	0.111111	0.090577
2020-08-31	CTG	0.088866	0.000000	0.000000	0.062077	0.111111	0.101343
2020-08-31	MBB	0.047681	0.000000	0.000000	0.000000	0.111111	0.118399
2020-08-31	ACB	0.056874	0.004559	0.300000	0.296788	0.111111	0.136027
2020-08-31	VIB	0.020279	0.000000	0.000000	0.000000	0.111111	0.112517
2020-08-31	STB	0.032484	0.000000	0.000000	0.041135	0.111111	0.101895
2020-08-31	SHB	0.032185	0.300000	0.160703	0.300000	0.111111	0.087421
2020-08-31	KLB	0.003515	0.139442	0.300000	0.300000	0.111111	0.128499
2020-09-30	VCB	0.455432	0.300000	0.246477	0.000000	0.111111	0.122445
2020-09-30	BID	0.246771	0.144144	0.000000	0.000000	0.111111	0.090502
2020-09-30	CTG	0.096041	0.000000	0.000000	0.000000	0.111111	0.101076
2020-09-30	MBB	0.048444	0.000000	0.000000	0.000000	0.111111	0.118361
2020-09-30	ACB	0.062104	0.060901	0.300000	0.100000	0.111111	0.135591
2020-09-30	VIB	0.022110	0.000000	0.000000	0.000000	0.111111	0.112176
2020-09-30	STB	0.032111	0.000000	0.000000	0.300000	0.111111	0.101555
2020-09-30	SHB	0.033760	0.300000	0.153523	0.300000	0.111111	0.086662
2020-09-30	KLB	0.003228	0.194955	0.300000	0.300000	0.111111	0.131632
2020-10-31	VCB	0.409789	0.300000	0.300000	0.000000	0.100000	0.113448
2020-10-31	BID	0.218408	0.121667	0.000000	0.000000	0.100000	0.083748
2020-10-31	CTG	0.088126	0.000000	0.000000	0.000000	0.100000	0.091362
2020-10-31	MBB	0.047310	0.000000	0.000000	0.000000	0.100000	0.107570
2020-10-31	ACB	0.058557	0.000000	0.240280	0.100000	0.100000	0.120093
2020-10-31	VPB	0.081657	0.000000	0.000000	0.000000	0.100000	0.086028

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2020-10-31	VIB	0.024706	0.043184	0.000000	0.000000	0.100000	0.100957
2020-10-31	STB	0.035139	0.000000	0.000000	0.300000	0.100000	0.094401
2020-10-31	SHB	0.033224	0.300000	0.159720	0.300000	0.100000	0.079652
2020-10-31	KLB	0.003084	0.235149	0.300000	0.300000	0.100000	0.122740
2020-11-30	VCB	0.404752	0.300000	0.300000	0.105812	0.100000	0.114370
2020-11-30	BID	0.206215	0.013720	0.000000	0.000000	0.100000	0.084757
2020-11-30	CTG	0.096313	0.000000	0.000000	0.000000	0.100000	0.092055
2020-11-30	MBB	0.049074	0.000000	0.000000	0.000000	0.100000	0.107561
2020-11-30	ACB	0.062761	0.000000	0.240837	0.264574	0.100000	0.121131
2020-11-30	VPB	0.081175	0.000000	0.000000	0.029614	0.100000	0.086776
2020-11-30	VIB	0.029273	0.146524	0.000000	0.000000	0.100000	0.097078
2020-11-30	STB	0.033889	0.000000	0.000000	0.000000	0.100000	0.091718
2020-11-30	SHB	0.033246	0.269958	0.159163	0.300000	0.100000	0.079960
2020-11-30	KLB	0.003303	0.269798	0.300000	0.300000	0.100000	0.124595
2020-12-31	VCB	0.400213	0.300000	0.300000	0.038394	0.090909	0.104254
2020-12-31	BID	0.197126	0.000000	0.000000	0.300000	0.090909	0.076899
2020-12-31	CTG	0.098182	0.000000	0.000000	0.156124	0.090909	0.082941
2020-12-31	MBB	0.049490	0.000000	0.000000	0.101321	0.090909	0.097452
2020-12-31	ACB	0.062549	0.000000	0.240434	0.000000	0.090909	0.109236
2020-12-31	VPB	0.081282	0.000000	0.000000	0.140821	0.090909	0.079042
2020-12-31	VIB	0.026080	0.236992	0.000000	0.001454	0.090909	0.086133
2020-12-31	STB	0.032939	0.000000	0.000000	0.002530	0.090909	0.084303
2020-12-31	SHB	0.032392	0.252185	0.159566	0.000000	0.090909	0.072532
2020-12-31	LPB	0.016567	0.000000	0.000000	0.259355	0.090909	0.094387
2020-12-31	KLB	0.003180	0.210822	0.300000	0.000000	0.090909	0.112820
2021-01-31	VCB	0.385892	0.269339	0.300000	0.000000	0.090909	0.104911

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2021-01-31	BID	0.209155	0.000000	0.000000	0.000000	0.090909	0.077250
2021-01-31	CTG	0.093336	0.000000	0.000000	0.000000	0.090909	0.085707
2021-01-31	MBB	0.050850	0.000000	0.000000	0.236123	0.090909	0.096346
2021-01-31	ACB	0.058667	0.000000	0.221427	0.000000	0.090909	0.108619
2021-01-31	VPB	0.089583	0.000000	0.000000	0.000000	0.090909	0.077337
2021-01-31	VIB	0.027827	0.224999	0.000000	0.300000	0.090909	0.085355
2021-01-31	STB	0.034524	0.000000	0.000000	0.000000	0.090909	0.085217
2021-01-31	SHB	0.029430	0.205662	0.164210	0.000000	0.090909	0.072922
2021-01-31	LPB	0.016509	0.000000	0.014363	0.163877	0.090909	0.094501
2021-01-31	KLB	0.004227	0.300000	0.300000	0.300000	0.090909	0.111836
2021-02-28	VCB	0.391361	0.189547	0.255318	0.000000	0.083333	0.084670
2021-02-28	BID	0.186043	0.000000	0.000000	0.000000	0.083333	0.060766
2021-02-28	CTG	0.087982	0.000000	0.000000	0.000000	0.083333	0.068346
2021-02-28	MBB	0.055356	0.000000	0.000000	0.003297	0.083333	0.075686
2021-02-28	ACB	0.059791	0.000000	0.092282	0.086887	0.083333	0.086925
2021-02-28	VPB	0.088312	0.000000	0.000000	0.000000	0.083333	0.060042
2021-02-28	VIB	0.028990	0.300000	0.000000	0.300000	0.083333	0.066998
2021-02-28	STB	0.036864	0.000000	0.000000	0.000000	0.083333	0.066845
2021-02-28	SHB	0.027003	0.210453	0.121874	0.000000	0.083333	0.058239
2021-02-28	LPB	0.017974	0.000000	0.000000	0.152701	0.083333	0.074926
2021-02-28	BAB	0.016306	0.000000	0.300000	0.300000	0.083333	0.220327
2021-02-28	KLB	0.004019	0.300000	0.230526	0.157115	0.083333	0.076230
2021-03-31	VCB	0.355770	0.261641	0.300000	0.000000	0.076923	0.079908
2021-03-31	BID	0.175039	0.000000	0.000000	0.000000	0.076923	0.056312
2021-03-31	CTG	0.092474	0.000000	0.000000	0.000000	0.076923	0.060973
2021-03-31	MBB	0.056158	0.000000	0.000000	0.000000	0.076923	0.073280

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2021-03-31	ACB	0.063439	0.000000	0.048220	0.000000	0.076923	0.079084
2021-03-31	VPB	0.101383	0.000000	0.000000	0.000000	0.076923	0.056538
2021-03-31	HDB	0.028668	0.000000	0.000000	0.000000	0.076923	0.068571
2021-03-31	VIB	0.031333	0.300000	0.000000	0.300000	0.076923	0.063679
2021-03-31	STB	0.034455	0.000000	0.000000	0.000000	0.076923	0.063188
2021-03-31	SHB	0.025309	0.135702	0.161904	0.000000	0.076923	0.057594
2021-03-31	LPB	0.017971	0.000000	0.015017	0.100000	0.076923	0.076171
2021-03-31	BAB	0.014561	0.002657	0.300000	0.300000	0.076923	0.197437
2021-03-31	KLB	0.003442	0.300000	0.174860	0.300000	0.076923	0.067263
2021-04-30	VCB	0.329000	0.035307	0.300000	0.000000	0.076923	0.081472
2021-04-30	BID	0.165127	0.000000	0.000000	0.000000	0.076923	0.057198
2021-04-30	CTG	0.095499	0.000000	0.000000	0.000000	0.076923	0.061755
2021-04-30	MBB	0.054785	0.000000	0.000000	0.256608	0.076923	0.071828
2021-04-30	ACB	0.061210	0.000000	0.040075	0.000000	0.076923	0.076345
2021-04-30	VPB	0.108396	0.000000	0.000000	0.000000	0.076923	0.053904
2021-04-30	HDB	0.028801	0.000000	0.000000	0.000000	0.076923	0.068030
2021-04-30	VIB	0.036609	0.300000	0.000000	0.300000	0.076923	0.061689
2021-04-30	STB	0.038581	0.000000	0.000000	0.000000	0.076923	0.063291
2021-04-30	SHB	0.039178	0.098117	0.165276	0.000000	0.076923	0.057778
2021-04-30	LPB	0.020321	0.000000	0.006097	0.300000	0.076923	0.074917
2021-04-30	BAB	0.018600	0.266577	0.300000	0.035611	0.076923	0.203911
2021-04-30	KLB	0.003893	0.300000	0.188552	0.107781	0.076923	0.067882
2021-05-31	VCB	0.324410	0.004001	0.300000	0.095120	0.076923	0.089779
2021-05-31	BID	0.147356	0.000000	0.000000	0.000000	0.076923	0.063484
2021-05-31	CTG	0.090715	0.000000	0.000000	0.000000	0.076923	0.067636
2021-05-31	MBB	0.055244	0.000000	0.000000	0.300000	0.076923	0.078996

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2021-05-31	ACB	0.059559	0.000000	0.132389	0.000000	0.076923	0.084757
2021-05-31	VPB	0.132753	0.000000	0.000000	0.271848	0.076923	0.059045
2021-05-31	HDB	0.027801	0.000000	0.000000	0.033032	0.076923	0.074725
2021-05-31	VIB	0.039326	0.300000	0.000000	0.300000	0.076923	0.068381
2021-05-31	STB	0.040187	0.000000	0.000000	0.000000	0.076923	0.068193
2021-05-31	SHB	0.039055	0.131285	0.074042	0.000000	0.076923	0.054371
2021-05-31	LPB	0.023553	0.000000	0.000000	0.000000	0.076923	0.079760
2021-05-31	BAB	0.015663	0.300000	0.300000	0.000000	0.076923	0.138032
2021-05-31	KLB	0.004379	0.264714	0.193569	0.000000	0.076923	0.072840
2021-06-30	VCB	0.269885	0.249672	0.292735	0.000000	0.071429	0.085073
2021-06-30	BID	0.147687	0.000000	0.000000	0.000000	0.071429	0.058816
2021-06-30	CTG	0.099616	0.000000	0.000000	0.000000	0.071429	0.062739
2021-06-30	MBB	0.058322	0.000000	0.000000	0.000000	0.071429	0.073093
2021-06-30	ACB	0.060033	0.000000	0.000000	0.000000	0.071429	0.077584
2021-06-30	VPB	0.132490	0.000000	0.000000	0.170277	0.071429	0.052031
2021-06-30	HDB	0.029084	0.000000	0.000000	0.000000	0.071429	0.067594
2021-06-30	VIB	0.040871	0.300000	0.000000	0.000000	0.071429	0.062825
2021-06-30	TPB	0.025708	0.000000	0.230208	0.000000	0.071429	0.096516
2021-06-30	STB	0.047953	0.000000	0.000000	0.000000	0.071429	0.061774
2021-06-30	SHB	0.040204	0.236911	0.096537	0.300000	0.071429	0.049623
2021-06-30	LPB	0.027674	0.000000	0.000000	0.217232	0.071429	0.068165
2021-06-30	BAB	0.015926	0.071808	0.300000	0.185210	0.071429	0.118059
2021-06-30	KLB	0.004547	0.141609	0.080520	0.127281	0.071429	0.066107
2021-07-31	VCB	0.304913	0.288372	0.300000	0.000000	0.071429	0.090883
2021-07-31	BID	0.137137	0.000000	0.000000	0.000000	0.071429	0.062504
2021-07-31	CTG	0.094596	0.000000	0.000000	0.000000	0.071429	0.061947

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2021-07-31	MBB	0.063680	0.000000	0.000000	0.000000	0.071429	0.071966
2021-07-31	ACB	0.061927	0.000000	0.011659	0.000000	0.071429	0.078219
2021-07-31	VPB	0.124049	0.000000	0.000000	0.300000	0.071429	0.054760
2021-07-31	HDB	0.029248	0.000000	0.000000	0.000000	0.071429	0.067663
2021-07-31	VIB	0.039521	0.300000	0.000000	0.000000	0.071429	0.068067
2021-07-31	TPB	0.024145	0.004652	0.167387	0.000000	0.071429	0.085073
2021-07-31	STB	0.041546	0.000000	0.000000	0.000000	0.071429	0.060176
2021-07-31	SHB	0.036308	0.280543	0.122372	0.300000	0.071429	0.052677
2021-07-31	LPB	0.026484	0.000000	0.000000	0.000000	0.071429	0.067580
2021-07-31	BAB	0.012647	0.000000	0.300000	0.234981	0.071429	0.111472
2021-07-31	KLB	0.003799	0.126433	0.098583	0.165019	0.071429	0.067014
2021-08-31	VCB	0.254886	0.300000	0.300000	0.000000	0.066667	0.083184
2021-08-31	BID	0.118439	0.000000	0.000000	0.000000	0.066667	0.060272
2021-08-31	CTG	0.079140	0.000000	0.000000	0.000000	0.066667	0.059753
2021-08-31	MBB	0.056944	0.000000	0.000000	0.000000	0.066667	0.068713
2021-08-31	TCB	0.119425	0.000000	0.000000	0.034881	0.066667	0.060118
2021-08-31	ACB	0.062258	0.000000	0.054315	0.000000	0.066667	0.076169
2021-08-31	VPB	0.110969	0.000000	0.000000	0.000000	0.066667	0.051924
2021-08-31	HDB	0.027559	0.000000	0.000000	0.000000	0.066667	0.064352
2021-08-31	VIB	0.033019	0.300000	0.000000	0.078364	0.066667	0.065097
2021-08-31	TPB	0.022821	0.022450	0.137321	0.000000	0.066667	0.080784
2021-08-31	STB	0.040307	0.000000	0.000000	0.032210	0.066667	0.056445
2021-08-31	SHB	0.034945	0.300000	0.120161	0.000000	0.066667	0.050663
2021-08-31	LPB	0.025078	0.000000	0.000000	0.300000	0.066667	0.065151
2021-08-31	BAB	0.011077	0.000000	0.300000	0.254546	0.066667	0.095168
2021-08-31	KLB	0.003134	0.077550	0.088203	0.300000	0.066667	0.062208

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2021-09-30	VCB	0.267803	0.089171	0.300000	0.000000	0.066667	0.079395
2021-09-30	BID	0.116290	0.000000	0.000000	0.000000	0.066667	0.059862
2021-09-30	CTG	0.075773	0.000000	0.000000	0.000000	0.066667	0.058820
2021-09-30	MBB	0.056907	0.000000	0.000000	0.300000	0.066667	0.068566
2021-09-30	TCB	0.117428	0.000000	0.000000	0.000000	0.066667	0.061713
2021-09-30	ACB	0.057110	0.000000	0.065548	0.033406	0.066667	0.077441
2021-09-30	VPB	0.116065	0.000000	0.000000	0.000000	0.066667	0.052402
2021-09-30	HDB	0.027482	0.000000	0.000000	0.000000	0.066667	0.065155
2021-09-30	VIB	0.029790	0.300000	0.000000	0.300000	0.066667	0.062940
2021-09-30	TPB	0.023033	0.175842	0.167326	0.000000	0.066667	0.082789
2021-09-30	STB	0.038191	0.000000	0.000000	0.000000	0.066667	0.057308
2021-09-30	SHB	0.035807	0.300000	0.132744	0.066594	0.066667	0.051402
2021-09-30	LPB	0.023477	0.134987	0.000000	0.300000	0.066667	0.067647
2021-09-30	BAB	0.011567	0.000000	0.300000	0.000000	0.066667	0.094015
2021-09-30	KLB	0.003277	0.000000	0.034382	0.000000	0.066667	0.060544
2021-10-31	VCB	0.261601	0.107695	0.300000	0.000000	0.066667	0.079705
2021-10-31	BID	0.117629	0.000000	0.000000	0.000000	0.066667	0.063715
2021-10-31	CTG	0.072360	0.000000	0.000000	0.000000	0.066667	0.059165
2021-10-31	MBB	0.056760	0.000000	0.000000	0.000000	0.066667	0.068276
2021-10-31	TCB	0.119978	0.000000	0.000000	0.000000	0.066667	0.061707
2021-10-31	ACB	0.056255	0.000000	0.053615	0.029498	0.066667	0.075514
2021-10-31	VPB	0.124052	0.000000	0.000000	0.300000	0.066667	0.052542
2021-10-31	HDB	0.026883	0.000000	0.000000	0.000000	0.066667	0.065189
2021-10-31	VIB	0.029033	0.220406	0.000000	0.000000	0.066667	0.060967
2021-10-31	TPB	0.028073	0.118432	0.183153	0.000000	0.066667	0.082597
2021-10-31	STB	0.035852	0.000000	0.000000	0.251191	0.066667	0.056699

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2021-10-31	SHB	0.034598	0.300000	0.136118	0.300000	0.066667	0.051281
2021-10-31	LPB	0.022039	0.226779	0.000000	0.119312	0.066667	0.066871
2021-10-31	BAB	0.011555	0.000000	0.300000	0.000000	0.066667	0.093989
2021-10-31	KLB	0.003333	0.026688	0.027115	0.000000	0.066667	0.061782
2021-11-30	VCB	0.255940	0.089712	0.300000	0.000000	0.066667	0.079781
2021-11-30	BID	0.120088	0.000000	0.000000	0.000000	0.066667	0.063926
2021-11-30	CTG	0.072758	0.000000	0.000000	0.000000	0.066667	0.059175
2021-11-30	MBB	0.056347	0.000000	0.000000	0.000000	0.066667	0.068621
2021-11-30	TCB	0.121085	0.000000	0.000000	0.000000	0.066667	0.062363
2021-11-30	ACB	0.057094	0.000000	0.058296	0.000000	0.066667	0.075649
2021-11-30	VPB	0.125100	0.000000	0.000000	0.000000	0.066667	0.052633
2021-11-30	HDB	0.025934	0.000000	0.000000	0.000000	0.066667	0.065328
2021-11-30	VIB	0.029476	0.300000	0.000000	0.000000	0.066667	0.060954
2021-11-30	TPB	0.029365	0.300000	0.178534	0.000000	0.066667	0.078878
2021-11-30	STB	0.036006	0.000000	0.000000	0.286983	0.066667	0.057662
2021-11-30	SHB	0.035265	0.300000	0.136405	0.300000	0.066667	0.051284
2021-11-30	LPB	0.020548	0.010288	0.000000	0.300000	0.066667	0.066839
2021-11-30	BAB	0.011109	0.000000	0.300000	0.000000	0.066667	0.094272
2021-11-30	KLB	0.003884	0.000000	0.026766	0.113017	0.066667	0.062636
2021-12-31	VCB	0.250050	0.198537	0.300000	0.000000	0.066667	0.080619
2021-12-31	BID	0.124413	0.000000	0.000000	0.000000	0.066667	0.065129
2021-12-31	CTG	0.075031	0.000000	0.000000	0.000000	0.066667	0.059885
2021-12-31	MBB	0.055527	0.000000	0.000000	0.089245	0.066667	0.067943
2021-12-31	TCB	0.117718	0.000000	0.000000	0.113233	0.066667	0.061886
2021-12-31	ACB	0.055774	0.000000	0.086323	0.300000	0.066667	0.076545
2021-12-31	VPB	0.120011	0.000000	0.000000	0.078775	0.066667	0.054333

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2021-12-31	HDB	0.030305	0.000000	0.000000	0.110549	0.066667	0.066643
2021-12-31	VIB	0.032555	0.198610	0.000000	0.000000	0.066667	0.060892
2021-12-31	TPB	0.028595	0.300000	0.151726	0.096226	0.066667	0.077485
2021-12-31	STB	0.037873	0.000000	0.000000	0.069509	0.066667	0.057159
2021-12-31	SHB	0.035498	0.300000	0.152327	0.000000	0.066667	0.052020
2021-12-31	LPB	0.021410	0.000000	0.000000	0.142464	0.066667	0.066682
2021-12-31	BAB	0.011494	0.000000	0.300000	0.000000	0.066667	0.092681
2021-12-31	KLB	0.003745	0.002853	0.009623	0.000000	0.066667	0.060097
2022-01-31	VCB	0.252133	0.162588	0.300000	0.000000	0.066667	0.080777
2022-01-31	BID	0.128671	0.000000	0.000000	0.000000	0.066667	0.065175
2022-01-31	CTG	0.076101	0.000000	0.000000	0.000000	0.066667	0.059908
2022-01-31	MBB	0.054231	0.000000	0.000000	0.000000	0.066667	0.068159
2022-01-31	TCB	0.111282	0.000000	0.000000	0.300000	0.066667	0.062124
2022-01-31	ACB	0.056599	0.000000	0.083347	0.300000	0.066667	0.076816
2022-01-31	VPB	0.111626	0.000000	0.000000	0.004191	0.066667	0.054490
2022-01-31	HDB	0.030047	0.000000	0.000000	0.000000	0.066667	0.065351
2022-01-31	VIB	0.035747	0.237412	0.000000	0.000000	0.066667	0.060590
2022-01-31	TPB	0.034418	0.300000	0.153425	0.300000	0.066667	0.077905
2022-01-31	STB	0.040441	0.000000	0.000000	0.046411	0.066667	0.057183
2022-01-31	SHB	0.033004	0.300000	0.152971	0.000000	0.066667	0.052269
2022-01-31	LPB	0.020893	0.000000	0.000000	0.000000	0.066667	0.066682
2022-01-31	BAB	0.010921	0.000000	0.300000	0.000000	0.066667	0.092193
2022-01-31	KLB	0.003886	0.000000	0.010257	0.049398	0.066667	0.060376
2022-02-28	VCB	0.257855	0.131970	0.300000	0.300000	0.066667	0.081088
2022-02-28	BID	0.150601	0.000000	0.000000	0.011320	0.066667	0.065467
2022-02-28	CTG	0.075010	0.000000	0.000000	0.000000	0.066667	0.062216

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2022-02-28	MBB	0.057240	0.000000	0.000000	0.242635	0.066667	0.069452
2022-02-28	TCB	0.106439	0.000000	0.000000	0.000000	0.066667	0.061934
2022-02-28	ACB	0.051318	0.000000	0.085852	0.300000	0.066667	0.076871
2022-02-28	VPB	0.103484	0.000000	0.000000	0.060375	0.066667	0.054378
2022-02-28	HDB	0.027254	0.000000	0.000000	0.085670	0.066667	0.065307
2022-02-28	VIB	0.032537	0.268030	0.000000	0.000000	0.066667	0.060929
2022-02-28	TPB	0.031853	0.300000	0.145731	0.000000	0.066667	0.074283
2022-02-28	STB	0.041331	0.000000	0.000000	0.000000	0.066667	0.057047
2022-02-28	SHB	0.030138	0.300000	0.154746	0.000000	0.066667	0.052158
2022-02-28	LPB	0.021204	0.000000	0.000000	0.000000	0.066667	0.066540
2022-02-28	BAB	0.009849	0.000000	0.300000	0.000000	0.066667	0.091761
2022-02-28	KLB	0.003886	0.000000	0.013671	0.000000	0.066667	0.060567
2022-03-31	VCB	0.255927	0.157162	0.300000	0.037671	0.066667	0.080258
2022-03-31	BID	0.143447	0.000000	0.000000	0.000000	0.066667	0.062453
2022-03-31	CTG	0.070451	0.000000	0.000000	0.000000	0.066667	0.062300
2022-03-31	MBB	0.061096	0.000000	0.000000	0.000000	0.066667	0.069126
2022-03-31	TCB	0.106214	0.000000	0.000000	0.173545	0.066667	0.062168
2022-03-31	ACB	0.052653	0.000000	0.085562	0.000000	0.066667	0.077311
2022-03-31	VPB	0.112170	0.000000	0.000000	0.000000	0.066667	0.054668
2022-03-31	HDB	0.026544	0.000000	0.000000	0.000000	0.066667	0.065626
2022-03-31	VIB	0.033553	0.242838	0.000000	0.000000	0.066667	0.061373
2022-03-31	TPB	0.033537	0.300000	0.147568	0.300000	0.066667	0.074652
2022-03-31	STB	0.039925	0.000000	0.000000	0.000000	0.066667	0.056911
2022-03-31	SHB	0.030327	0.300000	0.155947	0.000000	0.066667	0.052449
2022-03-31	LPB	0.020087	0.000000	0.000000	0.000000	0.066667	0.067268
2022-03-31	BAB	0.010111	0.000000	0.300000	0.271999	0.066667	0.092178

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2022-03-31	KLB	0.003959	0.000000	0.010922	0.216784	0.066667	0.061259
2022-04-30	VCB	0.253889	0.102956	0.300000	0.300000	0.066667	0.080272
2022-04-30	BID	0.145820	0.000000	0.000000	0.300000	0.066667	0.062008
2022-04-30	CTG	0.070414	0.000000	0.000000	0.000000	0.066667	0.061614
2022-04-30	MBB	0.059731	0.000000	0.000000	0.000000	0.066667	0.069679
2022-04-30	TCB	0.106589	0.000000	0.000000	0.000000	0.066667	0.062246
2022-04-30	ACB	0.052409	0.000000	0.084006	0.000000	0.066667	0.077513
2022-04-30	VPB	0.112128	0.000000	0.000000	0.000000	0.066667	0.054957
2022-04-30	HDB	0.026687	0.000000	0.000000	0.000000	0.066667	0.065459
2022-04-30	VIB	0.034377	0.297044	0.000000	0.000000	0.066667	0.061496
2022-04-30	TPB	0.032538	0.300000	0.152237	0.291952	0.066667	0.075116
2022-04-30	STB	0.039338	0.000000	0.000000	0.000000	0.066667	0.056689
2022-04-30	SHB	0.030590	0.300000	0.153891	0.000000	0.066667	0.052476
2022-04-30	LPB	0.019605	0.000000	0.000000	0.000000	0.066667	0.066204
2022-04-30	BAB	0.010279	0.000000	0.300000	0.000000	0.066667	0.092534
2022-04-30	KLB	0.005605	0.000000	0.009866	0.108048	0.066667	0.061736
2022-05-31	VCB	0.272787	0.000000	0.300000	0.300000	0.066667	0.081617
2022-05-31	BID	0.136155	0.000000	0.000000	0.258623	0.066667	0.062757
2022-05-31	CTG	0.065632	0.000000	0.000000	0.000000	0.066667	0.062048
2022-05-31	MBB	0.058903	0.000000	0.000000	0.000000	0.066667	0.069648
2022-05-31	TCB	0.103230	0.000000	0.000000	0.041063	0.066667	0.062693
2022-05-31	ACB	0.056550	0.000000	0.091413	0.000000	0.066667	0.077649
2022-05-31	VPB	0.120598	0.000000	0.000000	0.000000	0.066667	0.055279
2022-05-31	HDB	0.025972	0.000000	0.000000	0.000000	0.066667	0.065763
2022-05-31	VIB	0.034251	0.284594	0.000000	0.000000	0.066667	0.061753
2022-05-31	TPB	0.029692	0.300000	0.126780	0.300000	0.066667	0.074773

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2022-05-31	STB	0.037477	0.000000	0.000000	0.000000	0.066667	0.056877
2022-05-31	SHB	0.025415	0.297768	0.142362	0.000000	0.066667	0.052679
2022-05-31	LPB	0.017901	0.000000	0.000000	0.000000	0.066667	0.066118
2022-05-31	BAB	0.010031	0.000000	0.300000	0.100314	0.066667	0.093008
2022-05-31	KLB	0.005406	0.117638	0.039445	0.000000	0.066667	0.057338
2022-06-30	VCB	0.286788	0.012871	0.300000	0.297747	0.066667	0.083319
2022-06-30	BID	0.139075	0.000000	0.000000	0.000000	0.066667	0.062499
2022-06-30	CTG	0.071132	0.000000	0.000000	0.000000	0.066667	0.061886
2022-06-30	MBB	0.059160	0.000000	0.000000	0.000000	0.066667	0.069885
2022-06-30	TCB	0.093965	0.000000	0.000000	0.000000	0.066667	0.063264
2022-06-30	ACB	0.058821	0.000000	0.187579	0.000000	0.066667	0.079346
2022-06-30	VPB	0.109821	0.039369	0.000000	0.000000	0.066667	0.056790
2022-06-30	HDB	0.028870	0.000000	0.000000	0.000000	0.066667	0.066442
2022-06-30	VIB	0.031072	0.300000	0.000000	0.000000	0.066667	0.062144
2022-06-30	TPB	0.030826	0.213955	0.047980	0.102253	0.066667	0.072131
2022-06-30	STB	0.032601	0.000000	0.000000	0.000000	0.066667	0.057024
2022-06-30	SHB	0.024883	0.235397	0.105450	0.000000	0.066667	0.050496
2022-06-30	LPB	0.017659	0.000000	0.000000	0.000000	0.066667	0.064888
2022-06-30	BAB	0.010489	0.000000	0.300000	0.300000	0.066667	0.092359
2022-06-30	KLB	0.004839	0.198409	0.058991	0.300000	0.066667	0.057527
2022-07-31	VCB	0.291893	0.014193	0.300000	0.300000	0.066667	0.083964
2022-07-31	BID	0.141710	0.000000	0.000000	0.300000	0.066667	0.063572
2022-07-31	CTG	0.071585	0.000000	0.000000	0.000000	0.066667	0.062688
2022-07-31	MBB	0.055346	0.000000	0.000000	0.172353	0.066667	0.070183
2022-07-31	TCB	0.096526	0.000000	0.000000	0.187960	0.066667	0.062174
2022-07-31	ACB	0.060027	0.181105	0.205502	0.000000	0.066667	0.080094

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2022-07-31	VPB	0.110311	0.000000	0.000000	0.000000	0.066667	0.055793
2022-07-31	HDB	0.028513	0.000000	0.000000	0.000000	0.066667	0.067213
2022-07-31	VIB	0.026748	0.300000	0.000000	0.039687	0.066667	0.060653
2022-07-31	TPB	0.027758	0.083043	0.039997	0.000000	0.066667	0.073175
2022-07-31	STB	0.033665	0.000000	0.000000	0.000000	0.066667	0.055198
2022-07-31	SHB	0.024347	0.275985	0.107880	0.000000	0.066667	0.050780
2022-07-31	LPB	0.016475	0.000000	0.000000	0.000000	0.066667	0.065587
2022-07-31	BAB	0.010376	0.000000	0.300000	0.000000	0.066667	0.092936
2022-07-31	KLB	0.004721	0.145674	0.046620	0.000000	0.066667	0.055989
2022-08-31	VCB	0.279114	0.000000	0.300000	0.300000	0.066667	0.084761
2022-08-31	BID	0.151057	0.000000	0.000000	0.300000	0.066667	0.064227
2022-08-31	CTG	0.071282	0.000000	0.000000	0.000000	0.066667	0.063374
2022-08-31	MBB	0.056202	0.000000	0.000000	0.191965	0.066667	0.069158
2022-08-31	TCB	0.098500	0.000000	0.000000	0.132088	0.066667	0.063707
2022-08-31	ACB	0.058911	0.256132	0.242274	0.000000	0.066667	0.080639
2022-08-31	VPB	0.104496	0.000000	0.000000	0.000000	0.066667	0.056192
2022-08-31	HDB	0.027471	0.000000	0.000000	0.000000	0.066667	0.067419
2022-08-31	VIB	0.031703	0.300000	0.000000	0.000000	0.066667	0.058583
2022-08-31	TPB	0.026055	0.000000	0.000000	0.075947	0.066667	0.071089
2022-08-31	STB	0.037035	0.000000	0.000000	0.000000	0.066667	0.055961
2022-08-31	SHB	0.025319	0.292083	0.104553	0.000000	0.066667	0.051104
2022-08-31	LPB	0.018389	0.000000	0.000000	0.000000	0.066667	0.064823
2022-08-31	BAB	0.010180	0.000000	0.300000	0.000000	0.066667	0.092779
2022-08-31	KLB	0.004285	0.151785	0.053173	0.000000	0.066667	0.056183
2022-09-30	VCB	0.291132	0.000000	0.300000	0.300000	0.062500	0.081789
2022-09-30	BID	0.150266	0.000000	0.000000	0.300000	0.062500	0.060475

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2022-09-30	CTG	0.068693	0.000000	0.000000	0.000000	0.062500	0.059699
2022-09-30	MBB	0.057938	0.000000	0.000000	0.247901	0.062500	0.065053
2022-09-30	TCB	0.093918	0.000000	0.000000	0.000000	0.062500	0.059790
2022-09-30	ACB	0.054757	0.251442	0.201363	0.152099	0.062500	0.075841
2022-09-30	VPB	0.107403	0.000000	0.000000	0.000000	0.062500	0.052874
2022-09-30	HDB	0.028174	0.000000	0.000000	0.000000	0.062500	0.063798
2022-09-30	VIB	0.027626	0.300000	0.000000	0.000000	0.062500	0.054117
2022-09-30	TPB	0.025435	0.000000	0.000000	0.000000	0.062500	0.066779
2022-09-30	STB	0.034491	0.000000	0.000000	0.000000	0.062500	0.052334
2022-09-30	SHB	0.025015	0.300000	0.087826	0.000000	0.062500	0.048127
2022-09-30	LPB	0.017718	0.000000	0.000000	0.000000	0.062500	0.060054
2022-09-30	BAB	0.009386	0.000000	0.300000	0.000000	0.062500	0.087195
2022-09-30	KLB	0.004279	0.148558	0.026055	0.000000	0.062500	0.052814
2022-09-30	VBB	0.003771	0.000000	0.084756	0.000000	0.062500	0.059261
2022-10-31	VCB	0.296212	0.000000	0.300000	0.300000	0.062500	0.079928
2022-10-31	BID	0.148705	0.000000	0.000000	0.228056	0.062500	0.060349
2022-10-31	CTG	0.065861	0.000000	0.000000	0.000000	0.062500	0.059513
2022-10-31	MBB	0.056920	0.000000	0.000000	0.000000	0.062500	0.064245
2022-10-31	TCB	0.091516	0.000000	0.000000	0.000000	0.062500	0.059558
2022-10-31	ACB	0.057558	0.300000	0.185046	0.300000	0.062500	0.075723
2022-10-31	VPB	0.106464	0.000000	0.000000	0.171944	0.062500	0.052566
2022-10-31	HDB	0.029476	0.000000	0.000000	0.000000	0.062500	0.063079
2022-10-31	VIB	0.028379	0.004408	0.000000	0.000000	0.062500	0.053652
2022-10-31	TPB	0.026509	0.000000	0.000000	0.000000	0.062500	0.066392
2022-10-31	STB	0.033448	0.000000	0.000000	0.000000	0.062500	0.052185
2022-10-31	SHB	0.023287	0.300000	0.084268	0.000000	0.062500	0.048082

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2022-10-31	LPB	0.016904	0.186458	0.000000	0.000000	0.062500	0.060279
2022-10-31	BAB	0.010298	0.000000	0.300000	0.000000	0.062500	0.086782
2022-10-31	KLB	0.004798	0.209134	0.007534	0.000000	0.062500	0.052504
2022-10-31	VBB	0.003665	0.000000	0.123152	0.000000	0.062500	0.065164
2022-11-30	VCB	0.314370	0.000000	0.300000	0.300000	0.062500	0.079523
2022-11-30	BID	0.160176	0.000000	0.000000	0.000000	0.062500	0.060227
2022-11-30	CTG	0.073815	0.000000	0.000000	0.000000	0.062500	0.058687
2022-11-30	MBB	0.053273	0.000000	0.000000	0.000000	0.062500	0.063591
2022-11-30	TCB	0.073188	0.000000	0.000000	0.000000	0.062500	0.059298
2022-11-30	ACB	0.061396	0.300000	0.201554	0.300000	0.062500	0.076004
2022-11-30	VPB	0.102197	0.000000	0.000000	0.300000	0.062500	0.052691
2022-11-30	HDB	0.026681	0.000000	0.000000	0.100000	0.062500	0.063769
2022-11-30	VIB	0.026740	0.091443	0.000000	0.000000	0.062500	0.054023
2022-11-30	TPB	0.023209	0.000000	0.000000	0.000000	0.062500	0.067017
2022-11-30	STB	0.027716	0.000000	0.000000	0.000000	0.062500	0.051952
2022-11-30	SHB	0.022651	0.286587	0.060825	0.000000	0.062500	0.047367
2022-11-30	LPB	0.016262	0.021969	0.000000	0.000000	0.062500	0.059033
2022-11-30	BAB	0.009837	0.000000	0.300000	0.000000	0.062500	0.088404
2022-11-30	KLB	0.004615	0.300000	0.033258	0.000000	0.062500	0.053734
2022-11-30	VBB	0.003873	0.000000	0.104362	0.000000	0.062500	0.064680
2022-12-31	VCB	0.319787	0.000000	0.300000	0.300000	0.062500	0.081312
2022-12-31	BID	0.175901	0.000000	0.000000	0.068573	0.062500	0.061219
2022-12-31	CTG	0.075501	0.000000	0.000000	0.031427	0.062500	0.059551
2022-12-31	MBB	0.048360	0.000000	0.000000	0.000000	0.062500	0.063449
2022-12-31	TCB	0.071129	0.000000	0.000000	0.000000	0.062500	0.056292
2022-12-31	ACB	0.055946	0.300000	0.203962	0.300000	0.062500	0.077232

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2022-12-31	VPB	0.096948	0.000000	0.000000	0.300000	0.062500	0.053100
2022-12-31	HDB	0.023363	0.000000	0.000000	0.000000	0.062500	0.063110
2022-12-31	VIB	0.025487	0.100000	0.000000	0.000000	0.062500	0.054134
2022-12-31	TPB	0.022758	0.000000	0.000000	0.000000	0.062500	0.065088
2022-12-31	STB	0.031687	0.000000	0.000000	0.000000	0.062500	0.050503
2022-12-31	SHB	0.021219	0.300000	0.055329	0.000000	0.062500	0.047798
2022-12-31	LPB	0.016820	0.000000	0.000000	0.000000	0.062500	0.059243
2022-12-31	BAB	0.008824	0.000000	0.300000	0.000000	0.062500	0.088009
2022-12-31	KLB	0.002975	0.300000	0.023589	0.000000	0.062500	0.054068
2022-12-31	VBB	0.003294	0.000000	0.117120	0.000000	0.062500	0.065891
2023-01-31	VCB	0.317999	0.000000	0.300000	0.300000	0.062500	0.080921
2023-01-31	BID	0.166343	0.000000	0.000000	0.300000	0.062500	0.060148
2023-01-31	CTG	0.076014	0.000000	0.000000	0.300000	0.062500	0.059978
2023-01-31	MBB	0.047811	0.000000	0.000000	0.000000	0.062500	0.064027
2023-01-31	TCB	0.071492	0.000000	0.000000	0.000000	0.062500	0.056665
2023-01-31	ACB	0.055805	0.300000	0.207299	0.100000	0.062500	0.078125
2023-01-31	VPB	0.104050	0.000000	0.000000	0.000000	0.062500	0.053774
2023-01-31	HDB	0.024116	0.000000	0.000000	0.000000	0.062500	0.064209
2023-01-31	VIB	0.024097	0.202438	0.000000	0.000000	0.062500	0.054722
2023-01-31	TPB	0.021929	0.000000	0.000000	0.000000	0.062500	0.065514
2023-01-31	STB	0.035890	0.000000	0.000000	0.000000	0.062500	0.049855
2023-01-31	SHB	0.021019	0.300000	0.059810	0.000000	0.062500	0.048573
2023-01-31	LPB	0.019386	0.157028	0.000000	0.000000	0.062500	0.059282
2023-01-31	BAB	0.008560	0.000000	0.300000	0.000000	0.062500	0.088540
2023-01-31	KLB	0.002337	0.040534	0.022268	0.000000	0.062500	0.049622
2023-01-31	VBB	0.003153	0.000000	0.110623	0.000000	0.062500	0.066046

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2023-02-28	VCB	0.318169	0.000000	0.300000	0.173711	0.062500	0.081291
2023-02-28	BID	0.169254	0.000000	0.000000	0.300000	0.062500	0.060719
2023-02-28	CTG	0.074083	0.000000	0.000000	0.300000	0.062500	0.059925
2023-02-28	MBB	0.047624	0.000000	0.000000	0.000000	0.062500	0.064278
2023-02-28	TCB	0.070808	0.000000	0.000000	0.000000	0.062500	0.056655
2023-02-28	ACB	0.057804	0.300000	0.198868	0.000000	0.062500	0.078101
2023-02-28	VPB	0.098462	0.072847	0.000000	0.000000	0.062500	0.053712
2023-02-28	HDB	0.024575	0.000000	0.000000	0.000000	0.062500	0.064204
2023-02-28	VIB	0.025994	0.027153	0.000000	0.000000	0.062500	0.054375
2023-02-28	TPB	0.022682	0.000000	0.000000	0.000000	0.062500	0.065435
2023-02-28	STB	0.037650	0.000000	0.000000	0.000000	0.062500	0.049557
2023-02-28	SHB	0.020255	0.300000	0.059475	0.107355	0.062500	0.048542
2023-02-28	LPB	0.019065	0.300000	0.000000	0.118934	0.062500	0.058702
2023-02-28	BAB	0.007794	0.000000	0.300000	0.000000	0.062500	0.088277
2023-02-28	KLB	0.002051	0.000000	0.021094	0.000000	0.062500	0.049526
2023-02-28	VBB	0.003730	0.000000	0.120563	0.000000	0.062500	0.066700
2023-03-31	VCB	0.339910	0.000000	0.300000	0.300000	0.062500	0.080403
2023-03-31	BID	0.173758	0.000000	0.000000	0.000000	0.062500	0.061039
2023-03-31	CTG	0.069895	0.000000	0.000000	0.190223	0.062500	0.061447
2023-03-31	MBB	0.044104	0.000000	0.000000	0.000000	0.062500	0.064379
2023-03-31	TCB	0.067025	0.000000	0.000000	0.000000	0.062500	0.057270
2023-03-31	ACB	0.056841	0.300000	0.216046	0.098525	0.062500	0.076742
2023-03-31	VPB	0.090619	0.000000	0.000000	0.000000	0.062500	0.054768
2023-03-31	HDB	0.023577	0.000000	0.000000	0.000000	0.062500	0.063989
2023-03-31	VIB	0.024898	0.138354	0.000000	0.000000	0.062500	0.053982
2023-03-31	TPB	0.022202	0.000000	0.000000	0.000000	0.062500	0.064890

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2023-03-31	STB	0.034647	0.000000	0.000000	0.176741	0.062500	0.049583
2023-03-31	SHB	0.018940	0.261646	0.072202	0.000000	0.062500	0.049560
2023-03-31	LPB	0.018886	0.300000	0.000000	0.234512	0.062500	0.059525
2023-03-31	BAB	0.008244	0.000000	0.300000	0.000000	0.062500	0.089336
2023-03-31	KLB	0.001908	0.000000	0.058678	0.000000	0.062500	0.051691
2023-03-31	VBB	0.004547	0.000000	0.053074	0.000000	0.062500	0.061395
2023-04-30	VCB	0.317308	0.006709	0.300000	0.300000	0.062500	0.081219
2023-04-30	BID	0.173867	0.000000	0.000000	0.100000	0.062500	0.061877
2023-04-30	CTG	0.071110	0.000000	0.000000	0.300000	0.062500	0.060733
2023-04-30	MBB	0.044597	0.000000	0.000000	0.000000	0.062500	0.063209
2023-04-30	TCB	0.068495	0.000000	0.000000	0.000000	0.062500	0.056723
2023-04-30	ACB	0.055646	0.300000	0.208118	0.300000	0.062500	0.076416
2023-04-30	VPB	0.106821	0.000000	0.000000	0.000000	0.062500	0.055010
2023-04-30	HDB	0.025420	0.000000	0.000000	0.000000	0.062500	0.063295
2023-04-30	VIB	0.024766	0.300000	0.000000	0.000000	0.062500	0.053433
2023-04-30	TPB	0.022412	0.047374	0.000000	0.000000	0.062500	0.064361
2023-04-30	STB	0.036499	0.000000	0.000000	0.000000	0.062500	0.049179
2023-04-30	SHB	0.019859	0.045917	0.080770	0.000000	0.062500	0.054659
2023-04-30	LPB	0.019911	0.300000	0.000000	0.000000	0.062500	0.059102
2023-04-30	BAB	0.007759	0.000000	0.300000	0.000000	0.062500	0.089289
2023-04-30	KLB	0.001787	0.000000	0.052841	0.000000	0.062500	0.051399
2023-04-30	VBB	0.003741	0.000000	0.058272	0.000000	0.062500	0.060097
2023-05-31	VCB	0.320670	0.149606	0.300000	0.300000	0.062500	0.084141
2023-05-31	BID	0.167057	0.000000	0.000000	0.156299	0.062500	0.064744
2023-05-31	CTG	0.071719	0.000000	0.000000	0.254238	0.062500	0.064222
2023-05-31	MBB	0.045973	0.000000	0.000000	0.000000	0.062500	0.068858

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2023-05-31	TCB	0.072724	0.000000	0.000000	0.000000	0.062500	0.057588
2023-05-31	ACB	0.054968	0.300000	0.300000	0.241506	0.062500	0.085659
2023-05-31	VPB	0.103090	0.130830	0.000000	0.000000	0.062500	0.058118
2023-05-31	HDB	0.025333	0.068560	0.030592	0.000000	0.062500	0.071602
2023-05-31	VIB	0.024780	0.051004	0.000000	0.000000	0.062500	0.052449
2023-05-31	TPB	0.024526	0.000000	0.000000	0.000000	0.062500	0.060195
2023-05-31	STB	0.035969	0.000000	0.000000	0.000576	0.062500	0.050783
2023-05-31	SHB	0.021095	0.000000	0.000000	0.000000	0.062500	0.049802
2023-05-31	LPB	0.018473	0.300000	0.000000	0.047380	0.062500	0.055446
2023-05-31	BAB	0.007803	0.000000	0.300000	0.000000	0.062500	0.078827
2023-05-31	KLB	0.001928	0.000000	0.064249	0.000000	0.062500	0.045498
2023-05-31	VBB	0.003893	0.000000	0.005159	0.000000	0.062500	0.052067
2023-06-30	VCB	0.325965	0.139853	0.300000	0.300000	0.062500	0.083745
2023-06-30	BID	0.165383	0.000000	0.000000	0.106952	0.062500	0.064621
2023-06-30	CTG	0.068139	0.000000	0.000000	0.199169	0.062500	0.064528
2023-06-30	MBB	0.045404	0.000000	0.000000	0.000000	0.062500	0.069785
2023-06-30	TCB	0.072279	0.000000	0.000000	0.000000	0.062500	0.057516
2023-06-30	ACB	0.056457	0.300000	0.300000	0.120888	0.062500	0.086207
2023-06-30	VPB	0.097861	0.071718	0.000000	0.000000	0.062500	0.058291
2023-06-30	HDB	0.025599	0.011254	0.039531	0.000000	0.062500	0.071888
2023-06-30	VIB	0.025311	0.156098	0.000000	0.000000	0.062500	0.051843
2023-06-30	TPB	0.025216	0.104135	0.000000	0.000000	0.062500	0.059129
2023-06-30	STB	0.038546	0.000000	0.000000	0.000000	0.062500	0.051263
2023-06-30	SHB	0.021668	0.000000	0.000000	0.000000	0.062500	0.051712
2023-06-30	LPB	0.018721	0.216942	0.000000	0.272992	0.062500	0.055572
2023-06-30	BAB	0.007864	0.000000	0.300000	0.000000	0.062500	0.077756

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2023-06-30	KLB	0.001920	0.000000	0.060469	0.000000	0.062500	0.044745
2023-06-30	VBB	0.003668	0.000000	0.000000	0.000000	0.062500	0.051401
2023-07-31	VCB	0.331378	0.011376	0.300000	0.256949	0.062500	0.091596
2023-07-31	BID	0.155696	0.000000	0.000000	0.000000	0.062500	0.064280
2023-07-31	CTG	0.068591	0.000000	0.000000	0.048401	0.062500	0.063893
2023-07-31	MBB	0.048290	0.000000	0.000000	0.000000	0.062500	0.068637
2023-07-31	TCB	0.074593	0.000000	0.000000	0.000000	0.062500	0.058167
2023-07-31	ACB	0.056082	0.300000	0.300000	0.146211	0.062500	0.085829
2023-07-31	VPB	0.096136	0.018919	0.000000	0.000000	0.062500	0.057599
2023-07-31	HDB	0.024708	0.000000	0.071517	0.000000	0.062500	0.072630
2023-07-31	VIB	0.026643	0.243322	0.000000	0.000000	0.062500	0.051051
2023-07-31	TPB	0.024149	0.139942	0.000000	0.000000	0.062500	0.059273
2023-07-31	STB	0.039623	0.000000	0.000000	0.248438	0.062500	0.050422
2023-07-31	SHB	0.022023	0.000000	0.000000	0.000000	0.062500	0.051331
2023-07-31	LPB	0.018688	0.286441	0.000000	0.300000	0.062500	0.054836
2023-07-31	BAB	0.007568	0.000000	0.300000	0.000000	0.062500	0.076053
2023-07-31	KLB	0.002190	0.000000	0.028483	0.000000	0.062500	0.044116
2023-07-31	VBB	0.003641	0.000000	0.000000	0.000000	0.062500	0.050287
2023-08-31	VCB	0.334119	0.179641	0.300000	0.300000	0.062500	0.092067
2023-08-31	BID	0.157990	0.000000	0.000000	0.126176	0.062500	0.064491
2023-08-31	CTG	0.064987	0.000000	0.000000	0.133906	0.062500	0.063957
2023-08-31	MBB	0.048297	0.000000	0.000000	0.000000	0.062500	0.068062
2023-08-31	TCB	0.073702	0.000000	0.000000	0.000000	0.062500	0.058101
2023-08-31	ACB	0.054397	0.300000	0.300000	0.300000	0.062500	0.085773
2023-08-31	VPB	0.099980	0.184233	0.000000	0.000000	0.062500	0.059106
2023-08-31	HDB	0.024712	0.000000	0.079869	0.015573	0.062500	0.072481

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2023-08-31	VIB	0.026301	0.180189	0.000000	0.000000	0.062500	0.050824
2023-08-31	TPB	0.023446	0.027478	0.000000	0.000000	0.062500	0.059171
2023-08-31	STB	0.035877	0.000000	0.000000	0.000000	0.062500	0.050248
2023-08-31	SHB	0.024428	0.000000	0.000000	0.000000	0.062500	0.051418
2023-08-31	LPB	0.018793	0.128459	0.000000	0.124346	0.062500	0.054722
2023-08-31	BAB	0.007053	0.000000	0.300000	0.000000	0.062500	0.075896
2023-08-31	KLB	0.002055	0.000000	0.020131	0.000000	0.062500	0.043349
2023-08-31	VBB	0.003862	0.000000	0.000000	0.000000	0.062500	0.050334
2023-09-30	VCB	0.324600	0.300000	0.300000	0.246128	0.058824	0.088216
2023-09-30	BID	0.156965	0.000000	0.000000	0.180050	0.058824	0.061490
2023-09-30	CTG	0.070231	0.000000	0.000000	0.183906	0.058824	0.061343
2023-09-30	MBB	0.047316	0.000000	0.000000	0.146603	0.058824	0.065590
2023-09-30	TCB	0.074029	0.000000	0.000000	0.050742	0.058824	0.056028
2023-09-30	ACB	0.053487	0.300000	0.300000	0.000000	0.058824	0.082388
2023-09-30	VPB	0.094491	0.197311	0.000000	0.063640	0.058824	0.056284
2023-09-30	HDB	0.023968	0.000000	0.079231	0.000000	0.058824	0.069504
2023-09-30	VIB	0.025701	0.060356	0.000000	0.000000	0.058824	0.048648
2023-09-30	TPB	0.024475	0.000000	0.000000	0.000000	0.058824	0.056678
2023-09-30	STB	0.040412	0.000000	0.000000	0.074431	0.058824	0.048323
2023-09-30	SHB	0.024104	0.000000	0.000000	0.000000	0.058824	0.048697
2023-09-30	LPB	0.023732	0.142333	0.000000	0.054499	0.058824	0.052347
2023-09-30	BAB	0.006994	0.000000	0.300000	0.000000	0.058824	0.072697
2023-09-30	KLB	0.002039	0.000000	0.020769	0.000000	0.058824	0.041538
2023-09-30	VBB	0.003609	0.000000	0.000000	0.000000	0.058824	0.047591
2023-09-30	BVB	0.003848	0.000000	0.000000	0.000000	0.058824	0.042637
2023-10-31	VCB	0.329213	0.265677	0.300000	0.176175	0.058824	0.087415

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2023-10-31	BID	0.153150	0.000000	0.000000	0.000000	0.058824	0.061257
2023-10-31	CTG	0.067140	0.000000	0.000000	0.000000	0.058824	0.062686
2023-10-31	MBB	0.048864	0.000000	0.000000	0.000000	0.058824	0.065456
2023-10-31	TCB	0.074582	0.000000	0.000000	0.000000	0.058824	0.056430
2023-10-31	ACB	0.053391	0.300000	0.300000	0.000000	0.058824	0.088038
2023-10-31	VPB	0.101025	0.078970	0.000000	0.189753	0.058824	0.055798
2023-10-31	HDB	0.025489	0.000000	0.095045	0.057828	0.058824	0.070834
2023-10-31	VIB	0.025229	0.000000	0.000000	0.000000	0.058824	0.048959
2023-10-31	TPB	0.023018	0.055353	0.000000	0.000000	0.058824	0.055992
2023-10-31	STB	0.039178	0.000000	0.000000	0.036175	0.058824	0.047410
2023-10-31	SHB	0.022191	0.000000	0.000000	0.300000	0.058824	0.048758
2023-10-31	LPB	0.021146	0.300000	0.000000	0.098145	0.058824	0.049854
2023-10-31	BAB	0.007119	0.000000	0.300000	0.133611	0.058824	0.071600
2023-10-31	KLB	0.001934	0.000000	0.004955	0.008312	0.058824	0.040849
2023-10-31	VBB	0.003432	0.000000	0.000000	0.000000	0.058824	0.046709
2023-10-31	BVB	0.003898	0.000000	0.000000	0.000000	0.058824	0.041953
2023-11-30	VCB	0.347333	0.235729	0.300000	0.300000	0.058824	0.087045
2023-11-30	BID	0.147656	0.000000	0.000000	0.000000	0.058824	0.060807
2023-11-30	CTG	0.065836	0.000000	0.000000	0.000000	0.058824	0.061896
2023-11-30	MBB	0.048057	0.000000	0.000000	0.000000	0.058824	0.065889
2023-11-30	TCB	0.065318	0.000000	0.000000	0.000000	0.058824	0.056412
2023-11-30	ACB	0.055641	0.300000	0.300000	0.000000	0.058824	0.087678
2023-11-30	VPB	0.099044	0.300000	0.000000	0.000000	0.058824	0.055717
2023-11-30	HDB	0.026884	0.000000	0.099727	0.000000	0.058824	0.071626
2023-11-30	VIB	0.024815	0.000000	0.000000	0.000000	0.058824	0.050591
2023-11-30	TPB	0.021663	0.000000	0.000000	0.000000	0.058824	0.055113

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2023-11-30	STB	0.036708	0.000000	0.000000	0.294513	0.058824	0.048389
2023-11-30	SHB	0.021463	0.000000	0.000000	0.105487	0.058824	0.048371
2023-11-30	LPB	0.023784	0.164271	0.000000	0.300000	0.058824	0.049891
2023-11-30	BAB	0.006964	0.000000	0.300000	0.000000	0.058824	0.071547
2023-11-30	KLB	0.001812	0.000000	0.000273	0.000000	0.058824	0.040678
2023-11-30	VBB	0.002997	0.000000	0.000000	0.000000	0.058824	0.046471
2023-11-30	BVB	0.004025	0.000000	0.000000	0.000000	0.058824	0.041879
2023-12-31	VCB	0.327663	0.300000	0.300000	0.227479	0.052632	0.079399
2023-12-31	BID	0.155413	0.000000	0.000000	0.000000	0.052632	0.054902
2023-12-31	CTG	0.067536	0.000000	0.000000	0.000000	0.052632	0.056263
2023-12-31	MBB	0.048370	0.000000	0.000000	0.000000	0.052632	0.059196
2023-12-31	TCB	0.067229	0.000000	0.000000	0.000000	0.052632	0.049295
2023-12-31	ACB	0.055147	0.170580	0.300000	0.000000	0.052632	0.080287
2023-12-31	VPB	0.095964	0.288639	0.000000	0.000000	0.052632	0.050198
2023-12-31	HDB	0.027187	0.000000	0.053398	0.000000	0.052632	0.065236
2023-12-31	VIB	0.025068	0.000000	0.000000	0.000000	0.052632	0.046765
2023-12-31	TPB	0.022544	0.000000	0.000000	0.000000	0.052632	0.049278
2023-12-31	STB	0.036142	0.000000	0.000000	0.208437	0.052632	0.043413
2023-12-31	SHB	0.022193	0.000000	0.000000	0.000000	0.052632	0.043701
2023-12-31	LPB	0.024012	0.240781	0.000000	0.264084	0.052632	0.045630
2023-12-31	NAB	0.007090	0.000000	0.000000	0.300000	0.052632	0.046208
2023-12-31	BAB	0.006786	0.000000	0.300000	0.000000	0.052632	0.064472
2023-12-31	KLB	0.001831	0.000000	0.000000	0.000000	0.052632	0.036790
2023-12-31	VBB	0.003195	0.000000	0.000000	0.000000	0.052632	0.041427
2023-12-31	SGB	0.002582	0.000000	0.046602	0.000000	0.052632	0.048951
2023-12-31	BVB	0.004047	0.000000	0.000000	0.000000	0.052632	0.038588

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2024-01-31	VCB	0.303513	0.226881	0.300000	0.300000	0.052632	0.080546
2024-01-31	BID	0.169631	0.000000	0.000000	0.000000	0.052632	0.054158
2024-01-31	CTG	0.067995	0.000000	0.000000	0.037261	0.052632	0.056899
2024-01-31	MBB	0.049503	0.000000	0.000000	0.000000	0.052632	0.059721
2024-01-31	TCB	0.070785	0.000000	0.000000	0.000000	0.052632	0.049215
2024-01-31	ACB	0.058673	0.173119	0.300000	0.212688	0.052632	0.082126
2024-01-31	VPB	0.094245	0.300000	0.000000	0.000000	0.052632	0.050182
2024-01-31	HDB	0.029957	0.000000	0.087687	0.000000	0.052632	0.066485
2024-01-31	VIB	0.025669	0.000000	0.000000	0.000000	0.052632	0.046137
2024-01-31	TPB	0.022548	0.000000	0.000000	0.000000	0.052632	0.049961
2024-01-31	STB	0.035889	0.000000	0.000000	0.150052	0.052632	0.043092
2024-01-31	SHB	0.021683	0.000000	0.000000	0.000000	0.052632	0.043339
2024-01-31	LPB	0.024152	0.300000	0.000000	0.300000	0.052632	0.045079
2024-01-31	NAB	0.007629	0.000000	0.000000	0.000000	0.052632	0.045871
2024-01-31	BAB	0.006630	0.000000	0.300000	0.000000	0.052632	0.063698
2024-01-31	KLB	0.001771	0.000000	0.000000	0.000000	0.052632	0.036455
2024-01-31	VBB	0.003253	0.000000	0.000000	0.000000	0.052632	0.040634
2024-01-31	SGB	0.002483	0.000000	0.012313	0.000000	0.052632	0.048329
2024-01-31	BVB	0.003992	0.000000	0.000000	0.000000	0.052632	0.038071
2024-02-29	VCB	0.299357	0.049873	0.300000	0.300000	0.045455	0.069171
2024-02-29	BID	0.166889	0.000000	0.000000	0.000000	0.045455	0.047376
2024-02-29	CTG	0.070939	0.000000	0.000000	0.000000	0.045455	0.049094
2024-02-29	MBB	0.051671	0.000000	0.000000	0.000000	0.045455	0.052405
2024-02-29	TCB	0.068823	0.000000	0.000000	0.000000	0.045455	0.045858
2024-02-29	ACB	0.056584	0.300000	0.300000	0.273400	0.045455	0.069684
2024-02-29	VPB	0.085221	0.058465	0.000000	0.000000	0.045455	0.044877

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2024-02-29	HDB	0.028781	0.259594	0.065650	0.000000	0.045455	0.056419
2024-02-29	VIB	0.025026	0.000000	0.000000	0.000000	0.045455	0.040588
2024-02-29	TPB	0.020696	0.000000	0.000000	0.000000	0.045455	0.044596
2024-02-29	STB	0.034362	0.000000	0.000000	0.126600	0.045455	0.037650
2024-02-29	SHB	0.020844	0.000000	0.000000	0.000000	0.045455	0.037337
2024-02-29	LPB	0.024220	0.300000	0.000000	0.300000	0.045455	0.039004
2024-02-29	MSB	0.013821	0.000000	0.000000	0.000000	0.045455	0.052534
2024-02-29	NAB	0.007003	0.032068	0.000000	0.000000	0.045455	0.039381
2024-02-29	BAB	0.006278	0.000000	0.300000	0.000000	0.045455	0.059088
2024-02-29	KLB	0.001657	0.000000	0.034350	0.000000	0.045455	0.034753
2024-02-29	PGB	0.004209	0.000000	0.000000	0.000000	0.045455	0.026372
2024-02-29	VBB	0.003016	0.000000	0.000000	0.000000	0.045455	0.035009
2024-02-29	SGB	0.002328	0.000000	0.000000	0.000000	0.045455	0.041795
2024-02-29	ABB	0.004669	0.000000	0.000000	0.000000	0.045455	0.043322
2024-02-29	BVB	0.003607	0.000000	0.000000	0.000000	0.045455	0.033689
2024-03-31	VCB	0.295866	0.229590	0.300000	0.300000	0.043478	0.064554
2024-03-31	BID	0.166670	0.000000	0.000000	0.045852	0.043478	0.046818
2024-03-31	CTG	0.071739	0.000000	0.000000	0.000000	0.043478	0.046158
2024-03-31	MBB	0.051561	0.000000	0.000000	0.000000	0.043478	0.047958
2024-03-31	TCB	0.075567	0.000000	0.000000	0.000000	0.043478	0.042984
2024-03-31	ACB	0.055308	0.300000	0.300000	0.300000	0.043478	0.065892
2024-03-31	VPB	0.078559	0.000000	0.000000	0.000000	0.043478	0.042909
2024-03-31	HDB	0.027659	0.081629	0.039021	0.054148	0.043478	0.053644
2024-03-31	VIB	0.023736	0.000000	0.000000	0.000000	0.043478	0.038206
2024-03-31	TPB	0.020687	0.000000	0.000000	0.000000	0.043478	0.042489
2024-03-31	STB	0.032592	0.000000	0.000000	0.000000	0.043478	0.035507

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2024-03-31	SHB	0.019070	0.000000	0.000000	0.000000	0.043478	0.035960
2024-03-31	LPB	0.021100	0.255029	0.000000	0.300000	0.043478	0.036594
2024-03-31	MSB	0.014518	0.000000	0.000000	0.000000	0.043478	0.050638
2024-03-31	OCB	0.014984	0.000000	0.022387	0.000000	0.043478	0.047706
2024-03-31	NAB	0.006618	0.042630	0.000000	0.000000	0.043478	0.038319
2024-03-31	BAB	0.005691	0.000000	0.300000	0.000000	0.043478	0.055333
2024-03-31	KLB	0.001501	0.000000	0.033466	0.000000	0.043478	0.033064
2024-03-31	PGB	0.004091	0.091121	0.000000	0.000000	0.043478	0.026980
2024-03-31	VBB	0.002738	0.000000	0.000000	0.000000	0.043478	0.033102
2024-03-31	SGB	0.002107	0.000000	0.005125	0.000000	0.043478	0.040660
2024-03-31	ABB	0.004305	0.000000	0.000000	0.000000	0.043478	0.041621
2024-03-31	BVB	0.003334	0.000000	0.000000	0.000000	0.043478	0.032905
2024-04-30	VCB	0.287435	0.300000	0.300000	0.300000	0.043478	0.062450
2024-04-30	BID	0.163195	0.000000	0.000000	0.000000	0.043478	0.045776
2024-04-30	CTG	0.071464	0.000000	0.000000	0.000000	0.043478	0.047042
2024-04-30	MBB	0.054028	0.000000	0.000000	0.000000	0.043478	0.048199
2024-04-30	TCB	0.084841	0.000000	0.000000	0.000000	0.043478	0.042283
2024-04-30	ACB	0.056078	0.300000	0.300000	0.300000	0.043478	0.072331
2024-04-30	VPB	0.077694	0.000000	0.089093	0.000000	0.043478	0.046905
2024-04-30	HDB	0.028491	0.083667	0.006275	0.000000	0.043478	0.053561
2024-04-30	VIB	0.026572	0.000000	0.000000	0.000000	0.043478	0.039355
2024-04-30	TPB	0.019829	0.000000	0.000000	0.000000	0.043478	0.041649
2024-04-30	STB	0.032518	0.000000	0.000000	0.100000	0.043478	0.034951
2024-04-30	SHB	0.018354	0.000000	0.000000	0.000000	0.043478	0.035484
2024-04-30	LPB	0.021573	0.120295	0.000000	0.300000	0.043478	0.036367
2024-04-30	MSB	0.013241	0.000000	0.000000	0.000000	0.043478	0.048634

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2024-04-30	OCB	0.014725	0.000000	0.000000	0.000000	0.043478	0.047056
2024-04-30	NAB	0.006653	0.098804	0.000000	0.000000	0.043478	0.037807
2024-04-30	BAB	0.005669	0.000000	0.300000	0.000000	0.043478	0.054532
2024-04-30	KLB	0.001483	0.000000	0.004632	0.000000	0.043478	0.032455
2024-04-30	PGB	0.003821	0.097234	0.000000	0.000000	0.043478	0.026613
2024-04-30	VBB	0.002623	0.000000	0.000000	0.000000	0.043478	0.033159
2024-04-30	SGB	0.002241	0.000000	0.000000	0.000000	0.043478	0.040062
2024-04-30	ABB	0.004181	0.000000	0.000000	0.000000	0.043478	0.040981
2024-04-30	BVB	0.003291	0.000000	0.000000	0.000000	0.043478	0.032345
2024-05-31	VCB	0.281444	0.300000	0.300000	0.300000	0.041667	0.057934
2024-05-31	BID	0.157029	0.015931	0.000000	0.300000	0.041667	0.042356
2024-05-31	CTG	0.067202	0.000000	0.000000	0.000000	0.041667	0.043843
2024-05-31	MBB	0.048229	0.000000	0.000000	0.000000	0.041667	0.044511
2024-05-31	TCB	0.084718	0.000000	0.000000	0.000000	0.041667	0.038495
2024-05-31	ACB	0.053825	0.300000	0.300000	0.300000	0.041667	0.066942
2024-05-31	VPB	0.074355	0.000000	0.016635	0.000000	0.041667	0.044129
2024-05-31	HDB	0.028445	0.230827	0.000000	0.000000	0.041667	0.049642
2024-05-31	VIB	0.023910	0.000000	0.000000	0.000000	0.041667	0.037466
2024-05-31	TPB	0.018941	0.000000	0.000000	0.000000	0.041667	0.038428
2024-05-31	STB	0.029569	0.000000	0.000000	0.000000	0.041667	0.032945
2024-05-31	SHB	0.018769	0.000000	0.000000	0.000000	0.041667	0.042911
2024-05-31	LPB	0.026169	0.000000	0.000000	0.000000	0.041667	0.034371
2024-05-31	SSB	0.030650	0.002775	0.083365	0.000000	0.041667	0.043047
2024-05-31	MSB	0.012610	0.000000	0.000000	0.000000	0.041667	0.046011
2024-05-31	OCB	0.013701	0.000000	0.000000	0.100000	0.041667	0.044917
2024-05-31	NAB	0.006840	0.149295	0.000000	0.000000	0.041667	0.035045

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2024-05-31	BAB	0.005637	0.000000	0.300000	0.000000	0.041667	0.064605
2024-05-31	KLB	0.001422	0.000000	0.000000	0.000000	0.041667	0.030762
2024-05-31	PGB	0.003651	0.001171	0.000000	0.000000	0.041667	0.024782
2024-05-31	VBB	0.002966	0.000000	0.000000	0.000000	0.041667	0.030657
2024-05-31	SGB	0.002478	0.000000	0.000000	0.000000	0.041667	0.037204
2024-05-31	ABB	0.004053	0.000000	0.000000	0.000000	0.041667	0.038828
2024-05-31	BVB	0.003384	0.000000	0.000000	0.000000	0.041667	0.030169
2024-06-30	VCB	0.270118	0.198848	0.300000	0.300000	0.041667	0.057436
2024-06-30	BID	0.150904	0.163613	0.000000	0.300000	0.041667	0.041983
2024-06-30	CTG	0.065590	0.000000	0.000000	0.039031	0.041667	0.043041
2024-06-30	MBB	0.048372	0.000000	0.000000	0.022493	0.041667	0.043019
2024-06-30	TCB	0.088421	0.000000	0.000000	0.000000	0.041667	0.038217
2024-06-30	ACB	0.059070	0.106564	0.300000	0.300000	0.041667	0.065441
2024-06-30	VPB	0.076268	0.000000	0.005910	0.000000	0.041667	0.049741
2024-06-30	HDB	0.028119	0.294089	0.000000	0.000000	0.041667	0.049219
2024-06-30	VIB	0.024730	0.000000	0.000000	0.000000	0.041667	0.037319
2024-06-30	TPB	0.018745	0.000000	0.000000	0.000000	0.041667	0.037945
2024-06-30	STB	0.029309	0.000000	0.000000	0.000000	0.041667	0.032587
2024-06-30	SHB	0.018770	0.000000	0.000000	0.000000	0.041667	0.042953
2024-06-30	LPB	0.032435	0.064773	0.000000	0.038476	0.041667	0.034646
2024-06-30	SSB	0.030064	0.114681	0.094090	0.000000	0.041667	0.042968
2024-06-30	MSB	0.013309	0.000000	0.000000	0.000000	0.041667	0.045317
2024-06-30	OCB	0.014852	0.000000	0.000000	0.000000	0.041667	0.044031
2024-06-30	NAB	0.006497	0.041063	0.000000	0.000000	0.041667	0.035241
2024-06-30	BAB	0.005610	0.000000	0.300000	0.000000	0.041667	0.065994
2024-06-30	KLB	0.001517	0.000000	0.000000	0.000000	0.041667	0.031523

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2024-06-30	PGB	0.003564	0.016369	0.000000	0.000000	0.041667	0.024645
2024-06-30	VBB	0.002977	0.000000	0.000000	0.000000	0.041667	0.030228
2024-06-30	SGB	0.002326	0.000000	0.000000	0.000000	0.041667	0.036692
2024-06-30	ABB	0.004541	0.000000	0.000000	0.000000	0.041667	0.039839
2024-06-30	BVB	0.003888	0.000000	0.000000	0.000000	0.041667	0.029971
2024-07-31	VCB	0.268964	0.300000	0.096232	0.300000	0.041667	0.049314
2024-07-31	BID	0.141208	0.000000	0.000000	0.158870	0.041667	0.037588
2024-07-31	CTG	0.064951	0.000000	0.000000	0.000000	0.041667	0.042018
2024-07-31	MBB	0.050305	0.000000	0.000000	0.000000	0.041667	0.040331
2024-07-31	TCB	0.089551	0.000000	0.000000	0.000000	0.041667	0.036417
2024-07-31	ACB	0.058096	0.300000	0.241881	0.300000	0.041667	0.061764
2024-07-31	VPB	0.080525	0.000000	0.000000	0.000000	0.041667	0.045919
2024-07-31	HDB	0.028357	0.100000	0.000000	0.000000	0.041667	0.046705
2024-07-31	VIB	0.024338	0.000000	0.000000	0.000000	0.041667	0.034075
2024-07-31	TPB	0.019134	0.000000	0.000000	0.000000	0.041667	0.037459
2024-07-31	STB	0.030889	0.000000	0.000000	0.000000	0.041667	0.031857
2024-07-31	SHB	0.019129	0.000000	0.000000	0.000000	0.041667	0.039976
2024-07-31	LPB	0.036115	0.300000	0.000000	0.241130	0.041667	0.032040
2024-07-31	SSB	0.029636	0.000000	0.264903	0.000000	0.041667	0.054747
2024-07-31	MSB	0.013564	0.000000	0.000000	0.000000	0.041667	0.043420
2024-07-31	OCB	0.014477	0.000000	0.000000	0.000000	0.041667	0.040557
2024-07-31	NAB	0.006645	0.000000	0.000000	0.000000	0.041667	0.037511
2024-07-31	BAB	0.005717	0.000000	0.300000	0.000000	0.041667	0.068722
2024-07-31	KLB	0.001559	0.000000	0.000000	0.000000	0.041667	0.028357
2024-07-31	PGB	0.003590	0.000000	0.000000	0.000000	0.041667	0.023825
2024-07-31	VBB	0.002790	0.000000	0.000000	0.000000	0.041667	0.033718

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2024-07-31	SGB	0.002388	0.000000	0.096984	0.000000	0.041667	0.050050
2024-07-31	ABB	0.004358	0.000000	0.000000	0.000000	0.041667	0.043624
2024-07-31	BVB	0.003712	0.000000	0.000000	0.000000	0.041667	0.040005
2024-08-31	VCB	0.268548	0.000000	0.065913	0.213667	0.041667	0.051852
2024-08-31	BID	0.148183	0.155839	0.000000	0.000000	0.041667	0.036259
2024-08-31	CTG	0.063949	0.000000	0.000000	0.000000	0.041667	0.040956
2024-08-31	MBB	0.052739	0.000000	0.000000	0.000000	0.041667	0.040640
2024-08-31	TCB	0.085014	0.000000	0.000000	0.000000	0.041667	0.035554
2024-08-31	ACB	0.057157	0.297073	0.238980	0.300000	0.041667	0.061074
2024-08-31	VPB	0.078211	0.000000	0.000000	0.000000	0.041667	0.044654
2024-08-31	HDB	0.032184	0.016297	0.000000	0.000000	0.041667	0.045761
2024-08-31	VIB	0.023436	0.000000	0.000000	0.000000	0.041667	0.033231
2024-08-31	TPB	0.019250	0.000000	0.000000	0.000000	0.041667	0.036577
2024-08-31	STB	0.029661	0.000000	0.000000	0.053596	0.041667	0.031377
2024-08-31	SHB	0.018617	0.000000	0.000000	0.000000	0.041667	0.039155
2024-08-31	LPB	0.035607	0.300000	0.000000	0.300000	0.041667	0.031073
2024-08-31	SSB	0.029620	0.000000	0.265745	0.000000	0.041667	0.053274
2024-08-31	MSB	0.013245	0.000000	0.000000	0.000000	0.041667	0.043320
2024-08-31	OCB	0.014194	0.000000	0.000000	0.132737	0.041667	0.039763
2024-08-31	NAB	0.007521	0.230790	0.000000	0.000000	0.041667	0.037969
2024-08-31	BAB	0.005364	0.000000	0.300000	0.000000	0.041667	0.080372
2024-08-31	KLB	0.001474	0.000000	0.000000	0.000000	0.041667	0.028000
2024-08-31	PGB	0.003250	0.000000	0.000000	0.000000	0.041667	0.023295
2024-08-31	VBB	0.002661	0.000000	0.000000	0.000000	0.041667	0.032796
2024-08-31	SGB	0.002261	0.000000	0.129363	0.000000	0.041667	0.050915
2024-08-31	ABB	0.004105	0.000000	0.000000	0.000000	0.041667	0.042718

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2024-08-31	BVB	0.003749	0.000000	0.000000	0.000000	0.041667	0.039414
2024-09-30	VCB	0.268808	0.247946	0.164723	0.215077	0.040000	0.054208
2024-09-30	BID	0.148251	0.088752	0.000000	0.000000	0.040000	0.034863
2024-09-30	CTG	0.068195	0.000000	0.000000	0.000000	0.040000	0.040777
2024-09-30	MBB	0.052286	0.000000	0.000000	0.000000	0.040000	0.038354
2024-09-30	TCB	0.083150	0.000000	0.000000	0.000000	0.040000	0.033390
2024-09-30	ACB	0.056223	0.000000	0.133702	0.184923	0.040000	0.057125
2024-09-30	VPB	0.075943	0.000000	0.000000	0.000000	0.040000	0.042885
2024-09-30	HDB	0.033006	0.063970	0.000000	0.000000	0.040000	0.040581
2024-09-30	VIB	0.023102	0.000000	0.000000	0.000000	0.040000	0.032402
2024-09-30	TPB	0.018478	0.000000	0.000000	0.000000	0.040000	0.034301
2024-09-30	STB	0.030424	0.000000	0.000000	0.000000	0.040000	0.029445
2024-09-30	SHB	0.017161	0.000000	0.000000	0.000000	0.040000	0.036690
2024-09-30	LPB	0.037352	0.299332	0.000000	0.300000	0.040000	0.029274
2024-09-30	SSB	0.028855	0.000000	0.242411	0.000000	0.040000	0.049702
2024-09-30	MSB	0.013281	0.000000	0.000000	0.000000	0.040000	0.040569
2024-09-30	OCB	0.013612	0.000000	0.000000	0.000000	0.040000	0.037255
2024-09-30	NAB	0.007976	0.300000	0.000000	0.000000	0.040000	0.035173
2024-09-30	BAB	0.005182	0.000000	0.300000	0.000000	0.040000	0.083904
2024-09-30	KLB	0.001567	0.000000	0.000000	0.000000	0.040000	0.027002
2024-09-30	PGB	0.003106	0.000000	0.000000	0.000000	0.040000	0.021920
2024-09-30	VAB	0.001697	0.000000	0.000000	0.300000	0.040000	0.038938
2024-09-30	VBB	0.002516	0.000000	0.000000	0.000000	0.040000	0.031223
2024-09-30	SGB	0.002185	0.000000	0.159164	0.000000	0.040000	0.050229
2024-09-30	ABB	0.003948	0.000000	0.000000	0.000000	0.040000	0.040968
2024-09-30	BVB	0.003696	0.000000	0.000000	0.000000	0.040000	0.038820

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2024-10-31	VCB	0.262621	0.082972	0.147731	0.300000	0.040000	0.053775
2024-10-31	BID	0.146713	0.181170	0.000000	0.174852	0.040000	0.034893
2024-10-31	CTG	0.069998	0.000000	0.000000	0.000000	0.040000	0.040439
2024-10-31	MBB	0.052642	0.000000	0.000000	0.000000	0.040000	0.038222
2024-10-31	TCB	0.083914	0.000000	0.000000	0.000000	0.040000	0.033296
2024-10-31	ACB	0.056836	0.000000	0.159565	0.150890	0.040000	0.061457
2024-10-31	VPB	0.078442	0.000000	0.000000	0.000000	0.040000	0.042611
2024-10-31	HDB	0.032718	0.179389	0.000000	0.000000	0.040000	0.040411
2024-10-31	VIB	0.023662	0.000000	0.000000	0.000000	0.040000	0.033099
2024-10-31	TPB	0.020753	0.000000	0.000000	0.000000	0.040000	0.034100
2024-10-31	STB	0.032342	0.000000	0.000000	0.095848	0.040000	0.029441
2024-10-31	SHB	0.017422	0.000000	0.000000	0.000000	0.040000	0.036311
2024-10-31	LPB	0.037003	0.300000	0.000000	0.278410	0.040000	0.029482
2024-10-31	SSB	0.024952	0.000000	0.238152	0.000000	0.040000	0.049408
2024-10-31	MSB	0.014538	0.000000	0.000000	0.000000	0.040000	0.040359
2024-10-31	OCB	0.013483	0.000000	0.000000	0.000000	0.040000	0.037212
2024-10-31	NAB	0.007858	0.256468	0.000000	0.000000	0.040000	0.034659
2024-10-31	BAB	0.005130	0.000000	0.300000	0.000000	0.040000	0.083664
2024-10-31	KLB	0.001586	0.000000	0.000000	0.000000	0.040000	0.026570
2024-10-31	PGB	0.003284	0.000000	0.000000	0.000000	0.040000	0.021842
2024-10-31	VAB	0.001783	0.000000	0.000000	0.000000	0.040000	0.038615
2024-10-31	VBB	0.002545	0.000000	0.000000	0.000000	0.040000	0.031019
2024-10-31	SGB	0.002144	0.000000	0.154552	0.000000	0.040000	0.049889
2024-10-31	ABB	0.003940	0.000000	0.000000	0.000000	0.040000	0.040696
2024-10-31	BVB	0.003692	0.000000	0.000000	0.000000	0.040000	0.038529
2024-11-30	VCB	0.269394	0.176749	0.145661	0.300000	0.040000	0.054110

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2024-11-30	BID	0.142140	0.015066	0.000000	0.072775	0.040000	0.034970
2024-11-30	CTG	0.068212	0.000000	0.000000	0.000000	0.040000	0.040600
2024-11-30	MBB	0.051749	0.000000	0.000000	0.000000	0.040000	0.038260
2024-11-30	TCB	0.083547	0.000000	0.000000	0.000000	0.040000	0.033329
2024-11-30	ACB	0.056536	0.172242	0.169369	0.162632	0.040000	0.061526
2024-11-30	VPB	0.080492	0.000000	0.000000	0.000000	0.040000	0.042826
2024-11-30	HDB	0.031460	0.193681	0.000000	0.000000	0.040000	0.040618
2024-11-30	VIB	0.023425	0.000000	0.000000	0.000000	0.040000	0.033100
2024-11-30	TPB	0.021048	0.000000	0.000000	0.000000	0.040000	0.035267
2024-11-30	STB	0.034419	0.000000	0.000000	0.035397	0.040000	0.029425
2024-11-30	SHB	0.016934	0.000000	0.000000	0.000000	0.040000	0.036250
2024-11-30	LPB	0.038472	0.300000	0.000000	0.300000	0.040000	0.029810
2024-11-30	SSB	0.024422	0.000000	0.227986	0.129196	0.040000	0.047375
2024-11-30	MSB	0.013877	0.000000	0.000000	0.000000	0.040000	0.039330
2024-11-30	OCB	0.012866	0.000000	0.000000	0.000000	0.040000	0.037492
2024-11-30	NAB	0.007596	0.142262	0.000000	0.000000	0.040000	0.034797
2024-11-30	BAB	0.005173	0.000000	0.300000	0.000000	0.040000	0.083292
2024-11-30	KLB	0.001432	0.000000	0.000000	0.000000	0.040000	0.026586
2024-11-30	PGB	0.003032	0.000000	0.000000	0.000000	0.040000	0.021842
2024-11-30	VAB	0.001722	0.000000	0.000000	0.000000	0.040000	0.038098
2024-11-30	VBB	0.002518	0.000000	0.000000	0.000000	0.040000	0.030991
2024-11-30	SGB	0.002096	0.000000	0.156984	0.000000	0.040000	0.050056
2024-11-30	ABB	0.003779	0.000000	0.000000	0.000000	0.040000	0.040896
2024-11-30	BVB	0.003658	0.000000	0.000000	0.000000	0.040000	0.039153
2024-12-31	VCB	0.274011	0.300000	0.134575	0.300000	0.040000	0.053946
2024-12-31	BID	0.141995	0.000000	0.000000	0.066425	0.040000	0.034769

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2024-12-31	CTG	0.069690	0.000000	0.000000	0.000000	0.040000	0.040388
2024-12-31	MBB	0.050901	0.000000	0.000000	0.000000	0.040000	0.038087
2024-12-31	TCB	0.084207	0.000000	0.000000	0.000000	0.040000	0.033308
2024-12-31	ACB	0.057228	0.103517	0.173917	0.119741	0.040000	0.061580
2024-12-31	VPB	0.076929	0.000000	0.000000	0.000000	0.040000	0.042845
2024-12-31	HDB	0.030264	0.278888	0.000000	0.000000	0.040000	0.040186
2024-12-31	VIB	0.023967	0.000000	0.000000	0.000000	0.040000	0.033069
2024-12-31	TPB	0.020108	0.000000	0.000000	0.000000	0.040000	0.035503
2024-12-31	STB	0.033226	0.000000	0.000000	0.081165	0.040000	0.029291
2024-12-31	SHB	0.016785	0.000000	0.000000	0.000000	0.040000	0.036333
2024-12-31	LPB	0.038436	0.300000	0.000000	0.300000	0.040000	0.029881
2024-12-31	SSB	0.025598	0.000000	0.241182	0.132669	0.040000	0.048572
2024-12-31	MSB	0.013183	0.000000	0.000000	0.000000	0.040000	0.039008
2024-12-31	OCB	0.012320	0.000000	0.000000	0.000000	0.040000	0.038134
2024-12-31	NAB	0.007628	0.017595	0.000000	0.000000	0.040000	0.034583
2024-12-31	BAB	0.005101	0.000000	0.300000	0.000000	0.040000	0.082941
2024-12-31	KLB	0.001450	0.000000	0.000000	0.000000	0.040000	0.027323
2024-12-31	PGB	0.003035	0.000000	0.000000	0.000000	0.040000	0.021680
2024-12-31	VAB	0.001720	0.000000	0.000000	0.000000	0.040000	0.038042
2024-12-31	VBB	0.002925	0.000000	0.000000	0.000000	0.040000	0.030901
2024-12-31	SGB	0.002037	0.000000	0.150325	0.000000	0.040000	0.049831
2024-12-31	ABB	0.003654	0.000000	0.000000	0.000000	0.040000	0.040761
2024-12-31	BVB	0.003605	0.000000	0.000000	0.000000	0.040000	0.039038
2025-01-31	VCB	0.263005	0.300000	0.124959	0.032437	0.040000	0.052915
2025-01-31	BID	0.135537	0.000000	0.000000	0.000000	0.040000	0.034239
2025-01-31	CTG	0.072372	0.000000	0.000000	0.000000	0.040000	0.039935

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2025-01-31	MBB	0.051952	0.000000	0.000000	0.000000	0.040000	0.037261
2025-01-31	TCB	0.086365	0.000000	0.000000	0.000000	0.040000	0.032690
2025-01-31	ACB	0.057561	0.300000	0.150359	0.000000	0.040000	0.060360
2025-01-31	VPB	0.075711	0.000000	0.000000	0.000000	0.040000	0.041660
2025-01-31	HDB	0.035868	0.000000	0.000000	0.000000	0.040000	0.042021
2025-01-31	VIB	0.024403	0.000000	0.000000	0.000000	0.040000	0.033338
2025-01-31	TPB	0.020295	0.000000	0.000000	0.000000	0.040000	0.034600
2025-01-31	STB	0.036156	0.000000	0.000000	0.300000	0.040000	0.028827
2025-01-31	SHB	0.016398	0.000000	0.000000	0.000000	0.040000	0.035819
2025-01-31	LPB	0.042713	0.300000	0.000000	0.300000	0.040000	0.029338
2025-01-31	SSB	0.024768	0.000000	0.260570	0.179310	0.040000	0.048335
2025-01-31	MSB	0.013122	0.000000	0.000000	0.000000	0.040000	0.039401
2025-01-31	OCB	0.012829	0.000000	0.000000	0.188253	0.040000	0.037441
2025-01-31	NAB	0.007583	0.100000	0.000000	0.000000	0.040000	0.034361
2025-01-31	BAB	0.005139	0.000000	0.300000	0.000000	0.040000	0.086509
2025-01-31	KLB	0.001352	0.000000	0.000000	0.000000	0.040000	0.026823
2025-01-31	PGB	0.002869	0.000000	0.004336	0.000000	0.040000	0.024902
2025-01-31	VAB	0.001746	0.000000	0.000000	0.000000	0.040000	0.037496
2025-01-31	VBB	0.002872	0.000000	0.000000	0.000000	0.040000	0.030004
2025-01-31	SGB	0.002083	0.000000	0.159777	0.000000	0.040000	0.049562
2025-01-31	ABB	0.003637	0.000000	0.000000	0.000000	0.040000	0.042021
2025-01-31	BVB	0.003666	0.000000	0.000000	0.000000	0.040000	0.040144
2025-02-28	VCB	0.262105	0.300000	0.122524	0.278878	0.040000	0.052565
2025-02-28	BID	0.141699	0.000000	0.000000	0.119912	0.040000	0.034131
2025-02-28	CTG	0.071390	0.000000	0.000000	0.300000	0.040000	0.039625
2025-02-28	MBB	0.052330	0.000000	0.000000	0.018560	0.040000	0.037039

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2025-02-28	TCB	0.085092	0.000000	0.000000	0.000000	0.040000	0.032525
2025-02-28	ACB	0.055500	0.156307	0.140167	0.000000	0.040000	0.060275
2025-02-28	VPB	0.073177	0.000000	0.000000	0.000000	0.040000	0.041731
2025-02-28	HDB	0.031283	0.186809	0.000000	0.111966	0.040000	0.038840
2025-02-28	VIB	0.025107	0.000000	0.000000	0.000000	0.040000	0.033864
2025-02-28	TPB	0.019677	0.000000	0.000000	0.079175	0.040000	0.037367
2025-02-28	STB	0.035584	0.000000	0.000000	0.091510	0.040000	0.028611
2025-02-28	SHB	0.016178	0.000000	0.000000	0.000000	0.040000	0.035778
2025-02-28	LPB	0.048095	0.300000	0.000000	0.000000	0.040000	0.028910
2025-02-28	SSB	0.027213	0.000000	0.295831	0.000000	0.040000	0.052323
2025-02-28	MSB	0.012322	0.000000	0.000000	0.000000	0.040000	0.040492
2025-02-28	OCB	0.012378	0.000000	0.000000	0.000000	0.040000	0.036964
2025-02-28	NAB	0.007891	0.056883	0.000000	0.000000	0.040000	0.034393
2025-02-28	BAB	0.005044	0.000000	0.300000	0.000000	0.040000	0.084975
2025-02-28	KLB	0.001327	0.000000	0.000000	0.000000	0.040000	0.026857
2025-02-28	PGB	0.002778	0.000000	0.000000	0.000000	0.040000	0.025125
2025-02-28	VAB	0.001658	0.000000	0.000000	0.000000	0.040000	0.037194
2025-02-28	VBB	0.002728	0.000000	0.000000	0.000000	0.040000	0.029939
2025-02-28	SGB	0.002141	0.000000	0.141478	0.000000	0.040000	0.048957
2025-02-28	ABB	0.003427	0.000000	0.000000	0.000000	0.040000	0.041751
2025-02-28	BVB	0.003876	0.000000	0.000000	0.000000	0.040000	0.039769
2025-03-31	VCB	0.256873	0.100000	0.115961	0.283763	0.040000	0.055151
2025-03-31	BID	0.141092	0.000000	0.000000	0.158483	0.040000	0.037908
2025-03-31	CTG	0.075829	0.000000	0.000000	0.000000	0.040000	0.039925
2025-03-31	MBB	0.052259	0.000000	0.000000	0.000000	0.040000	0.038469
2025-03-31	TCB	0.087283	0.000000	0.000000	0.000000	0.040000	0.032443

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2025-03-31	ACB	0.055356	0.300000	0.179171	0.000000	0.040000	0.059653
2025-03-31	VPB	0.072668	0.000000	0.000000	0.000000	0.040000	0.041497
2025-03-31	HDB	0.030948	0.000000	0.000000	0.000000	0.040000	0.037239
2025-03-31	VIB	0.024537	0.000000	0.000000	0.000000	0.040000	0.033489
2025-03-31	TPB	0.019440	0.000000	0.000000	0.000000	0.040000	0.037164
2025-03-31	STB	0.035778	0.000000	0.000000	0.257754	0.040000	0.028890
2025-03-31	SHB	0.016945	0.000000	0.000000	0.000000	0.040000	0.035530
2025-03-31	LPB	0.047163	0.300000	0.000000	0.300000	0.040000	0.028546
2025-03-31	SSB	0.027385	0.000000	0.268444	0.000000	0.040000	0.049902
2025-03-31	MSB	0.012204	0.000000	0.000000	0.000000	0.040000	0.040388
2025-03-31	OCB	0.012088	0.000000	0.000000	0.000000	0.040000	0.036727
2025-03-31	NAB	0.008036	0.300000	0.000000	0.000000	0.040000	0.034039
2025-03-31	BAB	0.005205	0.000000	0.300000	0.000000	0.040000	0.084285
2025-03-31	KLB	0.001405	0.000000	0.000000	0.000000	0.040000	0.027078
2025-03-31	PGB	0.002791	0.000000	0.000000	0.000000	0.040000	0.025046
2025-03-31	VAB	0.001843	0.000000	0.000000	0.000000	0.040000	0.036919
2025-03-31	VBB	0.002802	0.000000	0.000000	0.000000	0.040000	0.029930
2025-03-31	SGB	0.002068	0.000000	0.136424	0.000000	0.040000	0.048136
2025-03-31	ABB	0.003569	0.000000	0.000000	0.000000	0.040000	0.041863
2025-03-31	BVB	0.004436	0.000000	0.000000	0.000000	0.040000	0.039785
2025-04-30	VCB	0.263224	0.220671	0.134200	0.242212	0.040000	0.056256
2025-04-30	BID	0.133396	0.000000	0.000000	0.000000	0.040000	0.038846
2025-04-30	CTG	0.075692	0.000000	0.000000	0.000000	0.040000	0.040702
2025-04-30	MBB	0.054730	0.000000	0.000000	0.000000	0.040000	0.038616
2025-04-30	TCB	0.091910	0.000000	0.000000	0.000000	0.040000	0.032554
2025-04-30	ACB	0.055314	0.300000	0.159560	0.000000	0.040000	0.060044

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2025-04-30	VPB	0.071473	0.000000	0.000000	0.000000	0.040000	0.041784
2025-04-30	HDB	0.029655	0.068048	0.000000	0.000000	0.040000	0.037909
2025-04-30	VIB	0.023455	0.000000	0.000000	0.000000	0.040000	0.033639
2025-04-30	TPB	0.016635	0.000000	0.000000	0.000000	0.040000	0.037300
2025-04-30	STB	0.035798	0.000000	0.000000	0.300000	0.040000	0.029222
2025-04-30	SHB	0.021085	0.000000	0.000000	0.000000	0.040000	0.035268
2025-04-30	LPB	0.043351	0.300000	0.000000	0.300000	0.040000	0.029234
2025-04-30	SSB	0.027576	0.000000	0.278663	0.100955	0.040000	0.051726
2025-04-30	MSB	0.013110	0.000000	0.000000	0.000000	0.040000	0.040496
2025-04-30	OCB	0.011968	0.000000	0.000000	0.056833	0.040000	0.036953
2025-04-30	NAB	0.007553	0.111280	0.000000	0.000000	0.040000	0.034144
2025-04-30	BAB	0.005201	0.000000	0.300000	0.000000	0.040000	0.080482
2025-04-30	KLB	0.001324	0.000000	0.000000	0.000000	0.040000	0.026860
2025-04-30	PGB	0.002843	0.000000	0.000000	0.000000	0.040000	0.025329
2025-04-30	VAB	0.001858	0.000000	0.000000	0.000000	0.040000	0.035606
2025-04-30	VBB	0.002949	0.000000	0.000000	0.000000	0.040000	0.029894
2025-04-30	SGB	0.002081	0.000000	0.127577	0.000000	0.040000	0.048344
2025-04-30	ABB	0.003659	0.000000	0.000000	0.000000	0.040000	0.041214
2025-04-30	BVB	0.004161	0.000000	0.000000	0.000000	0.040000	0.037575
2025-05-31	VCB	0.255298	0.300000	0.126505	0.000000	0.040000	0.056516
2025-05-31	BID	0.128853	0.000000	0.000000	0.279713	0.040000	0.038521
2025-05-31	CTG	0.073572	0.000000	0.000000	0.000000	0.040000	0.040770
2025-05-31	MBB	0.057943	0.000000	0.000000	0.000000	0.040000	0.038638
2025-05-31	TCB	0.095199	0.000000	0.000000	0.000000	0.040000	0.032438
2025-05-31	ACB	0.055423	0.300000	0.158165	0.000000	0.040000	0.060374
2025-05-31	VPB	0.067453	0.000000	0.000000	0.000000	0.040000	0.041784

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2025-05-31	HDB	0.031181	0.091731	0.000000	0.000000	0.040000	0.037728
2025-05-31	VIB	0.023088	0.000000	0.000000	0.000000	0.040000	0.033529
2025-05-31	TPB	0.017126	0.000000	0.000000	0.000000	0.040000	0.035891
2025-05-31	STB	0.039588	0.000000	0.000000	0.300000	0.040000	0.029270
2025-05-31	SHB	0.023124	0.000000	0.000000	0.000000	0.040000	0.032403
2025-05-31	LPB	0.045965	0.300000	0.000000	0.298693	0.040000	0.028960
2025-05-31	SSB	0.028114	0.000000	0.288013	0.000000	0.040000	0.052898
2025-05-31	MSB	0.013028	0.000000	0.000000	0.000000	0.040000	0.040032
2025-05-31	OCB	0.012091	0.000000	0.000000	0.121594	0.040000	0.037041
2025-05-31	NAB	0.008131	0.008269	0.000000	0.000000	0.040000	0.033814
2025-05-31	BAB	0.005300	0.000000	0.300000	0.000000	0.040000	0.080393
2025-05-31	KLB	0.001495	0.000000	0.000000	0.000000	0.040000	0.031245
2025-05-31	PGB	0.002855	0.000000	0.000000	0.000000	0.040000	0.025437
2025-05-31	VAB	0.002111	0.000000	0.000000	0.000000	0.040000	0.035639
2025-05-31	VBB	0.003098	0.000000	0.000000	0.000000	0.040000	0.029770
2025-05-31	SGB	0.002204	0.000000	0.127317	0.000000	0.040000	0.048319
2025-05-31	ABB	0.003758	0.000000	0.000000	0.000000	0.040000	0.041141
2025-05-31	BVB	0.004001	0.000000	0.000000	0.000000	0.040000	0.037450
2025-06-30	VCB	0.241100	0.057003	0.089438	0.233078	0.040000	0.051950
2025-06-30	BID	0.128818	0.000000	0.000000	0.130218	0.040000	0.038528
2025-06-30	CTG	0.072701	0.138049	0.000000	0.000000	0.040000	0.041168
2025-06-30	MBB	0.057487	0.000000	0.000000	0.000000	0.040000	0.038579
2025-06-30	TCB	0.105768	0.000000	0.000000	0.000000	0.040000	0.032272
2025-06-30	ACB	0.055856	0.000000	0.133803	0.000000	0.040000	0.056765
2025-06-30	VPB	0.072126	0.000000	0.000000	0.000000	0.040000	0.039066
2025-06-30	HDB	0.030264	0.211354	0.000000	0.000000	0.040000	0.037914

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2025-06-30	VIB	0.022861	0.000000	0.000000	0.000000	0.040000	0.032643
2025-06-30	TPB	0.017183	0.000000	0.000000	0.000000	0.040000	0.036912
2025-06-30	STB	0.039542	0.000000	0.000000	0.271928	0.040000	0.029290
2025-06-30	SHB	0.023865	0.000000	0.000000	0.000000	0.040000	0.035982
2025-06-30	LPB	0.046262	0.300000	0.000000	0.199158	0.040000	0.029778
2025-06-30	SSB	0.026978	0.000000	0.281121	0.000000	0.040000	0.051132
2025-06-30	MSB	0.012890	0.000000	0.000000	0.000000	0.040000	0.039324
2025-06-30	OCB	0.011831	0.000000	0.000000	0.165617	0.040000	0.036046
2025-06-30	NAB	0.007901	0.293593	0.000000	0.000000	0.040000	0.034201
2025-06-30	BAB	0.005268	0.000000	0.300000	0.000000	0.040000	0.084631
2025-06-30	KLB	0.002142	0.000000	0.000000	0.000000	0.040000	0.030623
2025-06-30	PGB	0.002858	0.000000	0.000000	0.000000	0.040000	0.024742
2025-06-30	VAB	0.002613	0.000000	0.000000	0.000000	0.040000	0.035713
2025-06-30	VBB	0.003528	0.000000	0.000000	0.000000	0.040000	0.030012
2025-06-30	SGB	0.002099	0.000000	0.195638	0.000000	0.040000	0.050766
2025-06-30	ABB	0.003899	0.000000	0.000000	0.000000	0.040000	0.043896
2025-06-30	BVB	0.004159	0.000000	0.000000	0.000000	0.040000	0.038066
2025-07-31	VCB	0.234626	0.078296	0.081971	0.000000	0.040000	0.051640
2025-07-31	BID	0.125075	0.000000	0.000000	0.000000	0.040000	0.038459
2025-07-31	CTG	0.076580	0.000000	0.000000	0.000000	0.040000	0.040809
2025-07-31	MBB	0.058651	0.000000	0.000000	0.000000	0.040000	0.038787
2025-07-31	TCB	0.114387	0.010777	0.000000	0.000000	0.040000	0.032734
2025-07-31	ACB	0.054289	0.300000	0.113315	0.000000	0.040000	0.056409
2025-07-31	VPB	0.071571	0.000000	0.000000	0.000000	0.040000	0.040135
2025-07-31	HDB	0.029278	0.007236	0.000000	0.000000	0.040000	0.037637
2025-07-31	VIB	0.022499	0.000000	0.000000	0.000000	0.040000	0.034025

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2025-07-31	TPB	0.016793	0.000000	0.000000	0.000000	0.040000	0.036497
2025-07-31	STB	0.043685	0.000000	0.000000	0.000000	0.040000	0.031236
2025-07-31	SHB	0.022696	0.000000	0.000000	0.000000	0.040000	0.036173
2025-07-31	LPB	0.045180	0.300000	0.000000	0.111878	0.040000	0.030183
2025-07-31	SSB	0.025622	0.000000	0.288162	0.286412	0.040000	0.052048
2025-07-31	MSB	0.012901	0.000000	0.000000	0.000000	0.040000	0.040001
2025-07-31	OCB	0.012578	0.000000	0.000883	0.000000	0.040000	0.038243
2025-07-31	NAB	0.007607	0.300000	0.000000	0.000000	0.040000	0.035404
2025-07-31	BAB	0.005334	0.000000	0.300000	0.000000	0.040000	0.082635
2025-07-31	KLB	0.001959	0.000000	0.000000	0.087649	0.040000	0.025405
2025-07-31	PGB	0.002637	0.000000	0.000000	0.000000	0.040000	0.025601
2025-07-31	VAB	0.002693	0.003692	0.000000	0.300000	0.040000	0.032617
2025-07-31	VBB	0.003497	0.000000	0.000000	0.214062	0.040000	0.028988
2025-07-31	SGB	0.002006	0.000000	0.215669	0.000000	0.040000	0.052733
2025-07-31	ABB	0.003944	0.000000	0.000000	0.000000	0.040000	0.043298
2025-07-31	BVB	0.003912	0.000000	0.000000	0.000000	0.040000	0.038303
2025-08-31	VCB	0.218100	0.135595	0.087518	0.000000	0.040000	0.050685
2025-08-31	BID	0.114771	0.000000	0.000000	0.000000	0.040000	0.037504
2025-08-31	CTG	0.073824	0.070569	0.000000	0.000000	0.040000	0.039483
2025-08-31	MBB	0.072464	0.000000	0.000000	0.000000	0.040000	0.039455
2025-08-31	TCB	0.100086	0.000000	0.000000	0.000000	0.040000	0.031313
2025-08-31	ACB	0.051594	0.057543	0.118684	0.000000	0.040000	0.055561
2025-08-31	VPB	0.087833	0.000000	0.000000	0.000000	0.040000	0.039153
2025-08-31	HDB	0.031271	0.000000	0.000000	0.000000	0.040000	0.037151
2025-08-31	VIB	0.027338	0.000000	0.000000	0.000000	0.040000	0.036428
2025-08-31	TPB	0.017249	0.000000	0.000000	0.000000	0.040000	0.037683

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2025-08-31	STB	0.041123	0.000000	0.000000	0.000000	0.040000	0.029709
2025-08-31	SHB	0.030095	0.000000	0.000000	0.149902	0.040000	0.035714
2025-08-31	LPB	0.042164	0.300000	0.000000	0.000000	0.040000	0.030591
2025-08-31	SSB	0.024227	0.000000	0.272315	0.108702	0.040000	0.053331
2025-08-31	MSB	0.013058	0.000000	0.000000	0.000000	0.040000	0.042496
2025-08-31	OCB	0.014165	0.000000	0.000964	0.000000	0.040000	0.038783
2025-08-31	NAB	0.009426	0.300000	0.000000	0.000000	0.040000	0.036180
2025-08-31	BAB	0.005700	0.000000	0.300000	0.000000	0.040000	0.082136
2025-08-31	KLB	0.003656	0.000000	0.000000	0.300000	0.040000	0.024691
2025-08-31	PGB	0.002673	0.000000	0.000000	0.000000	0.040000	0.026554
2025-08-31	VAB	0.003961	0.136293	0.000000	0.300000	0.040000	0.032551
2025-08-31	VBB	0.003991	0.000000	0.000000	0.054734	0.040000	0.029743
2025-08-31	SGB	0.001848	0.000000	0.220520	0.086662	0.040000	0.051395
2025-08-31	ABB	0.005411	0.000000	0.000000	0.000000	0.040000	0.043626
2025-08-31	BVB	0.003974	0.000000	0.000000	0.000000	0.040000	0.038085
2025-09-30	VCB	0.208596	0.141576	0.108528	0.000000	0.040000	0.053076
2025-09-30	BID	0.109067	0.000000	0.000000	0.000000	0.040000	0.040416
2025-09-30	CTG	0.069264	0.063901	0.000000	0.000000	0.040000	0.041161
2025-09-30	MBB	0.081930	0.000000	0.000000	0.000000	0.040000	0.041637
2025-09-30	TCB	0.097831	0.000000	0.000000	0.162487	0.040000	0.033148
2025-09-30	ACB	0.052341	0.008688	0.105533	0.000000	0.040000	0.057494
2025-09-30	VPB	0.100008	0.000000	0.000000	0.000000	0.040000	0.033307
2025-09-30	HDB	0.033181	0.053700	0.000000	0.000000	0.040000	0.036709
2025-09-30	VIB	0.026738	0.000000	0.000000	0.000000	0.040000	0.039283
2025-09-30	TPB	0.019280	0.000000	0.000000	0.000000	0.040000	0.037824
2025-09-30	STB	0.038419	0.000000	0.000000	0.000000	0.040000	0.031933

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2025-09-30	SHB	0.031203	0.000000	0.000000	0.000000	0.040000	0.033112
2025-09-30	LPB	0.046557	0.300000	0.000000	0.039029	0.040000	0.032944
2025-09-30	SSB	0.022316	0.000000	0.253360	0.000000	0.040000	0.054817
2025-09-30	MSB	0.014057	0.000000	0.000000	0.000000	0.040000	0.043474
2025-09-30	OCB	0.013324	0.000000	0.000000	0.000000	0.040000	0.038918
2025-09-30	NAB	0.008408	0.300000	0.000000	0.000000	0.040000	0.037552
2025-09-30	BAB	0.004990	0.000000	0.300000	0.000000	0.040000	0.070948
2025-09-30	KLB	0.002975	0.000000	0.000000	0.198485	0.040000	0.024130
2025-09-30	PGB	0.002382	0.000000	0.000000	0.000000	0.040000	0.028898
2025-09-30	VAB	0.003845	0.132136	0.000000	0.300000	0.040000	0.033974
2025-09-30	VBB	0.003494	0.000000	0.000000	0.300000	0.040000	0.030875
2025-09-30	SGB	0.001668	0.000000	0.232579	0.000000	0.040000	0.053880
2025-09-30	ABB	0.004370	0.000000	0.000000	0.000000	0.040000	0.031115
2025-09-30	BVB	0.003758	0.000000	0.000000	0.000000	0.040000	0.039376
2025-10-31	VCB	0.200932	0.000000	0.174164	0.000000	0.040000	0.054919
2025-10-31	BID	0.108522	0.000000	0.000000	0.000000	0.040000	0.041401
2025-10-31	CTG	0.073095	0.074087	0.000000	0.000000	0.040000	0.042257
2025-10-31	MBB	0.082453	0.000000	0.000000	0.000000	0.040000	0.037330
2025-10-31	TCB	0.102321	0.000000	0.000000	0.000000	0.040000	0.033761
2025-10-31	ACB	0.051175	0.208478	0.000000	0.000000	0.040000	0.052878
2025-10-31	VPB	0.094728	0.000000	0.000000	0.000000	0.040000	0.031159
2025-10-31	HDB	0.032776	0.000000	0.000000	0.000000	0.040000	0.035756
2025-10-31	VIB	0.025096	0.000000	0.000000	0.000000	0.040000	0.039856
2025-10-31	TPB	0.018528	0.000000	0.000000	0.000000	0.040000	0.034448
2025-10-31	STB	0.041909	0.000000	0.000000	0.025275	0.040000	0.033102
2025-10-31	SHB	0.029904	0.000000	0.000000	0.142927	0.040000	0.032872

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2025-10-31	LPB	0.054598	0.300000	0.000000	0.000000	0.040000	0.031860
2025-10-31	SSB	0.021453	0.000000	0.258903	0.000000	0.040000	0.055223
2025-10-31	MSB	0.013358	0.000000	0.000000	0.000000	0.040000	0.039590
2025-10-31	OCB	0.013162	0.000000	0.000000	0.000000	0.040000	0.039965
2025-10-31	NAB	0.008211	0.300000	0.000000	0.000000	0.040000	0.039273
2025-10-31	BAB	0.004950	0.000000	0.300000	0.000000	0.040000	0.073383
2025-10-31	KLB	0.003571	0.000000	0.000000	0.281035	0.040000	0.025394
2025-10-31	PGB	0.002346	0.000000	0.000000	0.000000	0.040000	0.030233
2025-10-31	VAB	0.003700	0.117435	0.000000	0.261173	0.040000	0.034624
2025-10-31	VBB	0.003509	0.000000	0.000000	0.000000	0.040000	0.032324
2025-10-31	SGB	0.001628	0.000000	0.266933	0.000000	0.040000	0.055580
2025-10-31	ABB	0.004545	0.000000	0.000000	0.289590	0.040000	0.032512
2025-10-31	BVB	0.003530	0.000000	0.000000	0.000000	0.040000	0.040299
2025-11-30	VCB	0.192849	0.000000	0.210785	0.000000	0.040000	0.055668
2025-11-30	BID	0.101691	0.000000	0.000000	0.000000	0.040000	0.042872
2025-11-30	CTG	0.101902	0.298112	0.000000	0.000000	0.040000	0.046853
2025-11-30	MBB	0.073615	0.000000	0.000000	0.000000	0.040000	0.038754
2025-11-30	TCB	0.094343	0.000000	0.000000	0.000000	0.040000	0.035104
2025-11-30	ACB	0.050724	0.000000	0.000000	0.000000	0.040000	0.052328
2025-11-30	VPB	0.086642	0.000000	0.000000	0.099812	0.040000	0.030931
2025-11-30	HDB	0.047837	0.000000	0.000000	0.000000	0.040000	0.035293
2025-11-30	VIB	0.023187	0.000000	0.000000	0.000000	0.040000	0.038815
2025-11-30	TPB	0.017886	0.000000	0.000000	0.000000	0.040000	0.033662
2025-11-30	STB	0.040518	0.000000	0.000000	0.000000	0.040000	0.034606
2025-11-30	SHB	0.028839	0.000395	0.000000	0.240479	0.040000	0.034401
2025-11-30	LPB	0.055423	0.300000	0.056895	0.000000	0.040000	0.034058

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2025-11-30	SSB	0.019500	0.000000	0.217299	0.000000	0.040000	0.052141
2025-11-30	MSB	0.014559	0.000000	0.000000	0.000000	0.040000	0.039088
2025-11-30	OCB	0.013149	0.000000	0.000000	0.091431	0.040000	0.040393
2025-11-30	NAB	0.007886	0.288984	0.000000	0.000000	0.040000	0.039519
2025-11-30	BAB	0.004798	0.000000	0.300000	0.000000	0.040000	0.070758
2025-11-30	KLB	0.003765	0.000000	0.026435	0.246365	0.040000	0.024442
2025-11-30	PGB	0.002790	0.000000	0.016447	0.000000	0.040000	0.029830
2025-11-30	VAB	0.003588	0.021532	0.000000	0.214572	0.040000	0.033380
2025-11-30	VBB	0.004721	0.090977	0.000000	0.000000	0.040000	0.033105
2025-11-30	SGB	0.001650	0.000000	0.172138	0.000000	0.040000	0.052591
2025-11-30	ABB	0.004762	0.000000	0.000000	0.107342	0.040000	0.032358
2025-11-30	BVB	0.003375	0.000000	0.000000	0.000000	0.040000	0.039049
2025-12-31	VCB	0.189498	0.000000	0.176834	0.000000	0.040000	0.053238
2025-12-31	BID	0.102922	0.000000	0.000000	0.000000	0.040000	0.040832
2025-12-31	CTG	0.103969	0.000000	0.000000	0.000000	0.040000	0.044961
2025-12-31	MBB	0.073995	0.000000	0.000000	0.180838	0.040000	0.037745
2025-12-31	TCB	0.092534	0.170506	0.000000	0.080874	0.040000	0.040337
2025-12-31	ACB	0.049216	0.000000	0.000000	0.000000	0.040000	0.050392
2025-12-31	VPB	0.090092	0.000000	0.000000	0.000000	0.040000	0.029958
2025-12-31	HDB	0.048807	0.017732	0.000000	0.000000	0.040000	0.036257
2025-12-31	VIB	0.023590	0.000000	0.000000	0.000000	0.040000	0.038225
2025-12-31	TPB	0.018961	0.000000	0.000000	0.000000	0.040000	0.034347
2025-12-31	STB	0.036200	0.143214	0.051154	0.000000	0.040000	0.038504
2025-12-31	SHB	0.030060	0.000000	0.000000	0.000000	0.040000	0.033687
2025-12-31	LPB	0.054211	0.300000	0.073663	0.067354	0.040000	0.034080
2025-12-31	SSB	0.019446	0.000000	0.185126	0.000000	0.040000	0.049068

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2025-12-31	MSB	0.016210	0.000000	0.000000	0.000000	0.040000	0.036997
2025-12-31	OCB	0.012942	0.000000	0.000000	0.000000	0.040000	0.041647
2025-12-31	NAB	0.008080	0.257511	0.000000	0.000000	0.040000	0.039441
2025-12-31	BAB	0.004598	0.000000	0.300000	0.117064	0.040000	0.073699
2025-12-31	KLB	0.003519	0.000000	0.032932	0.300000	0.040000	0.023684
2025-12-31	PGB	0.002760	0.038384	0.034501	0.039620	0.040000	0.028706
2025-12-31	VAB	0.003500	0.006197	0.000000	0.010263	0.040000	0.033687
2025-12-31	VBB	0.004468	0.066456	0.000000	0.048374	0.040000	0.031914
2025-12-31	SGB	0.001672	0.000000	0.145791	0.000000	0.040000	0.050702
2025-12-31	ABB	0.005382	0.000000	0.000000	0.155613	0.040000	0.033264
2025-12-31	BVB	0.003367	0.000000	0.000000	0.000000	0.040000	0.044630
2026-01-31	VCB	0.185156	0.000000	0.154779	0.000000	0.040000	0.053947
2026-01-31	BID	0.105259	0.000000	0.000000	0.000000	0.040000	0.043454
2026-01-31	CTG	0.107008	0.000000	0.000000	0.054987	0.040000	0.045281
2026-01-31	MBB	0.078537	0.000000	0.000000	0.033939	0.040000	0.037280
2026-01-31	TCB	0.093342	0.167819	0.000000	0.136527	0.040000	0.039405
2026-01-31	ACB	0.047509	0.000000	0.000000	0.000000	0.040000	0.049047
2026-01-31	VPB	0.086071	0.000000	0.000000	0.000000	0.040000	0.029582
2026-01-31	HDB	0.057289	0.067623	0.000000	0.000000	0.040000	0.036249
2026-01-31	VIB	0.022078	0.000000	0.000000	0.000000	0.040000	0.037735
2026-01-31	TPB	0.018281	0.000000	0.000000	0.000000	0.040000	0.034007
2026-01-31	STB	0.042138	0.000000	0.000000	0.000000	0.040000	0.038704
2026-01-31	SHB	0.028523	0.000000	0.000000	0.000000	0.040000	0.033268
2026-01-31	LPB	0.045474	0.300000	0.056810	0.052926	0.040000	0.033280
2026-01-31	SSB	0.019132	0.000000	0.204459	0.000000	0.040000	0.048467
2026-01-31	MSB	0.014910	0.000000	0.000000	0.000000	0.040000	0.036188

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2026-01-31	OCB	0.012264	0.000000	0.000000	0.000000	0.040000	0.042800
2026-01-31	NAB	0.007881	0.300000	0.000000	0.016235	0.040000	0.039268
2026-01-31	BAB	0.004415	0.000000	0.300000	0.194698	0.040000	0.070862
2026-01-31	KLB	0.003769	0.000000	0.045887	0.300000	0.040000	0.026473
2026-01-31	PGB	0.002628	0.058273	0.028130	0.026802	0.040000	0.029561
2026-01-31	VAB	0.003272	0.048768	0.000000	0.000000	0.040000	0.034257
2026-01-31	VBB	0.004316	0.030685	0.000000	0.023565	0.040000	0.031577
2026-01-31	SGB	0.001619	0.000000	0.209935	0.000000	0.040000	0.052525
2026-01-31	ABB	0.005868	0.026833	0.000000	0.160322	0.040000	0.032593
2026-01-31	BVB	0.003260	0.000000	0.000000	0.000000	0.040000	0.044191
2026-02-28	VCB	0.205519	0.000000	0.148265	0.000000	0.040000	0.053706
2026-02-28	BID	0.136900	0.000000	0.000000	0.000000	0.040000	0.043634
2026-02-28	CTG	0.105003	0.000000	0.000000	0.000000	0.040000	0.044906
2026-02-28	MBB	0.076439	0.000000	0.000000	0.000000	0.040000	0.036944
2026-02-28	TCB	0.086925	0.182342	0.000000	0.000000	0.040000	0.039255
2026-02-28	ACB	0.043190	0.000000	0.000000	0.000000	0.040000	0.048741
2026-02-28	VPB	0.076148	0.000000	0.000000	0.169108	0.040000	0.029430
2026-02-28	HDB	0.049419	0.186305	0.000000	0.108614	0.040000	0.034425
2026-02-28	VIB	0.020047	0.000000	0.000000	0.000000	0.040000	0.037757
2026-02-28	TPB	0.016647	0.000000	0.000000	0.114602	0.040000	0.033975
2026-02-28	STB	0.041436	0.000000	0.007043	0.000000	0.040000	0.037217
2026-02-28	SHB	0.025277	0.000000	0.000000	0.071684	0.040000	0.033003
2026-02-28	LPB	0.041022	0.300000	0.044972	0.122961	0.040000	0.031320
2026-02-28	SSB	0.017023	0.000000	0.213821	0.000000	0.040000	0.049832
2026-02-28	MSB	0.013498	0.000000	0.000000	0.196805	0.040000	0.035791
2026-02-28	OCB	0.010963	0.000000	0.000000	0.000000	0.040000	0.045472

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2026-02-28	NAB	0.007057	0.240903	0.000000	0.000000	0.040000	0.039138
2026-02-28	BAB	0.004515	0.000000	0.300000	0.000000	0.040000	0.071130
2026-02-28	KLB	0.003087	0.033220	0.046682	0.000593	0.040000	0.028447
2026-02-28	PGB	0.002949	0.000000	0.023988	0.000000	0.040000	0.029335
2026-02-28	VAB	0.003218	0.000000	0.000000	0.000000	0.040000	0.036185
2026-02-28	VBB	0.003983	0.012059	0.000000	0.000000	0.040000	0.031586
2026-02-28	SGB	0.001586	0.000000	0.215229	0.000000	0.040000	0.052266
2026-02-28	ABB	0.005019	0.045171	0.000000	0.215633	0.040000	0.032007
2026-02-28	BVB	0.003130	0.000000	0.000000	0.000000	0.040000	0.044499
2026-03-31	VCB	0.193651	0.000000	0.069068	0.000000	0.040000	0.048334
2026-03-31	BID	0.124527	0.000000	0.000000	0.000000	0.040000	0.035332
2026-03-31	CTG	0.106090	0.000000	0.000000	0.000000	0.040000	0.044941
2026-03-31	MBB	0.081979	0.000000	0.000000	0.000000	0.040000	0.037167
2026-03-31	TCB	0.089833	0.178748	0.000000	0.000000	0.040000	0.039677
2026-03-31	ACB	0.045032	0.000000	0.002997	0.000000	0.040000	0.053368
2026-03-31	VPB	0.080577	0.000000	0.000000	0.162391	0.040000	0.029116
2026-03-31	HDB	0.049958	0.138152	0.000000	0.275245	0.040000	0.034485
2026-03-31	VIB	0.019935	0.000000	0.000000	0.000000	0.040000	0.041200
2026-03-31	TPB	0.018227	0.000000	0.000000	0.000000	0.040000	0.035148
2026-03-31	STB	0.044096	0.027699	0.060705	0.000000	0.040000	0.038660
2026-03-31	SHB	0.025790	0.000000	0.000000	0.000000	0.040000	0.032714
2026-03-31	LPB	0.043396	0.300000	0.065898	0.000000	0.040000	0.031036
2026-03-31	SSB	0.017170	0.000000	0.221723	0.000000	0.040000	0.049337
2026-03-31	MSB	0.013593	0.000000	0.000000	0.000000	0.040000	0.035564
2026-03-31	OCB	0.011079	0.000000	0.000000	0.000000	0.040000	0.044918
2026-03-31	NAB	0.007150	0.262583	0.045402	0.000000	0.040000	0.043600

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2026-03-31	BAB	0.004594	0.000000	0.300000	0.000000	0.040000	0.069478
2026-03-31	KLB	0.003170	0.000000	0.008083	0.300000	0.040000	0.027508
2026-03-31	PGB	0.003115	0.000000	0.025683	0.000000	0.040000	0.028926
2026-03-31	VAB	0.003148	0.000000	0.000000	0.000000	0.040000	0.035549
2026-03-31	VBB	0.004038	0.067016	0.068944	0.000000	0.040000	0.036464
2026-03-31	SGB	0.001624	0.000000	0.131497	0.000000	0.040000	0.051450
2026-03-31	ABB	0.005138	0.025803	0.000000	0.262364	0.040000	0.031512
2026-03-31	BVB	0.003089	0.000000	0.000000	0.000000	0.040000	0.044513
2026-04-30	VCB	0.191340	0.000000	0.034094	0.078654	0.040000	0.045718
2026-04-30	BID	0.113053	0.000000	0.000000	0.300000	0.040000	0.033365
2026-04-30	CTG	0.105919	0.000000	0.000000	0.092571	0.040000	0.046248
2026-04-30	MBB	0.083973	0.000000	0.000000	0.185321	0.040000	0.038349
2026-04-30	TCB	0.083984	0.192158	0.000000	0.000000	0.040000	0.040518
2026-04-30	ACB	0.047678	0.000000	0.042229	0.000000	0.040000	0.053642
2026-04-30	VPB	0.082023	0.000000	0.000000	0.000000	0.040000	0.029283
2026-04-30	HDB	0.050207	0.070429	0.000000	0.043454	0.040000	0.034504
2026-04-30	VIB	0.021560	0.000000	0.000000	0.000000	0.040000	0.041216
2026-04-30	TPB	0.017822	0.000000	0.000000	0.000000	0.040000	0.034538
2026-04-30	STB	0.046217	0.100852	0.072188	0.300000	0.040000	0.040231
2026-04-30	SHB	0.027759	0.000000	0.000000	0.000000	0.040000	0.033028
2026-04-30	LPB	0.046625	0.300000	0.055763	0.000000	0.040000	0.030759
2026-04-30	SSB	0.019006	0.000000	0.231462	0.000000	0.040000	0.048643
2026-04-30	MSB	0.014203	0.000000	0.000000	0.000000	0.040000	0.035803
2026-04-30	OCB	0.011703	0.000000	0.000000	0.000000	0.040000	0.047657
2026-04-30	NAB	0.007350	0.258381	0.051370	0.000000	0.040000	0.044595
2026-04-30	BAB	0.004691	0.000000	0.300000	0.000000	0.040000	0.067703

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2026-04-30	KLB	0.002994	0.000000	0.023740	0.000000	0.040000	0.027510
2026-04-30	PGB	0.003224	0.000000	0.012073	0.000000	0.040000	0.028279
2026-04-30	VAB	0.003250	0.000000	0.000000	0.000000	0.040000	0.036028
2026-04-30	VBB	0.004499	0.000000	0.027886	0.000000	0.040000	0.037161
2026-04-30	SGB	0.001749	0.000000	0.149194	0.000000	0.040000	0.050563
2026-04-30	ABB	0.006038	0.078180	0.000000	0.000000	0.040000	0.031555
2026-04-30	BVB	0.003132	0.000000	0.000000	0.000000	0.040000	0.043103
2026-05-31	VCB	0.191279	0.000000	0.000000	0.017306	0.040000	0.044190
2026-05-31	BID	0.111754	0.000000	0.000000	0.147149	0.040000	0.031357
2026-05-31	CTG	0.103916	0.000000	0.000000	0.300000	0.040000	0.044625
2026-05-31	MBB	0.080326	0.120225	0.000000	0.235545	0.040000	0.037771
2026-05-31	TCB	0.089926	0.014423	0.000000	0.000000	0.040000	0.037320
2026-05-31	ACB	0.046210	0.000000	0.051955	0.000000	0.040000	0.053378
2026-05-31	VPB	0.079088	0.000000	0.000000	0.000000	0.040000	0.030974
2026-05-31	HDB	0.050968	0.000000	0.000000	0.000000	0.040000	0.034366
2026-05-31	VIB	0.021240	0.000000	0.000000	0.000000	0.040000	0.041616
2026-05-31	TPB	0.017256	0.000000	0.000000	0.000000	0.040000	0.033777
2026-05-31	STB	0.048713	0.131561	0.077103	0.300000	0.040000	0.040531
2026-05-31	SHB	0.025589	0.000000	0.000000	0.000000	0.040000	0.033494
2026-05-31	LPB	0.051117	0.300000	0.045324	0.000000	0.040000	0.030975
2026-05-31	SSB	0.018188	0.000000	0.265335	0.000000	0.040000	0.049096
2026-05-31	MSB	0.014930	0.000000	0.000000	0.000000	0.040000	0.035901
2026-05-31	OCB	0.011469	0.000000	0.000000	0.000000	0.040000	0.047612
2026-05-31	NAB	0.007717	0.259867	0.016185	0.000000	0.040000	0.043596
2026-05-31	BAB	0.004638	0.000000	0.300000	0.000000	0.040000	0.065296
2026-05-31	KLB	0.003265	0.000000	0.000000	0.000000	0.040000	0.026757

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2026-05-31	PGB	0.003053	0.000000	0.028042	0.000000	0.040000	0.031467
2026-05-31	VAB	0.003203	0.000000	0.000000	0.000000	0.040000	0.035856
2026-05-31	VBB	0.005524	0.000000	0.009129	0.000000	0.040000	0.040016
2026-05-31	SGB	0.001685	0.000000	0.195197	0.000000	0.040000	0.053941
2026-05-31	ABB	0.005904	0.173923	0.011730	0.000000	0.040000	0.031777
2026-05-31	BVB	0.003042	0.000000	0.000000	0.000000	0.040000	0.044311

Phụ lục A – Lãi suất Phi rủi ro VN10Y

Lãi suất trái phiếu chính phủ Việt Nam kỳ hạn 10 năm (VN10Y) dùng làm lãi suất phi rủi ro time-varying. RF tháng = VN10Y_năm / 100 / 12.

Năm	VN10Y (%/năm)	RF tháng (%)
2014	6.50	0.5417
2015	6.50	0.5417
2016	6.20	0.5167
2017	5.80	0.4833
2018	4.58	0.3817
2019	4.58	0.3817
2020	3.00	0.2500
2021	2.53	0.2108
2022	4.00	0.3333
2023	3.50	0.2917
2024	3.06	0.2550
2025	3.10	0.2583
2026	4.35	0.3625

Phụ lục B – Tóm tắt Phương pháp luận

- Dữ liệu: Giá đóng cửa tháng và vốn hóa thị trường của 25 cổ phiếu ngân hàng niêm yết (HOSE/HNX), giai đoạn 06/2014 – 05/2026 (144 tháng).
- Cửa sổ trượt (rolling window): 36 tháng, tái cân bằng hàng tháng.
- Phân cụm K-means (k=4): feature = (idiosyncratic momentum 6-1 tháng, –vol). Tín hiệu tổng hợp theo Jegadeesh & Titman (1993) và Baker, Bradley & Wurgler (2011).
- Black-Litterman: $\Pi = \delta \cdot \Sigma \cdot w_{mkt}$ (Reverse-CAPM); $\Omega = \tau \cdot P \cdot \Sigma \cdot P^T$; $\mu_{BL} = M^{-1}[(\tau \Sigma)^{-1} \Pi + P^T \cdot \Omega^{-1} \cdot q]$; $\Sigma_{BL} = M^{-1} + \Sigma$; $\tau = 1/36$.
- Tối ưu hóa: SLSQP maximise Sharpe(μ_{BL} , Σ_{BL}), $\Sigma w = 1$, $0 \leq w \leq 30\%$, 10 điểm xuất phát.
- Đánh giá: OOS 106 tháng; Sharpe, Sortino, MDD, Calmar, Win-rate.

- Kiểm định: Paired t-test, Jobson-Korkie (1981), Jarque-Bera, Ljung-Box.
- Monte Carlo: 2000 mô phỏng nhiễu view q (khai thác tính tuyến tính của BL).